20 Managing Iatrogenic Events

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An iatrogenic event is defined as a procedure "induced inadvertently by a physician or surgeon" (Merriam-Webster dictionary). This is true for dentistry as well, with cases of endodontic malpractice being among the most prevalent in dentistry.[70,132,293] These occurrences are not limited to the United States, with other countries reporting similar percentages of endodontic malpractice claims.[257]

Although iatrogenic accidents cannot be prevented with absolute certainty, their occurrences can be greatly reduced with the aid of the "Three Ts": training, technique, and technology. Incumbent with the first (training) is expertise with the other two factors, especially regarding technology and, more specifically, the dental operating microscope (DOM) together with cone-beam computed tomography (CBCT).[12,182,272] They have proven themselves indispensable in treatment planning based on the diagnosis,[272] locating canals including complex anatomical spaces and periapical pathosis,[12,160,180,183,197,291,322,347] removing separated instruments,[324,346] repairing perforations,[268] preventing iatrogenic accidents,[182,364] and enhancing the delivery of quality care.[174] The superior visualization with the aid of CBCT imaging permits more accurate clinical evaluations including the identification of the canal root morphology that cannot be visualized with the DOM,[48] and allowing for a precise assessment of the challenge and the appropriate treatment plan for its resolution. Illumination and magnification along with the 3D imaging are indispensable for all of the treatment scenarios that follow, which are therefore assumed to be performed with the DOM and CBCT imaging, which has proven as accurate as direct visualization.[13]

Sodium Hypochlorite

Often happening without warning, the progression of events following sodium hypochlorite (NaOCl) extrusion into periradicular tissues is both rapid and startling to witness. In addition to the physical outcomes, the experience is likely to leave the patient wary of both the procedure and practitioner.

Although the properties and methods of delivering NaOCl are covered in Chapter 8, the gravity of the consequences of its misuse merits a brief highlight. NaOCl is an effective irrigating solution in concentrations ranging from

0.5% to 6%[149,373] and is able to disrupt the biofilm,[58] dissolve necrotic tissues,[211] and remove organic components of the smear layer.[20] However, it is extremely cytotoxic, irrespective of the tissue it contacts,[231] with concentrations as low as 0.01% lethal to fibroblasts in vitro.[138] NaOCl injection into vital tissue initiates hemolysis and ulceration, damages endothelial and fibroblast cells, and inhibits neutrophil migration.[107,109] A study investigating the physical effects of NaOCl on bone ex vivo found degradation of the organic matrix collagen and no sign of living cellular contents (Fig. 20.1).[163]

The majority of reported NaOCl complications appear to be due to an inaccurate working length, iatrogenic widening of the radicular foramen, lateral perforation of the root, or wedging of the irrigation needle.[171,316] The pattern of facial distention is comparable to that encountered with a cervicofacial emphysema (discussed later), dissecting fascial spaces in a similar fashion, with the concomitant tissue destruction magnifying the effect and impeding recovery. This swelling is further exacerbated by the presence of effervescing agents such as hydrogen peroxide, potentiating the effect of the hypochlorite and amplifying soft-tissue involvement.[23,186] The initial patient reaction to the fluid insult is severe and immediate pain, with marked edema or bruising that may continue to extend over the injured side of the face, cheek, or lips (Fig. 20.2). There may be spontaneous and profuse hemorrhaging from the canal space, and, in the case of maxillary posterior teeth, the patient may report periorbital pain,[24,107] chlorine taste, or irritation of the throat.[80] The amount that has been expressed into the periradicular area, the spatial location of the fluid introduction, and the proximity to sensitive anatomic structures will dictate the severity and duration of the reaction. Paresthesia, either transient[24] or permanent,[255] can occur, especially if interventional treatment is not rendered immediately. In addition to sensory deprivation, motor nerve

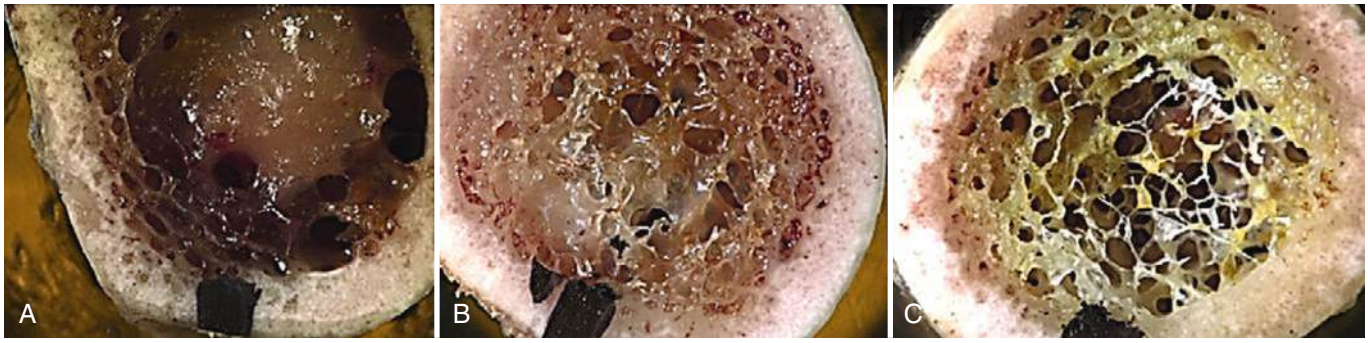


Fig. 20.1 Femur shaft cross sections (**A**) untreated bone section; (**B**) saline-treated bone; (**C**) sodium hypochlorite (NaOCl)-treated bone section. Grossly, NaOCl caused significant changes in cancellous structure, leaving large structural craters of apparent demineralization. (From Kerbl FM, DeVilliers P, Litaker M, Eleazer PD: Physical effects of NaOCl on bone: an ex vivo study. *J Endod* 38:357, 2012.)



Fig. 20.2 A, Clinical presentation 3 days following the extrusion of sodium hypochlorite (NaOCl) during treatment of the maxillary right canine. The patient had difficulty opening her right eye, and the swelling had extended to the submandibular/sublingual regions on the affected side; she also experienced limited mouth opening. There was altered sensation infraorbitally and in the region of the upper right lip, but no reported paresthesia of the alveolar or facial nerves. **B,** One-month postextrusion, the ecchymosis had resolved, with a return of sensation and full function. Therapy was completed without incident. (From de Sermeño RF, da Silva LA, Herrera H: Tissue damage after NaOCl extrusion during root canal treatment. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108:e46, 2009.)

dysfunction secondary to irreversible chemical damage has been reported (Fig. 20.3).[236] In this case, there was distinct loss of upper lip and cheek function, with problems in the musculature. In mandibular teeth, extension into the submandibular, submental, or sublingual regions can compromise the airway, compelling immediate hospitalization and surgical intervention to avert a life-threatening episode.[35]

Upon realizing that an extrusion event has occurred, it is imperative that the clinician cease further treatment and immediately engage in measures targeted at diminishing the effect of the chemical; ignoring the event reduces outcomes.[187,255] Treatment protocols are empirical and are predicated on cause and severity. With the emphasis on pain control, local anesthetics and analgesics can be administered, along with cold compresses, in an effort to moderate swelling and edema. Directed at diluting the effect of the extrusion, it has been suggested that the clinician immediately irrigate the canal with normal saline to encourage bleeding, both diluting the irritant and removing it from the site of the injury.[80,206] In addition, clinicians may immediately irrigate the canal and periradicular areas of the extrusion with saline, using a negative pressure irrigation device with the cannula placed at the radicular foramen. In one report, the subsequent reaction was short lived with this approach.[267]

After one day, warm compresses are substituted for the cold, and warm oral rinses are prescribed to stimulate the local microcirculation. Antibiotics are usually prescribed, and their administration commensurate with the severity of hard- and soft-tissue destruction and necrosis; milder cases can be treated with oral medications, whereas severe presentations are better controlled and modulated in a hospital setting with intravenous routes of delivery. The use of corticosteroids is equivocal; the control of the spreading edema must be balanced with the increased risk of infection. The patient should be monitored closely; a marked increase in the size or extent of the swelling or signs of impending airway obstruction mandate immediate referral to the hospital or a maxillofacial surgeon for more aggressive treatment and management. Lastly, reassuring patients that recovery, despite their current appearance, is usually complete and uneventful, will ease their anxiety. They should be informed as to why it occurred and what to expect during their recovery. Recovery should be monitored and documented daily until the resolution is imminent.

Prevention involves attention to detail and an appreciation for fluid dynamics. In summary, the clinician should do the following:

- Establish an accurate working length and avoid overinstrumentation/enlargement of the radicular foramen.
- If irrigating using positive pressure, employ a small side-vented needle placed no closer than 2 mm from the working length. Express the fluid slowly and observe that it is venting through the access cavity.[34]
- Carefully assess the canal integrity for signs of perforation or other large portals of fluid egress.
- Avoid wedging the needle tip in the canal space or inserting it beyond the working length.
- Confirm the identity of the solution prior to injection or irrigation.[124]

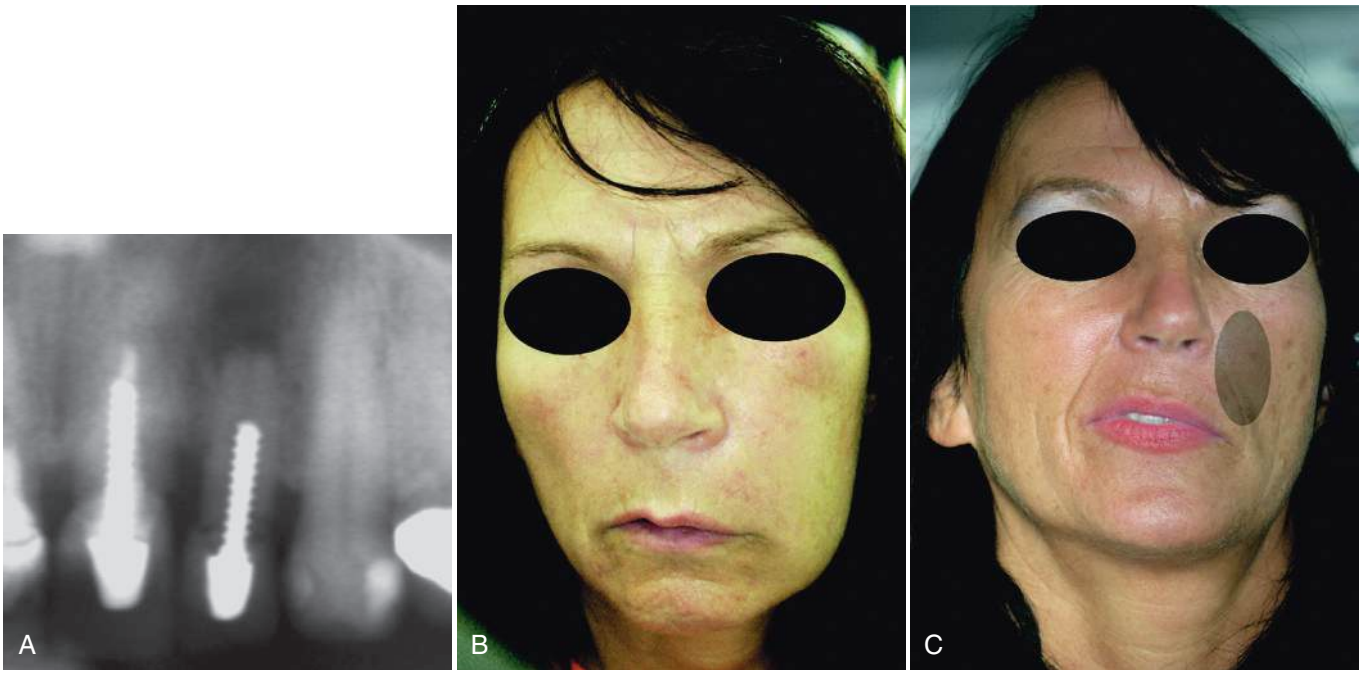


Fig. 20.3 A, Radiograph of the maxillary left lateral incisor (#10, #22) before post removal. Radicular root resection had been completed, and the canal space is empty. Prior to recementation, the canal was irrigated with 3% NaOCl, precipitating the extrusion event. **B,** Twenty-four hours postextrusion, there was altered sensation infraorbitally and from the upper left lip to the left lip corner. In addition, the buccal branch of the facial nerve was affected, as evidenced by a distinct loss of upper lip and cheek function (the corner of the mouth could not be pulled up by the mimic musculature). The mouth opening was limited to 20 mm (**C**), 3 years postincident. The weakness of the mimic musculature of the left face side is clearly visible. Attempts to laugh resulted in a hanging left lip corner secondary to the weakness of the motor innervation by the facial nerve. In the gray-marked area there is a permanent hypoesthesia. (From Pelka M, Petschelt A: Permanent mimic musculature and nerve damage caused by sodium hypochlorite: a case report. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 106:e80, 2008.)

Instrument Separation

The introduction of nickel titanium (NiTi) rotary instruments into endodontics revolutionized the way the root canal system is instrumented. However, with the advent of rotary NiTi files, there has also been a perceived increase in the occurrence of separated instruments,[101,281] with some reports indicating that separation of NiTi rotary instruments occurs more frequently than hand instruments.[343] On radiographic examination, separated instruments can usually be discerned by the distinct radiopacity within the canal. However, if the canal with a separated instrument is subsequently obturated with radiopaque materials, it may not be visualized on the radiograph. Upon removal of filling materials from the canal, a separated instrument may be visualized under the DOM or become apparent on radiographs for the first time (Fig. 20.4).

CAUSES OF INSTRUMENT SEPARATION

A common cause of instrument separation is improper use.[101,285] Typical examples of improper use include

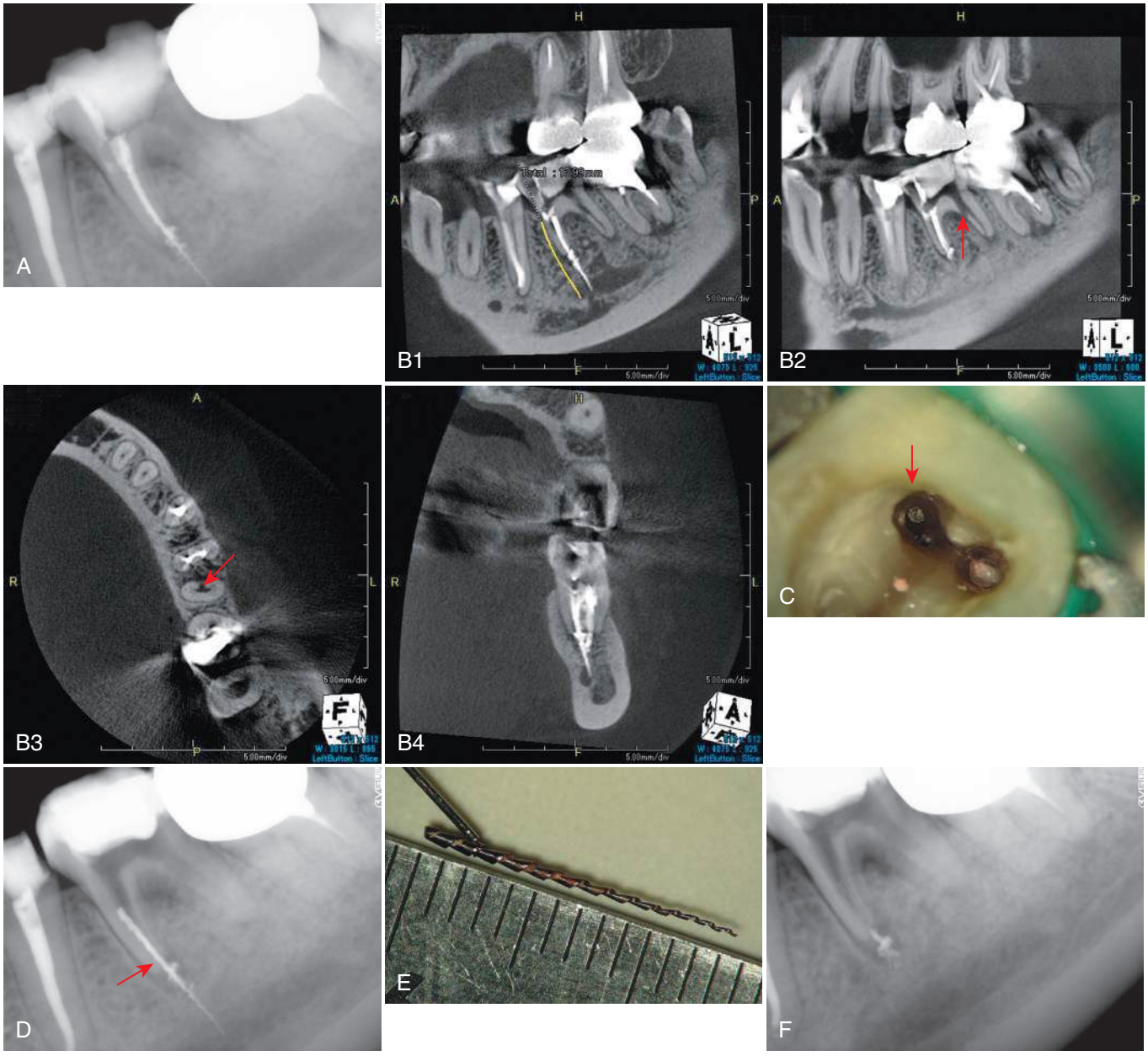


Fig. 20.4 **A,** Radiograph of the mandibular left first molar with a long indistinct separated instrument obviously longer than 4.5 mm in the mesial root that is extruded beyond the radicular foramen. **B1,** Sagittal view of cone-beam computed tomography (CBCT) showing the radicular end of the separated instrument is extended into the IAN and it measures 14 mm on it. **B2,** Sagittal view of CBCT showing root perforation in the distal canal *(red arrow)*. **B3,** Axial view of CBCT showing the perforation in the distal canal *(red arrow)*. **B4,** Coronal view of CBCT showing the radicular end of the separated instrument is in the IAN. **C,** Separated instrument in the mesiobuccal canal upon removal of root filling materials *(red arrow)*. **D,** Radiograph showing removal of the root fillings in the mesiobuccal canal revealed the overall length of the separated instrument *(red arrow)*. **E,** Retrieved separated instruments measuring 14 mm on the ruler. **F,** Radiograph showing the completion of the separated instrument removal.

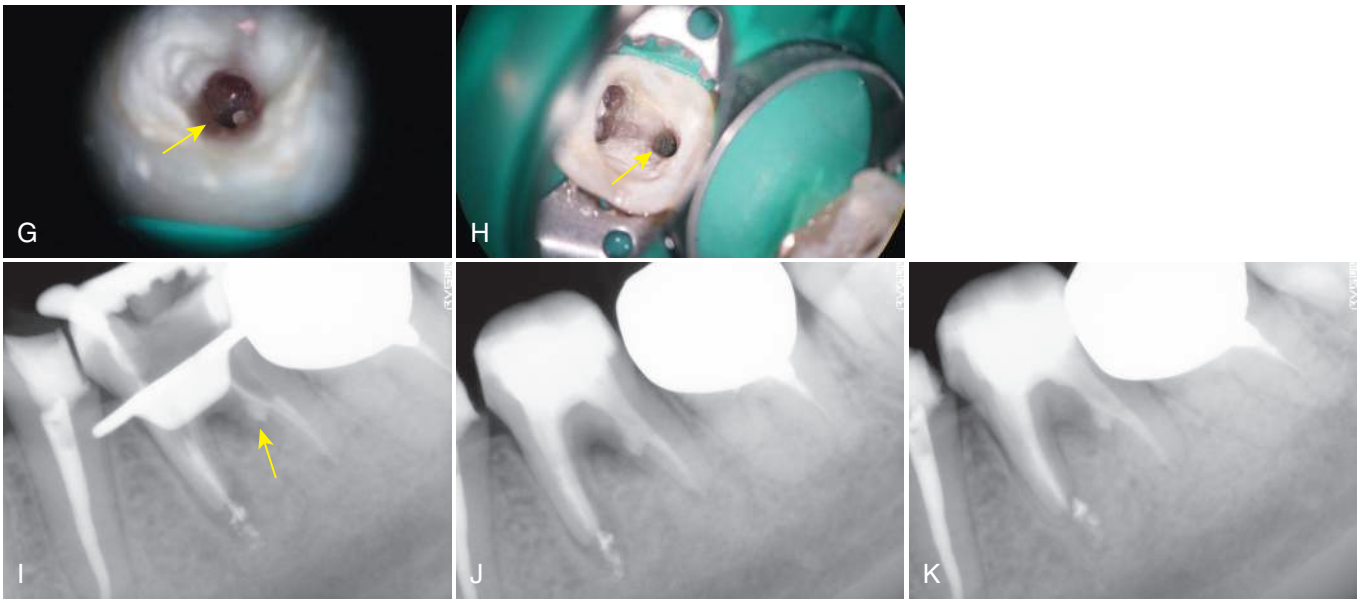


Fig. 20.4, cont'd G, Perforation site in the distal canal *(yellow arrow)*. **H,** Perforation repaired with mineral trioxide aggregate (MTA) *(yellow arrow)*. **I,** Postoperative radiograph showing root canal fillings in all the canals and perforation repair *(yellow arrow)*. **J,** One-month postoperative radiograph revealing the periradicular lesions still present. **K,** Three-month postoperative radiograph showing periradicular healing.

inadequate access, overuse of the instrument, too much radicular pressure during instrumentation,[287,339] and the continuous use of a large instrument in a curved canal.[127,251] Additional contributing factors are operator experience,[229,331] rotational speed,[66,194] canal curvature (radius),[195,251] instrument design and technique,[39,177] torque setting,[102] manufacturing process,[4] the type of NiTi alloys used,[106] the type of rotational motion (continuous rotations vs. reciprocating motions),[235] the type of tooth,[64,218,299] and absence of a glide path.[233] Based on these factors, it is not surprising that the reported incidence of separated NiTi files ranges from 0.4% to 23%.[4,6,240,253,288,317]

Separated instruments seem to occur more frequently in molars than in anterior teeth, especially in mandibular molars. This high incidence of instrument fracture in molars may be related to accessibility to the canal, diameter of the root canal, and root canal curvature.[64,218,299] Endodontists have reported the incidence of instrument separation to be about 5%.[4,229] The clinician must inform the patient of the instrument separation and treatment options as well as potential complications of both the separated instrument retention and its attempted removal.

Endodontic instruments fracture as a result of cyclic fatigue or torsional fatigue[60] or a combination of both types of fatigue.[285,297,345] Cyclic fatigue is caused by repetitive compressive and tensile stresses acting on the outer portion of a file rotating in a curved canal, leading to cyclic failure without previous signs of plastic deformation.[229] Torsional failure occurs when the tip of the instrument binds but the shank of the file (driven by the handpiece) continues to rotate.[285] Shear fracture of the material then occurs when the maximum strength of the material is exceeded.[179] Clinically, cyclic fatigue seems to be more prevalent in curved root canals, whereas torsional failure might happen even in a straight canal.[55,372] Cyclic-fatigued instruments are reported to become less resistant to torsional fatigue failure,[168] whereas the use of rotary instruments at a high torque setting is reported to increase the risk of cyclic fatigue failure.[101] In general, smaller sized instruments with smaller tapers are more resistant to cyclic fatigue and more susceptible to torsional fatigue, while larger sized instruments with greater tapers tend to be opposite to those types of fatigue. Therefore smaller instruments should be used to shape the canal where cyclic fatigue is expected to occur. Interestingly, the use of reciprocal motions (as opposed to rotational motions) is reported to extend the life span of an instrument[69,371] and increase cyclic fatigue resistance.[235] It is recommended that instruments be used in a reciprocating motion when either torsional or cyclic fatigue is expected to occur while shaping a severely curved canal and that the instruments used to prepare such curved root canals be discarded after use to prevent instrument fracture.

MANAGEMENT OF SEPARATED INSTRUMENTS

The management of separated instruments includes nonsurgical orthograde (mechanical or chemical) or surgical approaches. The mechanical removal involves the use of tools dedicated to separated instrument removal such as extractors, wire loops, post removal systems, ultrasonics, and laser irradiation (see also Chapter 10). The chemical removal involves the use of chemical solvents for instrument corrosion and electrochemical process for instrument dissolution. Surgical approaches include radicular surgery, intentional replantation, root amputation, or hemisection, which are in general considered relatively invasive due to the amount of dentin loss.

For a nonsurgical approach in removing a separated file, various methods and devices have been developed (see also Chapter 10). When a separated instrument extends above the root canal orifice, it can often be easily removed with a

hemostat, Steiglitz forceps, a modified Castroviejo needle holder,[96] or a Perry plier.[357] A spoon excavator or a Caulfield retriever, or EndoCowboy (Koehrer Medical Engineering, Neuss, Germany) can also be used to engage and remove the separated instrument with coronally directed pressure. In the event that a hand file can bypass the separated instrument, a "braiding technique" can be used whereby several Hedström files are inserted along the bypassed instrument, the files are twisted to grasp the separated instrument, and it is then extracted as one unit. When a fractured instrument is below the canal orifice and cannot be bypassed, one basic method for removing the separated instrument requires the exposure of approximately 2 mm of the coronal portion of the separated file. This allows the use of an extracting device for engaging the file, securing it, and removing it. The Masserann kit (Micro-Mega, Besancon, France) is a popular system for the removal of separated instruments located within the straight part of the canal. This kit involves the use of trephine burs to expose the coronal portion of the separated instrument and for creating a space for an extractor. An extractor is then used to grasp and remove the separated instrument. The limitation of this system is that it cannot be used for cases involving separated instruments located beyond the midroot or in a curved canal, as this technique involves a considerable amount of dentin sacrifice that can weaken the root structure and increase the risk of perforation.[369] An alternative to the Masserann kit is the Endo Safety System.[359] This system differs from the Masserann kit in that it uses smaller diameter trephine burs and the extractor has its own unique mechanism for grasping instruments. The Endo Extractor (Brasseler Inc., Savannah, GA) can also be used in these types of cases. This system consists of a trephine bur to expose the coronal portion of the separated instrument and a hollow tube extractor that is threaded over the exposed portion of the separated instrument and bonded to it with cyanoacrylate adhesive.[110] The Canal Finder System (FaSociété Endo Technique, Marseilles, France) or the Endo-Puls System (EndoTechnic, San Diego, CA) provides a different approach to a separated instrument. The system is composed of a handpiece and files that are designed exclusively for the system.[189] The system produces a vertical movement with a maximum amplitude of 1 to 2 mm, which decreases when the speed increases.[147] This vertical movement of the file will also help bypass an instrument fragment. The flutes of the file can mechanically engage with the separated instrument, and with the vertical vibration, the instrument fragment can be loosened or even retrieved.[148] The mechanical movement of the file can get aggressive, so the clinician must use caution to avoid root perforation, especially in a curved canal. These systems are technique sensitive and therefore results can vary between cases.

Ultrasonic instruments are very effective for the removal of separated instruments.[54,212,280] However, it is important to realize that separated NiTi files tend to fracture repeatedly when ultrasonics ia continuously applied to them, whereas stainless steel instruments are more resistant and are more easily removed with ultrasonics than NiTi instruments.[280,324,354] Small ultrasonic instruments allow continuous and improved vision of the field of operation. The use of ultrasonics, especially when performed under the DOM, enhances both preparation and removal procedures and can provide safety and accuracy in the process.[218,324,353] The ultrasonic tip is placed on the staging platform between the exposed portion of the separated file and the canal wall, and it is vibrated around it in a counterclockwise direction, applying an unscrewing force to the separated file. This technique will help with removing most rotary separated instruments that have a clockwise cutting action. If it is known that the separated file has a counterclockwise cutting action, then a clockwise rotation will be necessary. The energy applied will aid in loosening the file, and occasionally the file will suddenly exit out of the canal.

However, the use of ultrasonics does have some drawbacks. Upon ultrasonic activation, a separated file can be accidentally extruded into the bone, either radicularly or through a perforation when ultrasonics i applied to the top of the separated file or from the outer wall to the side of it. In addition, the ultrasonic activation of a separated instrument has been reported to separate into small fragments from the original fragment while troughing around it.[53] Smaller file fragments can easily get pushed into a more radicular level if they are pushed on the top by ultrasonics, making it more difficult to remove them due to the additional procedure, which would eventually increase the risk of perforation and extra dentin sacrifice, especially in the apical third or beyond the curve. At the same time, smaller file fragments can be considered easier and faster to remove than longer file fragments if they are visible simply because shorter file fragments requires less dentin sacrifice to loosen it than longer ones. On the other hand, such long instruments as separated at the shaft can also be easily removed as they can simply be grasped with a plier or forceps around the orifice and pulled out.[254] If the coronal end of the long file fragment is out of reach of the plier or the forceps, it will not be as easy. It all depends on how the separated instruments are managed to be successful in removing them as there are many variables involved in the procedures that affect the success in instrument retrieval.

The success rate for removing separated files varies from 33% to 95%[6,64,253,279,354] with the time required for using ultrasonic techniques varying between 3 and over 60 minutes.[8,212,331] The variations in success rates are due to the location of fractured instruments, the diameter of the root canal,[299] the degree of canal curvature,[8,252,313] the radius of curvature,[8,251] operator experience,[8,331] operator fatigue, and the length of the separated instrument.[330] Visualization and accessibility to the separated instrument play a major role in file retrieval.[64] In general, the success rate for removal is high for separated files located before the canal curvature, moderate for those located at the curvature, and low for those located beyond the curvature.[195,358] The success rate is also more favorable when the canal curvature is less and the radius of curvature is longer.[8] The combination of ultrasonic techniques and microscopes typically improves success rates in file removal.[64,100,218,354]

It has been recommended that removal attempts of separated instruments from root canals should not exceed 45 to 60 minutes because the success rates may drop with increased treatment time.[324] The reduced success rate could be related to operator fatigue or overenlargement of the canal, which compromises the integrity of the tooth and increases the risk of perforation. Any case where the separated

instrument can be visualized under the DOM may be considered less challenging to manage.

It would be the worst-case scenario if the separated file was decided to be retained only after a lot of dentin was sacrificed and the canal perforation occurred during the instrument retrieval attempts. Hence it is in the best interests of both the patient and the clinician to make an accurate diagnosis and a predictable treatment plan using CBCT to prevent any procedural mishaps. In other words, separated instrument removal attempts should be made only after the preparation is completed based on the treatment plan that always includes surgical interventions and the retention of the separated instrument, as the separated file can be incorporated as part of the obturation and kept under observation.[92,97,253,314]

CONDITIONS FOR SEPARATED INSTRUMENT REMOVAL ATTEMPTS

Root canal preparations for separated instrument removal followed by removal attempts can be made in either a dry or a wet environment. Dry conditions provide better visibility when using the DOM, preventing fewer additional procedural accidents.[281,313,324,353] For this reason, the root canal preparations for instrument removal should be conducted in a dry condition. However, heat generation by ultrasonics is inevitable,[115,134,181,199,299,313,324,353] and a temperature rise above 10°C on the external root surface can damage the periodontal tissues.[85,325] In addition, the separated instrument will be susceptible to secondary fracture when ultrasonic tips are in contact with the file.[332,353] Therefore it is recommended that pulsing and in/out motions be used to prevent the secondary fracture as well as the temperature rise during ultrasonic activation in preparation. At the same time, the ultrasonic tip should be activated at the lowest possible power setting.[42,88,200,353] Only after the loosening of the separated instrument is confirmed in preparation, the removal attempts can be made in a wet condition with a fluid to take advantage of cavitation and acoustic stream, which facilitates instrument removal. Ethylenediaminetetraacetic acid (EDTA) solution or oil with the lowest surface tension is recommended as the fluid for this purpose as EDTA solution removes hard-tissue debris produced during ultrasonic removal attempts and the oil enhances lubrication together with both cavitation and acoustic stream to remove the separated instrument.

ROOT CANAL PREPARATION TECHNIQUES

Root Canal Preparation for Visible Instrument Retrieval

In the root canal preparation stage for instrument retrieval, the goal is to make the separated instrument "dance" by enlarging the canal and loosening it with ultrasonics based on the diagnosis and the treatment plan whether ultrasonic oscillation or the loop is used to get the separated instrument out in the instrument retrieval stage. The majority of separated instruments are NiTi,[317] with the mean length of separated instruments being 2.5 to 3.5 mm.[157] Many NiTi rotary instruments used have a .04 to .08 taper range with a tip size of #20 to #30, resulting in a coronal diameter of about 0.30 to 0.58 mm. If the coronal diameter of the separated instrument is smaller than 0.45 mm, then a #3 modified Gates Glidden bur (#3 MGG bur from the Terauchi File Retrieval Kit [TFRK] [DELabs Santa Barbara, CA]) should be used to enlarge the canal to the separated instrument, followed by the micro-trephine bur (MT bur) (DELabs for exposing the coronal portion of the separated instrument, with less risk than ultrasonics for secondary fracture of the separated instrument (see Fig. 20.20, *A* and *D*).[331,332] The inner diameter of the micro-trephine bur is 0.45 mm and it is then used, in a counter clockwise rotation, with the intent of unscrewing the separated instrument. This bur should be used at 600 rpm in wet conditions; if the rotational speed exceeds 600 rpm, unintentional ledge formation may occur, especially in a curved canal greater than 15 degrees.

If the coronal diameter of the separated instrument is larger than 0.45 mm or the canal curvature is greater than 15 degrees, then a larger NiTi rotary instrument, with tip size 0.15 mm (three sizes) larger than the coronal diameter of the separated instrument, should be used to enlarge the canal coronal to the separated instrument, followed by ultrasonic instruments to expose the coronal semicircular portion.

First, a few drops of mineral based oil with lubricant additives is placed into the canal on the separated file to facilitate the separation of it from the canal wall, and then excess oil is removed with paper points or mild airblows for the visible root canal preparation with ultrasonic tips. The mineral oil almost instantly permeates the spaces between the canal walls and the tightly wedged file fragment and separates them just like loosening a rusty tightened screw from the metal plate (Fig. 20.5). Then the spoon tip with the concave portion facing the separated file (TFRK-6/12, DELabs) (see Fig. 20.20, *B1*, *C2*, and *D*) will be introduced into the thin gap between the separated file and the inside curve of the canal to create one quarter space of the circle (90-degree semicircular space) that is approximately one third of the separated file length. Second, the straight tip (TFRK-S, DELabs) (see Fig. 20.20, *B2*, *C2*, and *D*) is introduced to expand the space to complete the 180-degree semicircular space on the inside curve until the separated file gets loosened. When the depth of the space created is longer than one third of the file length and the semicircular space is greater than 180 degrees, the separated file will most likely get loosened (Fig. 20.6). The ultrasonic tip used for this purpose should be as thin as possible, allowing the clinician to both visualize the operative field and prevent both overenlargement of the canal wall and pushing the separated instrument radicularly (Fig. 20.7). In addition, the ultrasonic tip should always be applied to the area on the inner canal curve being extended in the radicular direction (Fig. 20.8) where the outer canal wall is in contact with the separated instrument to prevent secondary fracture, not where the outer canal wall is not in contact with the separated file (see Fig. 20.8, *H*).[332] If the ultrasonic tip is placed in the space outside the curve and ultrasonic energy is given to the separated instrument from the outer curve, the force applied will redirect the separated instrument in an apical direction (see Fig. 20.8, *D* and *E*), eventually pushing it more apically and making the instrument retrieval more difficult, whereas the force applied from the inner curve to the separated instrument will consequently reorient the separated instrument in a coronal direction

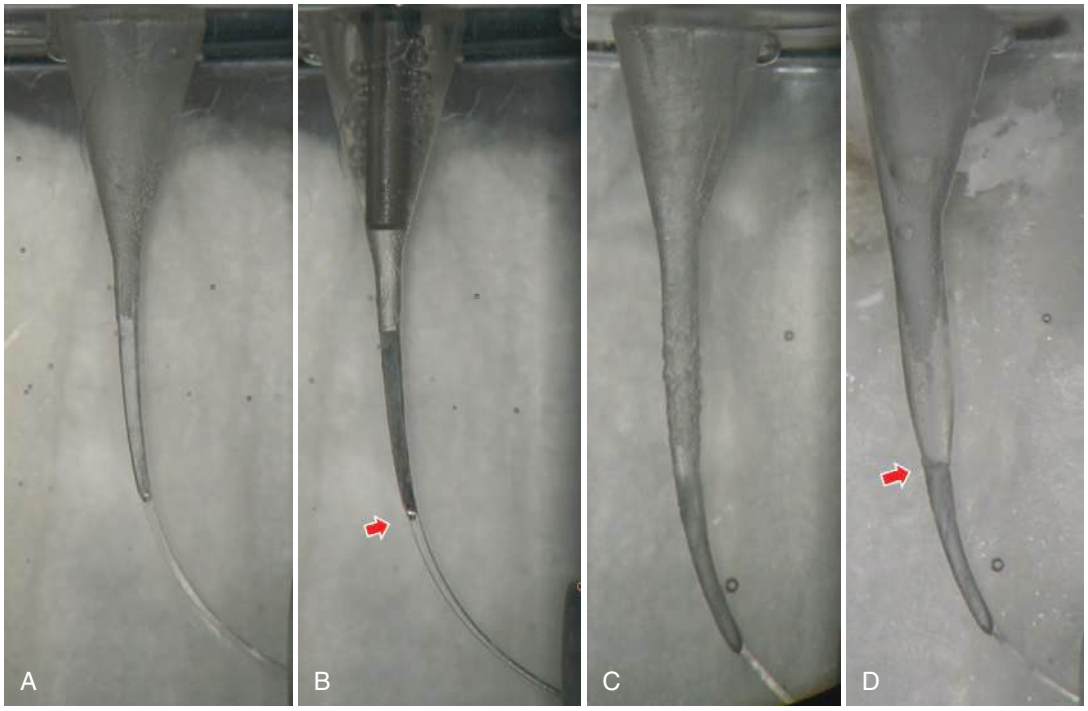


Fig. 20.5 **A,** Separated file covered with debris. **B,** Mineral oil immediately permeated the spaces between the canal walls and the separated file revealing the surface of the file and broke through the entire file. **C,** Separated file covered with debris. **D,** Water did not permeate those spaces at all.

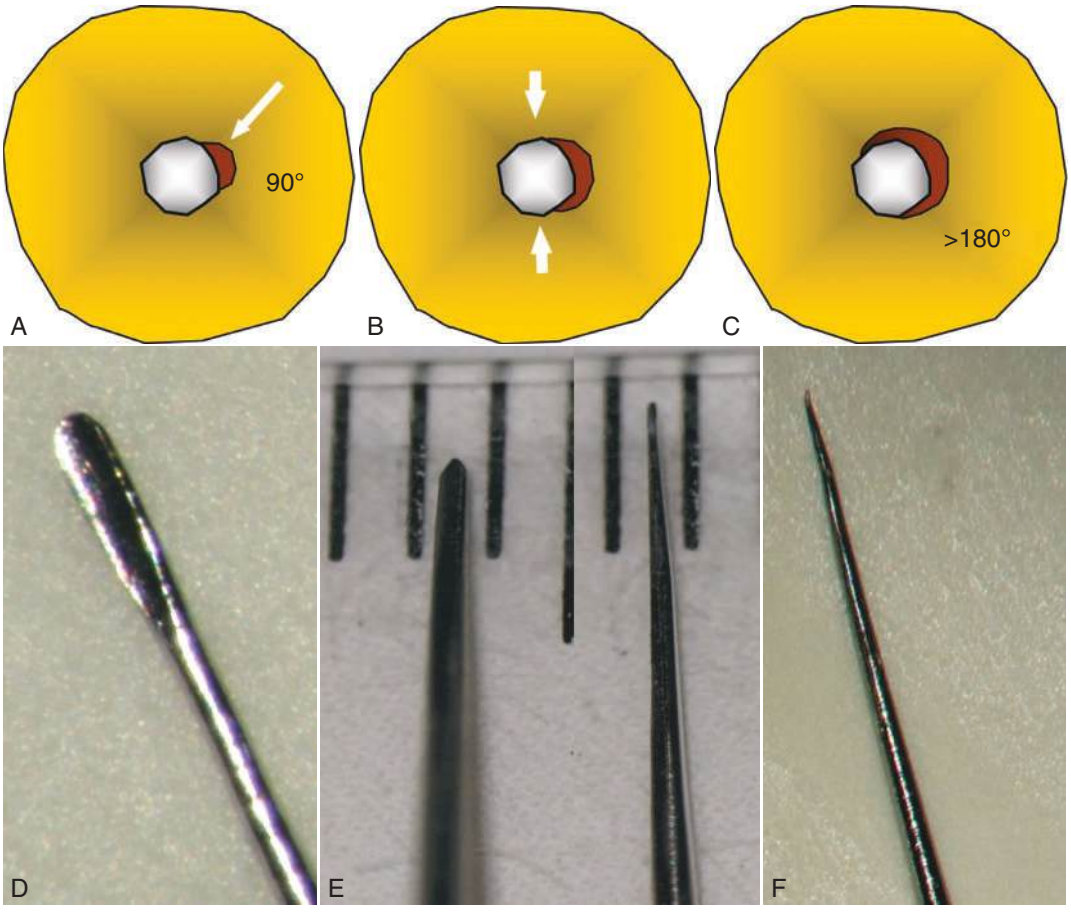


Fig. 20.6 **A**, Ninety-degree semicircular space created on the inner canal wall with the spoon tip on the microscopic view *(white arrow)*. **B**, A 150-degree semicircular space created on the inner canal wall with the straight tip, which is still not wide enough to make it "dance" because of the canal walls on both sides still holding the separated file *(white arrows)*. **C**, More than 180-degree semicircular space created on the inner canal wall with the straight tip, which results in loosening the separated file. **D**, Magnification of the spoon tip. **E**, Magnification of the Katana tip; a frontal view on the left (0.5 mm in width) and a side view on the right (0.1 mm in thickness). **F**, Magnification of the straight tip (0.1 mm in diameter).

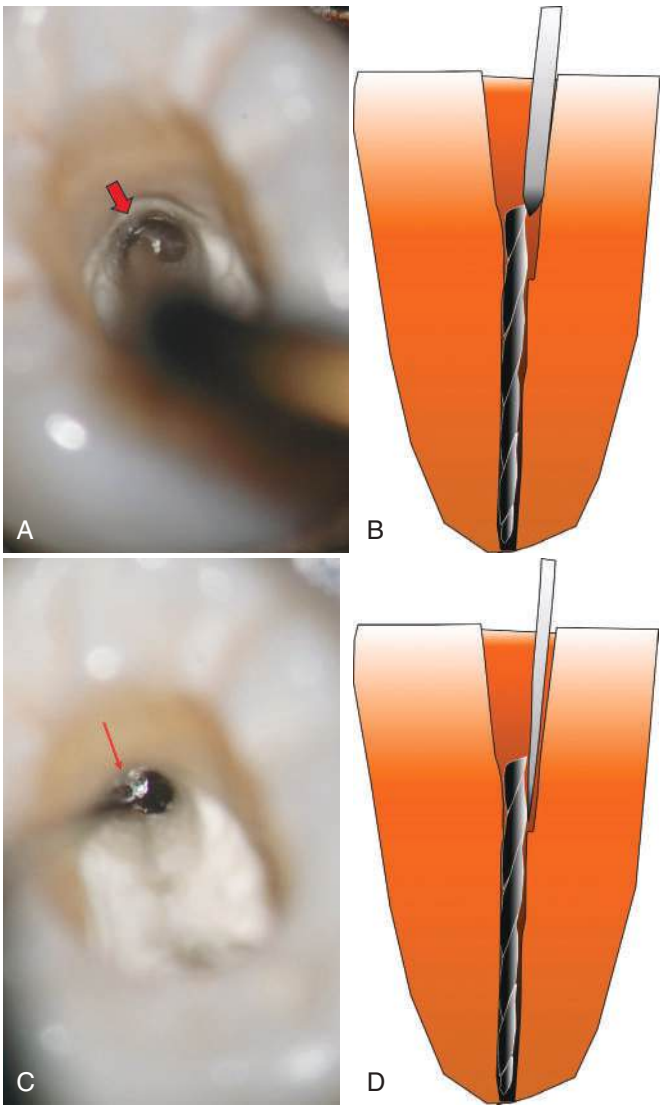


Fig. 20.7 **A**, Thicker ultrasonic tip touches the top of the separated instrument before being placed into the space on the canal wall *(red arrow)*. **B**, Side view shows the thicker ultrasonic tip in relation to the separated instrument, which may result in pushing it into a radicular direction. **C**, Thinner ultrasonic tip can be placed into the space on the canal wall without touching the top of the separated instrument *(red arrow)*. **D**, Side view shows the thinner ultrasonic tip in relation to the separated instrument.

(see Fig. 20.8, *F* and *G*). In most cases, the coronal one third of the separated file is the main portion engaged in the canal wall with more stress leaning toward the outer wall. This portion can get less resistant to mechanical force for disengagement when the mineral based oil injected permeates the tiny interfacial space between them to loosen the separated file. Therefore cutting the inner dentin wall contacting the coronal third of the separated file should typically be enough to disengage it (Fig. 20.9). For example, if the separated file fragment is 3 mm long, the 180-degree semicircular space needs to be extended radicularly about 1 mm on the inside curve to help the ultrasonic tip loosen it. Due to the force directed from the inner wall to the fragment during this ultrasonic preparation phase, the separated file may come out accidentally by ultrasonic activation at any moment (Fig. 20.10, *A*). However, a fatigued separated instrument is already extremely susceptible to secondary fracture by ultrasonic activation.[332,353] For this reason it is safer to create a semicircular space on the inner wall instead of troughing around it, and both pulsing and pecking motions should be applied to it while ultrasonics is activated to prevent instrument fracture and temperature rise, especially in dry conditions.[85,325,332,353] The depth of the semicircular space created on the inner curve must be not only longer than one third of the fragment length but also evenly the same length along the canal walls to allow the disengagement of the separated instrument. If the separated instrument is not loosened after the space creation, the depth of this space should be extended in a radicular direction until it is loosened. In general the longer the file fragment is, the more preparation time is needed to loosen it. The separated instrument must be seen "dancing," "not flexing," before making instrument retrieval attempts. The difference between the separated instrument dancing and flexing is that the separated instrument is considered dancing when it is seen moving from one place to another place, whereas it is still considered flexing when seen moving from one place to another and getting back to the original place (Fig. 20.11).

Root Canal Preparation for Nonvisible Instrument Retrieval

When the separated file is not visible beyond the curve, the mineral oil should be present all the time during the procedure to help it loosen the file fragment while preparing the root canal for instrument retrieval with ultrasonics. The key to success in loosening the separated file in nonvisible status is to get a sticky feeling with ultrasonic tips in the space on the inner canal wall. The ultrasonic tip used in this situation should be thin and sharp enough to feel the tiny gap between the inner canal wall and the separated instrument. First, a thin spoon-shaped tip (TFRK-6/12, DELabs) or a customized sword-shaped tip (any ultrasonic tip converted into a sword-shaped tip or a "Katana" tip) is precurved and placed in the space to grope for a sticky feeling while obtaining a tactile sensation on the fingers holding the handpiece (Fig. 20.12). If a larger sized tip is used, the tip will most likely hit on the top of the separated instrument before being placed into the gap between the fragment and the inner wall. It is also very typical to hear a clicking sound and get a sticky feeling when a sharp ultrasonic tip shifts from the separated file to the canal wall into the gap. When it is placed in the gap with a sticky feeling, a radiograph should be taken with the ultrasonic tip in it to confirm its position. Then the tip is activated in it to create a 90-degree semicircular space. The ultrasonic tip is simply activated directly on the file fragment from the inner curve in both pecking and pulsing motions to prevent secondary fracture and breakage of the tip without creating a 180-degree semicircular space or expanding the space laterally as it is impossible to do so in nonvisible status. The intensity of ultrasonic activation should be as low as practical for the same reasons. As this conservative root canal preparation continues with ultrasonic oscillation directed to the separated instrument from the inner wall, any dentin walls contacting the separated file will be eliminated by indirect ultrasonic oscillation transferred to it from the vibrating tip, which will eventually result in loosening it (Fig. 20.13).

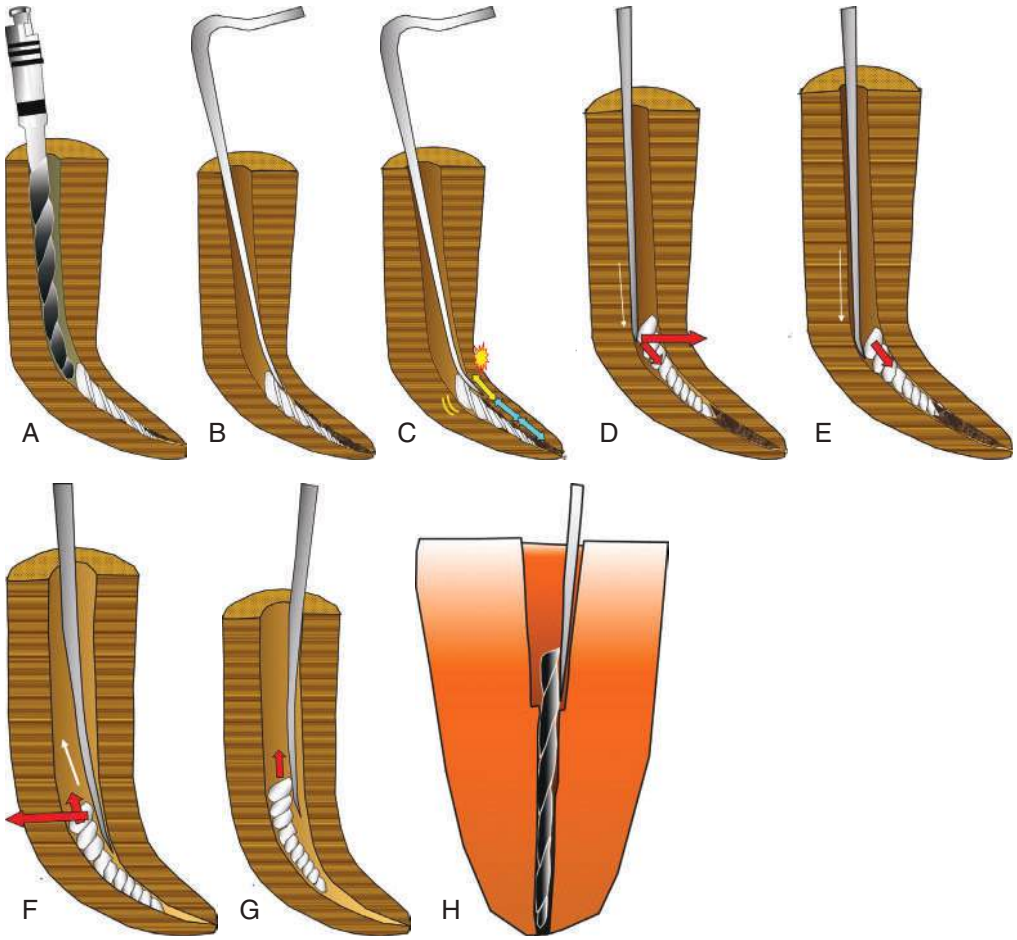


Fig. 20.8 A, Canal enlarged to the separated file with a large diameter rotary file. **B,** Space created on the inside curve with the spoon ultrasonic tip, followed by the straight tip. **C,** Space extended to about one third of the file length until the separated file dances. **D,** Ultrasonic activation from the outer curvature. **E,** Separated file redirected in a radicular direction by ultrasonic activation from the outer wall. **F,** Ultrasonic activation from the inner curvature. **G,** Separated file redirected in a coronal direction by ultrasonic activation from the inner wall. **H,** Ultrasonic tip placed on the separated file where there is no canal wall behind it, which may result in secondary fracture.



Fig. 20.9 The coronal one third of the file length is typically the part to be stuck in the canal walls, whereas the radicular portion of the file is typically not in contact with the canal walls. Stress is concentrated mostly on the coronal one third of the file length.

When the sticky feeling is lost from the ultrasonic tip or the separated file emerges from the nonvisible curve with ultrasonics, the root canal preparation for instrument retrieval can be considered complete, and then the instrument retrieval attempts should be made from this point (Fig. 20.14). The advantage of this conservative preparation technique is the clinician can be minimally invasive in preparation, while the disadvantage of this technique is that it is unpredictable when to make the separated instrument dance and the secondary fracture or breakage can occur if it is performed aggressively. It also requires the patience to be successful in loosening the fragment in this technique as it is not predictable.

When the separated file is found to be loose beyond the radicular foramen, it is not necessary to prepare the canal for its retrieval because it is not engaged in the canal wall, but it just lies in the radicular extent of the canal space (Fig. 20.15). The separated file in this status can be simply irrigated out of the canal using saline with ultrasonic irrigation devices such as ProUltra PiezoFlow (Dentsply, Tulsa, OK) and GentleWave (Sonendo, Inc, Laguna Hills, CA) (Fig. 20.16).

INSTRUMENT RETRIEVAL TECHNIQUES

Use of Ultrasonic Oscillation

Once the root canal preparation for instrument retrieval is completed, it is important to confirm that the canal wall is

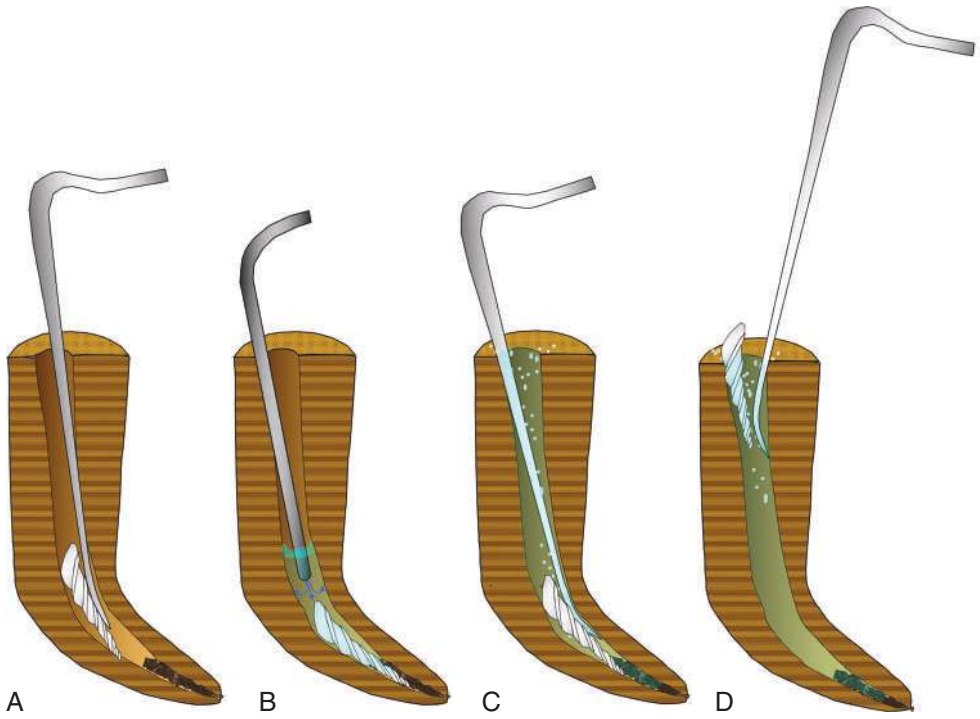


Fig. 20.10 A, Separated file may accidentally come out with ultrasonic activation during a preparation phase. **B,** EDTA solution is put into the canal to take advantage of cavitation and acoustic streaming effects for instrument retrieval. **C,** Ultrasonics must be activated within the space created. **D,** Separated file is removed from the canal with ultrasonics in the presence of EDTA solution or oil.

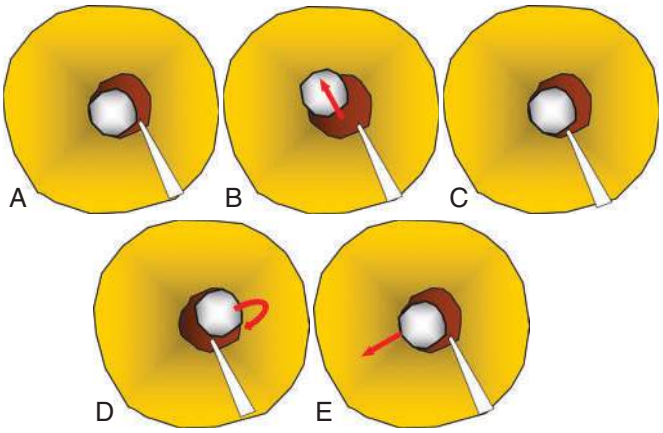


Fig. 20.11 A, Separated file in the canal with the space created on the inner wall *(right side)*. **B**, Separated file shifted to a different place in the space created *(red arrow)*. This movement shows the separated file is "dancing." **C**, Separated file in the canal with the space created on the inner wall *(right side)*. **D**, Separated file flexing to a different place *(red arrow)*. **E**. Separated file returning to the original place *(red arrow)*. This movement shows the separated file is "flexing."

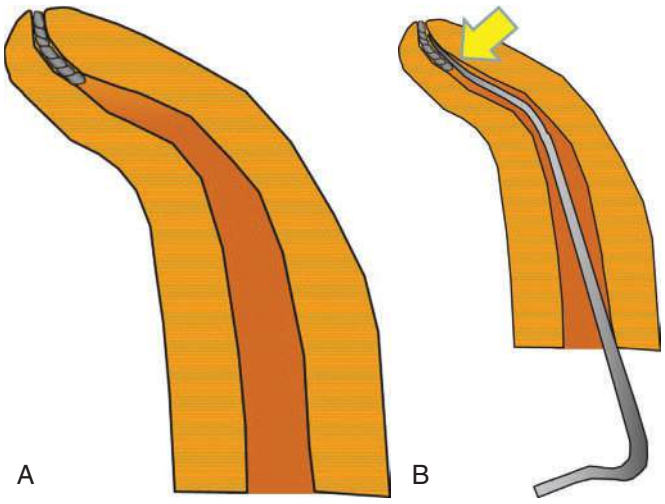


Fig. 20.12 A, Nonvisible separated file around the second curve, beyond the first curve. **B**, Thin ultrasonic tip placed in the space on the inner canal wall of the second curve.

smooth from the separated file to the coronal extent, with no overhangs on the outside canal wall. A retrospective study and the author's empirical findings show that when the separated instrument is shorter than 3 mm or between 3.1 and 4.4 mm with the curvature smaller than 30 degrees, the clinician will most likely be able to retrieve it with ultrasonics alone.[330] Then the canal is rinsed with the silicone based oil or EDTA solution and is filled with it up to the cavosurface, "not just in the root canal alone," in order to take advantage of cavitation and acoustic streaming for instrument retrieval in wet conditions (see Fig. 20.10, *B*). The ultrasonic tip placed is continuously activated at a power setting that is about 5% higher than it was in the preparation stage with short up-and-down strokes within the space created on the inner curve until the separated instrument is removed, which is typically completed in 10 seconds (see Fig. 20.10, *C* and *D*) (Fig. 20.17). If the separated instrument didn't come out of the canal in 10 seconds, there are three possibilities of the failure; the first one is that the space between the ultrasonic tip and the outer canal wall might be smaller than the diameter of the separated instrument (Fig. 20.18) or this space might be closed

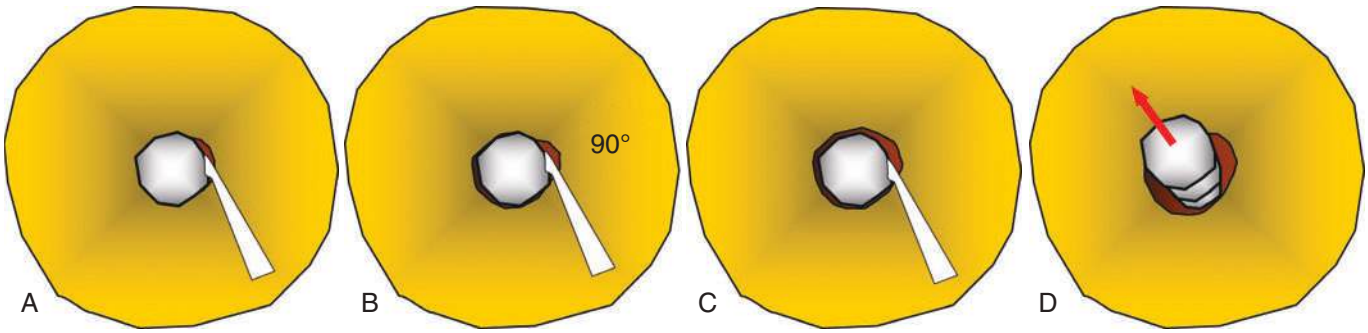


Fig. 20.13 **A**, Thin ultrasonic tip placed into the gap between the inner wall and the separated file and activated. **B,** Ninety-degree semicircular space created in this gap with ultrasonics. **C,** Ultrasonics activated directly on the separated file from this space. **D,** Canal walls contacting the separated file eliminated with ultrasonic oscillation transferred to the separated file from the vibrating ultrasonic tip. **D,** Separated file loosened and dislodged from the original position *(red arrow)*.

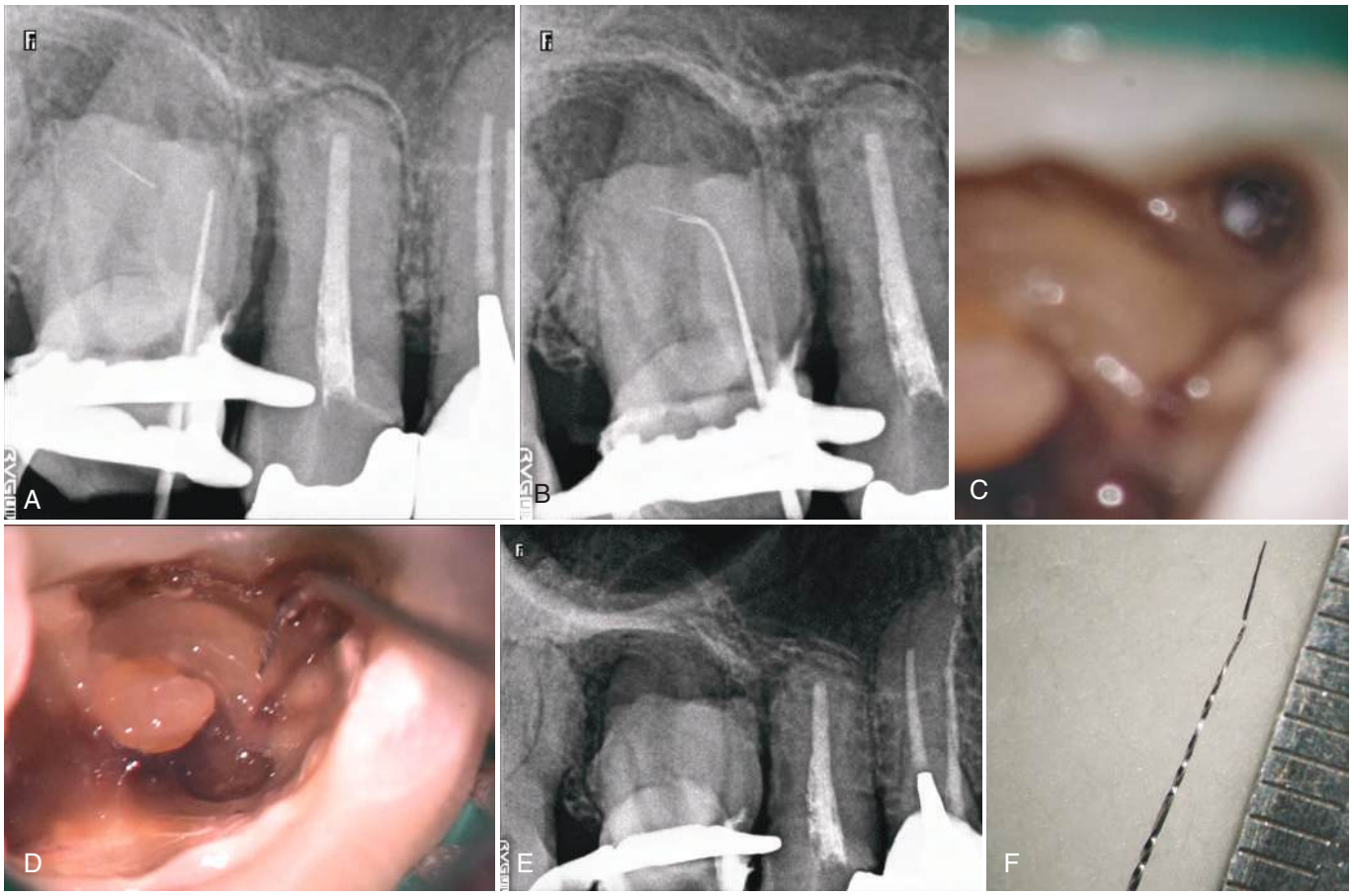


Fig. 20.14 **A,** Separated file beyond the severe curve in relation to the file placed in the mesiobuccal (MB) canal. **B,** Thin ultrasonic tip placed in the gap between the inner wall and the separated file. **C,** Separated file is not visible in the MB. **D,** Separated file coming out of the canal. **E,** Removal of the separated file confirmed on a radiograph. **F,** Retrieved separated file along with the original file

during the ultrasonic retrieval attempts, the second one is that the separated instrument might be pushed on the top back into the original position or from the outer wall by the ultrasonic tip (see Fig. 20.18, *F*), and the third one is that the amount of the fluid filled in the canal at the time of the ultrasonic removal attempts might not be sufficient enough to let the separated file flow out of the canal with the aid of acoustic streaming and cavitation (Fig. 20.19). If any of those possibilities is not applicable to the case and the instrument retrieval attempts were not successful with ultrasonics in wet conditions, or when the separated instrument is longer than 4.5 mm or between 3.1 and 4.4 mm with the curvature greater than 30 degrees, the loop (Yoshi Loop, DELabs) (Fig. 20.20, *C1*, *C2*, and *D*) should be used to retrieve the separated instrument.[330]

Longer separated instruments, especially around a severe curve, are apt to be touching all over the canal walls, which creates more friction as it is reoriented in a coronal direction

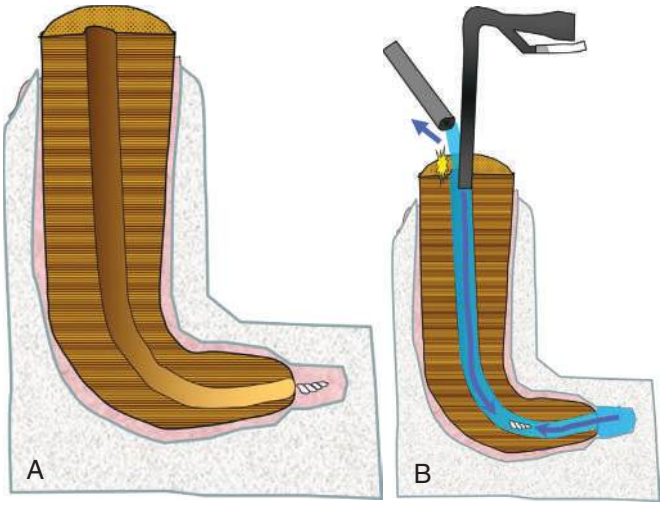


Fig. 20.15 A, Separated file loose beyond the radicular foramen. **B,** Ultrasonic irrigation such as ProUltra PiezoFlow using saline may help retrieve the separated file which is not engaged in the canal wall.

and makes it more resistant to instrument retrieval with ultrasonics alone, requiring more mechanical force to pull it out of the canal. This is why the loop is necessary to retrieve those types of separated instruments.

Use of the Loop

It requires at least 0.4-mm-diameter space to place the loop in the canal, and the coronal portion of the separated instrument needs to be peripherally exposed by at least 0.7 mm to be grasped by the Yoshi loop (Fig. 20.21, *A*), The loop size must be adjusted to fit the coronal diameter of the separated file using an endodontic explorer, such as a DG16 Endo Explorer (Hu-Friedy, Chicago, IL) (see Fig. 20.21, *G*). The tip of a DG16 is placed in the loop and the loop is contracted around it. The apical portion of the DG16 is used to make the loop size smaller and the coronal portion is used to make it larger when the loop is tightened around it. The loop is then pre-bent to 45 degrees instead of 90 degrees to both facilitate the placement of the loop over the separated instrument and be minimally invasive as bending it to 90 degrees requires more room compared to 45 degrees (see Fig. 20.21, *E* and *F*). The loop is then brought into the canal and placed over the exposed portion of the separated file as the loop is pushed back to 90 degrees upon it (see Fig. 20.21, *B* and *C*). Subsequently, the loop is tightened over the separated instrument by sliding down the handle. The separated instrument will be retrieved by pulling the loop out of the canal in various directions with a swaying motion to dislodge it from the canal walls (see Figs. 20.21, *D* and 20.20, *C2*). A couple of pull motions in several directions will usually dislodge the file fragment and complete the case for separated file removal.

Surgical approaches for the management of separated instruments represent a second category of procedures for dealing with this iatrogenic event (see also Chapter 11). Surgical approaches should be considered in the following situations[201]:

- As a last resort if other nonsurgical approaches fail, posttreatment disease develops, and the tooth is strategically important to retain
- As a first approach when periradicular pathosis is present at the time of instrument separation, especially if the separated instrument is found in the apical third of the canal coexistent, presumably with persistent extraradicular infection
- As a first approach when the radicular portion of the separated instrument is already outside the root, when an orthograde approach is unsuccessful, or when a significant amount of dentin must be sacrificed for instrument removal

PROGNOSIS

When an endodontic instrument separates in a canal during cleaning and shaping procedures, and if it hinders adequate cleaning of the canal, then the prognosis may be affected. Moreover, attempts to remove a separated instrument from a curved canal may cause additional complications such as excessive removal of dentin structure, secondary instrument fracture, root perforation, or even root fracture.[134,286,313,317] However, there are cases in which the prognosis is not compromised, depending on what stage of instrumentation the file fracture occurred, the preoperative diagnosis of the pulp and periradicular tissues, and whether or not the separated file can be removed or bypassed.[317] The presence of a separated instrument per se in a canal or outside the root does not necessarily condemn the case to postoperative disease. Rather, it is the residual presence of any infected tissue that determines the prognosis. The closer the root canal preparation is to the terminus at the time of instrument separation, the better the outcome will be.[339] It has been suggested that instrument separation occurring in a later stage of canal instrumentation, especially if it is at the apex, has the best prognosis because the canal is probably well debrided and may be relatively free from microorganisms.[338] If the preoperative canal is not infected, and no radicular periodontitis is associated with the root, then the presence of the separated instrument should not affect the prognosis[63] and the separated instrument can be incorporated into root canal filling materials. If the separated file can be removed predictably without sacrificing a significant amount of dentin, the prognosis will not be reduced. In other words, the treatment prognosis depends more on the removal of microbial infection than on instrument removal.

Ledge Formation

Ledges may be created as a result of instrumentation causing radicular transportation, which also causes zipping of the canal and possible canal perforation (Fig. 20.22). The *Glossary of Endodontic Terms* of the American Association of Endodontists[10] defines a ledge as "an artificial irregularity created on the surface of the root canal wall that impedes the placement of an instrument to the apex of an otherwise patent canal." Canal transportation[10] is defined as "the removal of canal wall structure on the outside curve in the radicular half of the canal due to the tendency of files to restore themselves to their original linear shape during canal preparation." The presence of a ledge hinders adequate shaping and cleaning of the canal in the areas radicular to the ledge. Consequently, there is a possible causal

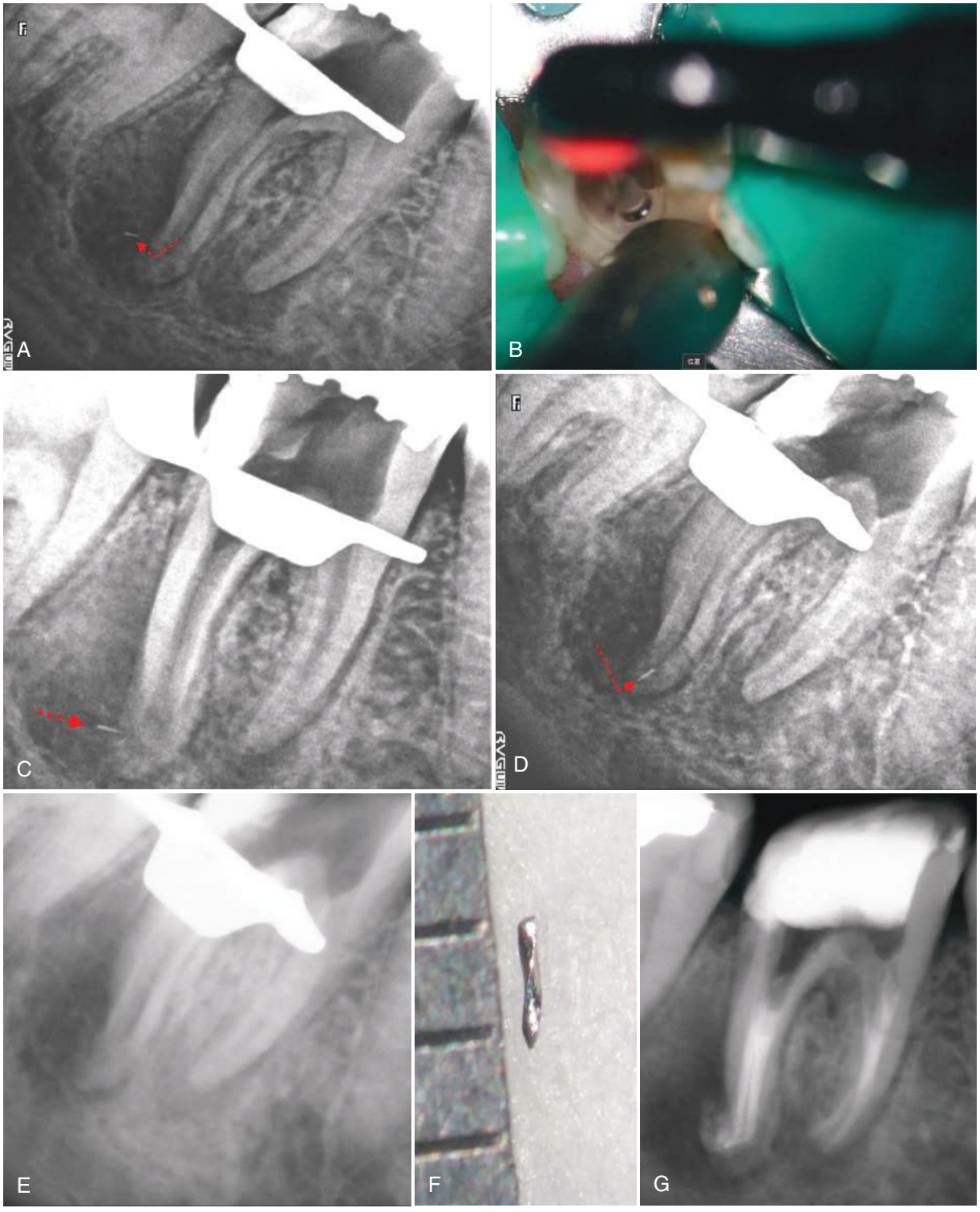


Fig. 20.16 **A,** Separated file pushed beyond the radicular foramen. **B,** PiezoFlow activated using saline to create cavitation and acoustic stream. **C,** Separated file approaching the radicular foramen with the PiezoFlow. **D,** Separated file brought back into the canal. **E,** Separated file retrieval confirmed on the radiograph. **F,** Retrieved separated file on a ruler. **G,** Root canals obturated with mineral trioxide aggregate (MTA).

relationship between ledge formation and an unfavorable endodontic treatment outcome.[61,62,118,125,133,159,215,377]

CAUSES OF LEDGE FORMATION

Ledge formation can occur during biomechanical preparation of the canal system, especially when the canals are more curved. There are a number of factors associated with ledge formation, such as the instrumentation technique, instrument type, root canal curvature, tooth type, working length, master radicular file size, clinician's level of expertise, and canal location. Failure to precurve the instruments, the inability to achieve a proper glide path to the apex, and forcing large files into curved canals are perhaps the most common reasons for ledge formation.[153] The incidence of ledge formation was reported to increase significantly when the curvature of the canal was greater than 20 degrees; more than 50% of the canals in the study were ledged when the canal curvature was greater than 30 degrees.[118] Jafarzadeh[156] described 14 possible causes of ledge formation (Box 20.1).

MANAGEMENT OF LEDGE FORMATION

Once a canal is ledged, the endodontic treatment becomes difficult and the prognosis may be compromised. When a

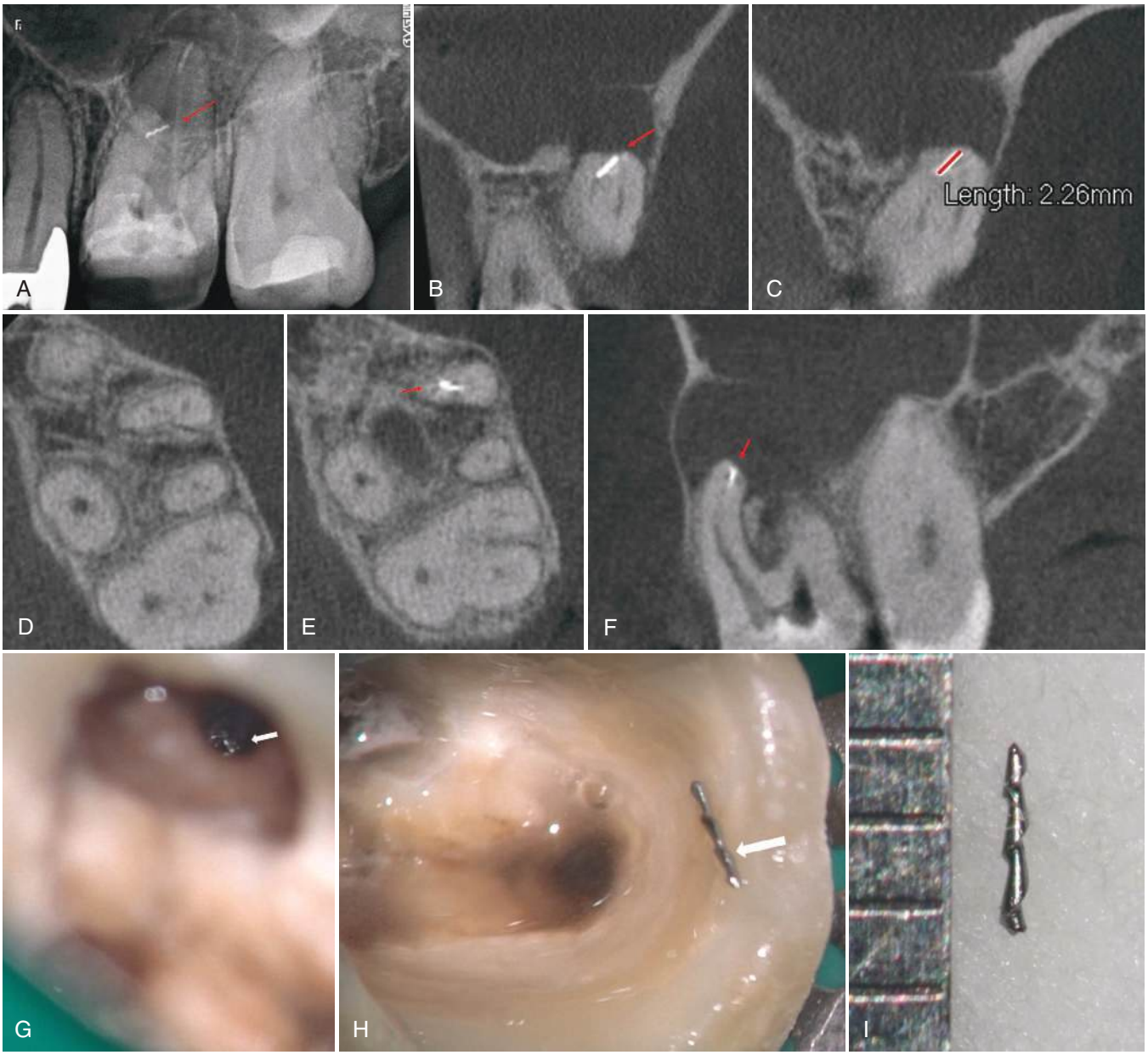


Fig. 20.17 **A**, Preoperative radiograph of a maxillary left first molar showing a separated instrument beyond the curve in the mesiobuccal root canal with large periapical lesions *(red arrow)*. **B**, Coronal view of cone-beam computed tomography (CBCT) shows the separated instrument in the buccally curved MB2 canal with a large periapical lesion developing into the sinus floor *(red arrow)*. **C**, Separated instrument measures 2.26 mm on the coronal view. **D** and **E**, Axial view shows the separated instrument in the MB2 canal directed toward the MB1 canal. **F**, Sagittal view shows the separated instrument in the MB2 canal lying in a distal direction with a large periapical lesion *(red arrow)*. **G**, Magnification of a partially visible portion of the separated instrument in the MB2 canal *(white arrow)*. **H**, Magnification of the separated instrument that came out of the canal with ultraspnics *(white arrow)*. **I**, Retrieved separated instrument measured 2.3 mm on a ruler.

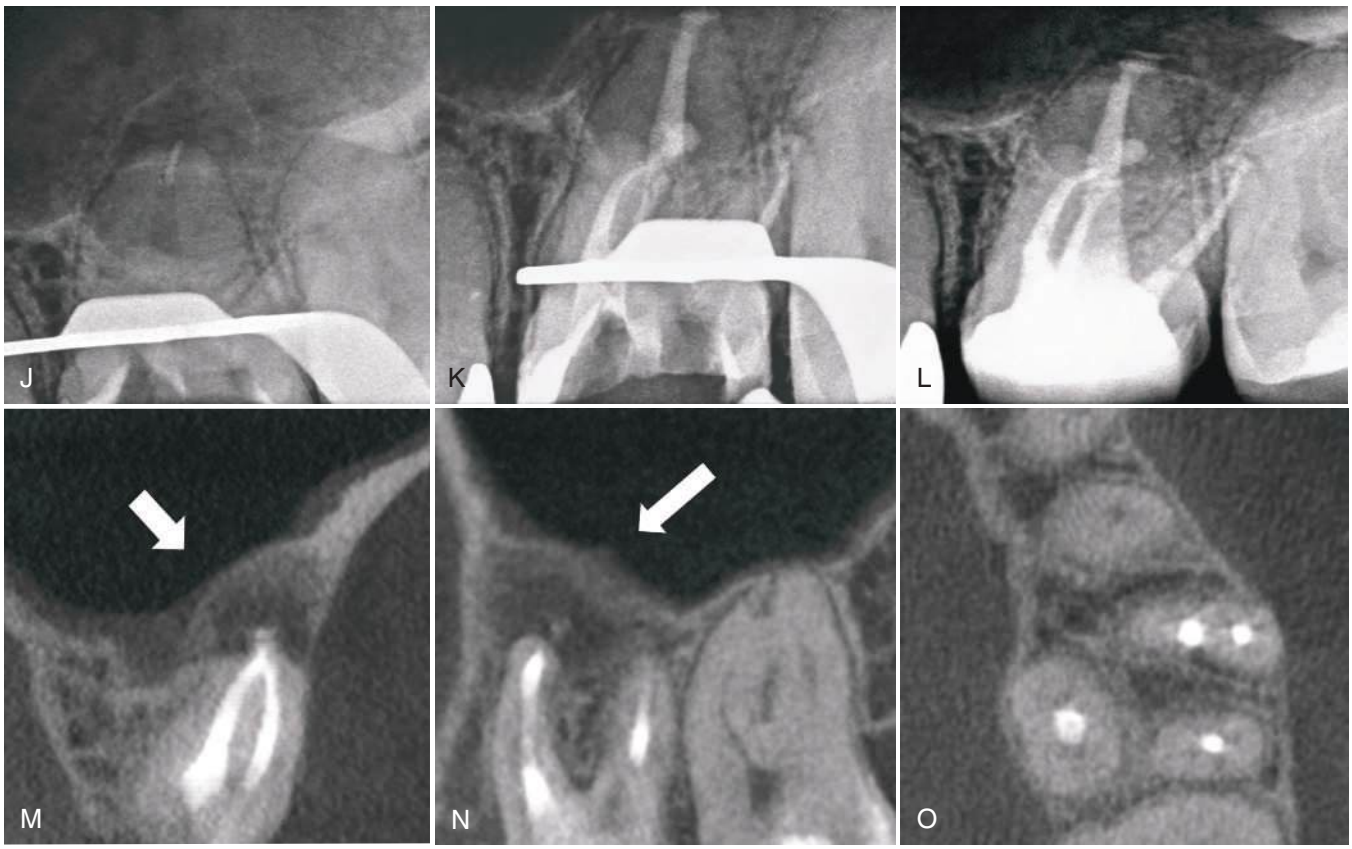


Fig. 20.17, cont'd J, Interoperative radiograph confirming removal of the separated instrument from the MB2. **K,** Postoperative radiograph showing the final root fillings with mineral trioxide aggregate (MTA), which reveals dentin sacrifice for removal of the separated instrument was minimum. **L**, Three-month postoperative radiograph showing nice periapical healing. **M**, Three-month postoperative coronal view of CBCT showing periapical bone reformation associated with the mesiobuccal root *(white arrow)*. **N**, Three-month postoperative sagittal view of CBCT showing periapical bone reformation associated with the mesiobuccal root *(white arrow)*. **O**, Three-month postoperative axial view of CBCT showing root fillings in MB, MB2, DB, and P root canals.

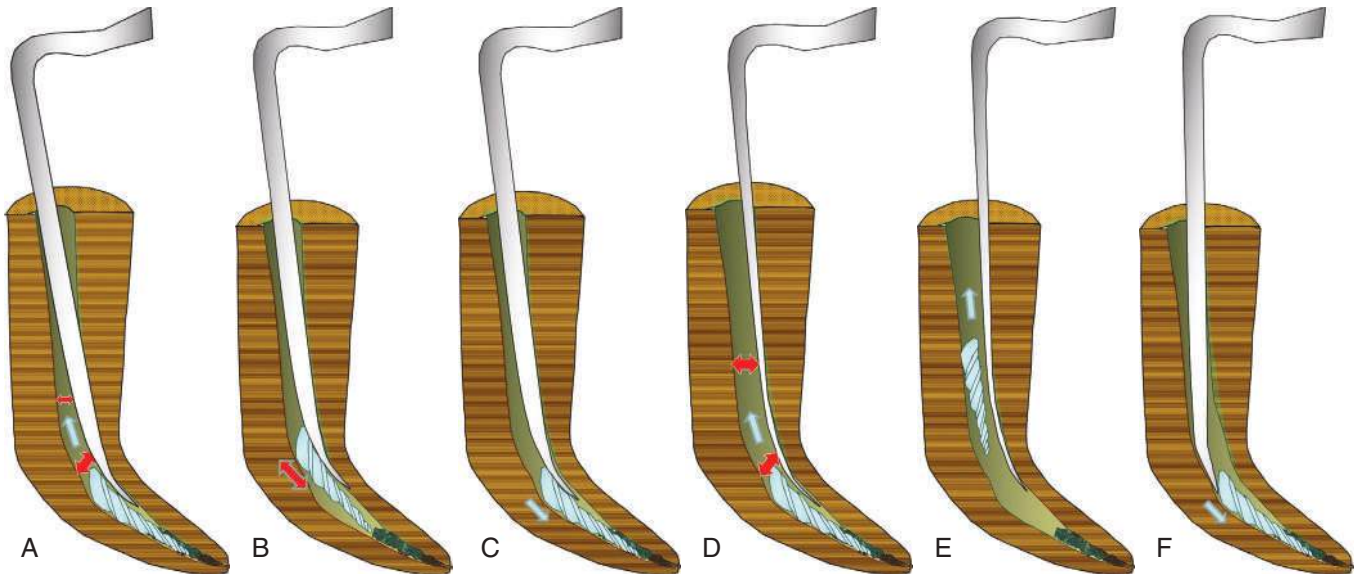


Fig. 20.18 A, Space between the ultrasonic tip and the canal wall is smaller than the diameter of the separated file. **B,** Separated file comes out and bumps against the ultrasonic tip. **C**, Separated file bounces back into the original position. **D**, Space between the ultrasonic tip and the canal wall is wider than the diameter of the separated file. **E**, Separated file comes out through the space. **F,** Ultrasonic tip placed on the top of the separated file in the space on the outer wall.

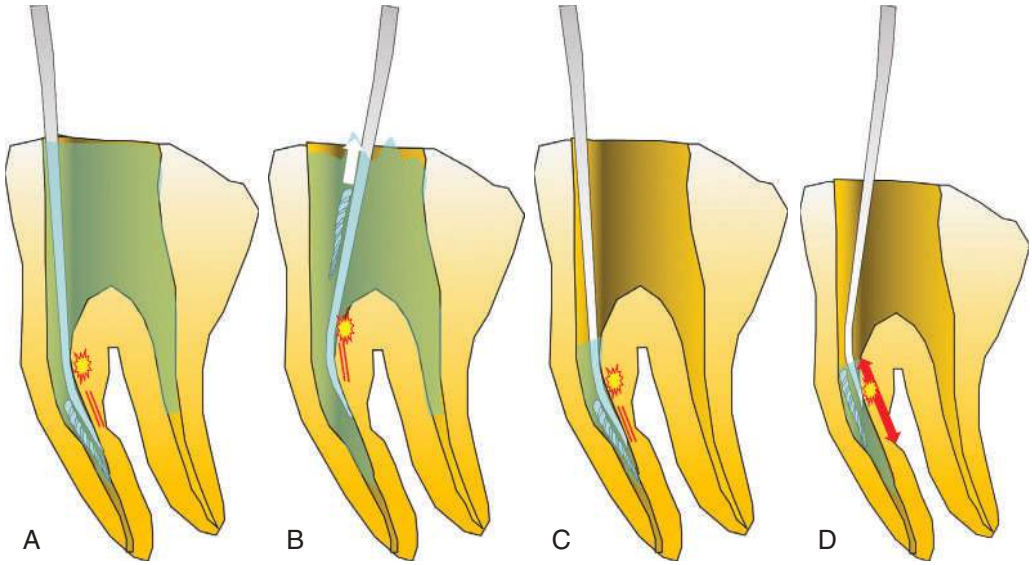


Fig. 20.19 A, Fluid filled in the canal and the pulp chamber. **B,** Separated file is coming out of the canal to the pulp chamber in the fluid with ultrasonics *(white arrow)*. **C,** Fluid filled only in the canal. **D,** Separated file fluctuating within the canal space filled with fluid *(red arrows)*.

Box 20.1 14 Possible Causes of Ledge Formation

1. Not extending the access cavity sufficiently to allow adequate access to the radicular part of the root canal[61,125,153,159,185,352]
2. Loss of instrument control if endodontic treatment is attempted via a proximal surface cavity or through a proximal restoration[153]
3. Incorrect assessment of the root canal direction[61,159,185]
4. Incorrect root canal length determination[61,159,185]
5. Forcing the instrument into the canal wall[356]
6. Using a noncurved stainless steel instrument that is too large for a curved canal[61,153,159,250,352]
7. Failing to use the instruments in sequential order[61,125,159,185]
8. Rotating the file excessively at the working length[125,356]
9. Inadequate irrigation or lubrication during instrumentation[352]
10. Over-relying on chelating agents[356]
11. Attempting to retrieve separated instruments[61,159,185]
12. Removing root filling materials during endodontic retreatment[61,159,185]
13. Attempting to negotiate calcified root canals[61,159,185]
14. Inadvertently packing debris in the radicular portion of the canal during instrumentation[61,352]

ledge is formed, early recognition of its location, using radiographs including CBCT and magnification, will facilitate its management.[125] The canal is usually "straightened" at the point where the file stops negotiating the curve, and there is suddenly a looser feeling of the file within the canal, with no more tactile sensation of the tip of the instrument binding in the canal. If the radiograph reveals that the tip of the instrument deviates away from the canal curvature, then it is highly likely that a ledge formation has occurred on the canal wall. The tip of the instrument placed in the canal should appear at the midpoint of the root on the radiographs because the canal is always in the center of the single-canaled root. If there are more than two canals in one root, then it is natural that the instrument placed in the canal appears off-center, which makes it essential to refer to CBCT for the number of the canals in the root before the instrumentation is initiated (Fig. 20.23).

When a ledge formation is identified, the shortest file possible that can reach the working length should be selected to bypass the ledge. Shorter instruments have more stiffness and allow the clinician's fingers to be placed closer to the tip of the instrument, which consequently provides greater tactile sensation and more control over the instrument.[156] It usually requires determination, perseverance, and patience to bypass a ledge once it is formed.[61] The successful management of a ledge depends on whether both the ledge and the orifice of the original pathway can be visualized, reached, or felt with instruments.[191] Any case where the ledge can be visualized under the DOM may be considered less complex to treat. Cases in which the ledge is not directly visible are more difficult to treat, as tactile sensation and negotiation require patience and experience.[61] However, 3D radiographic images obtained from CBCT reveal the locations and the directions of both the ledge and the original pathway to the canal terminus as the ledge always forms on the outer wall of the curved root canal in the straight line direction from the access. CBCT helps the clinician locate the entrance of the original pathway to the radicular foramen. First the ledge needs to be recognized under the DOM and both the ledge and the original canal need to be located on the 3D radiographic images. The maximum curvature of the root apical to the ledge can be identified on the axial view of CBCT as it is exactly the same as what is seen under the DOM. The maximum curvature can be located simply by viewing the axial slices from the coronal aspect before the ledge to the radicular aspect past the ledge in parallel with the coronal straight line to the ledge, and then a sagittal slice or a coronal slice is cut over the center of the curved root on the axial slice with the maximum curvature. On the sagittal view or the coronal view, the relationship between the ledge and the original pathway can be recognized, and with this view, the entrance to the original canal can be measured to locate its position under the DOM in reference to the ledge and the orifice from several reference points to

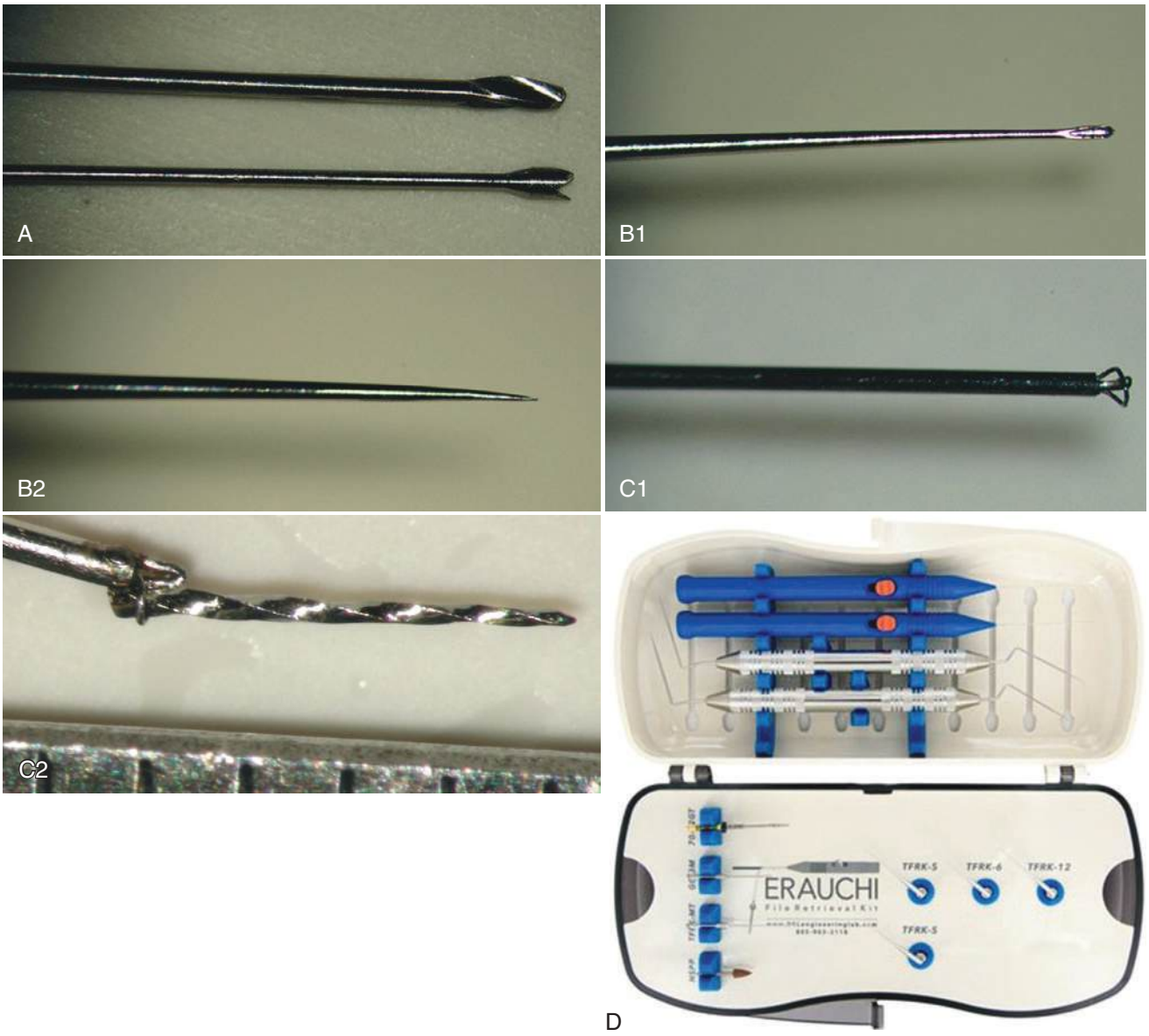


Fig. 20.20 Terauchi File Retrieval Kit (TFRK). **A,** Modified #3 Gates Glidden bur that is also called GG-3M *(upper)* and microtrephine bur that is also called TFRK-MT *(lower)*. **B1,** TFRK-6 ultrasonic tip (the tip portion looks like a spoon and is also called a spoon tip. It has a micro-concave on the spoon, which turns toward the handpiece at the 6 o'clock position. **B2,** TFRK-S ultrasonic tip (the tip portion looks like a sharp spear and is also called a straight tip). **C1,** Yoshi loop (TFRK-L), which captures a separated instrument. This micro-lasso is comprised of a tiny wire loop at the end of a stainless steel cannula, with a sliding handle that tightens the loop when pulled. **C2,** Yoshi loop holding a separated file from an actual case. **D,** Terauchi File Retrieval Kit (TFRK) in an autoclavable cassette case. The kit includes a GG-3M, TFRK-MT, TFRK-6 ultrasonic tip, TFRK-12 ultrasonic tip, TFRK-S ultrasonic tip, Yoshi loop, TFRK- GPR, and TFRK-ME, High Speed Polishing Point (HSPP), NiTi GT #70/.12 Rotary File (GT70.12).

it as the original canal is always in the center of the single-canaled root and the ledge is always in the straight line. CBCT without doubt, compared to 2D radiography, provides additional information for making an accurate diagnosis and therefore helps the clinician manage the ledge formation with a more predictable outcome and negotiate the original canal.[232,331]

BYPASSING A LEDGE

Using Hand Instruments

Based on the information from CBCT, the original pathway can be predictably identified and opened up under the DOM using micro-hand instruments. In addition, the ledge can always be visualized under the DOM as it is created in the straight-line direction. In addition, caries detector dye (Kuraray America, Inc., NY) or methylene blue dye can stain the orifice of the original pathway to locate it in comparison with dentin wall. Small hand files dedicated to microscopic use (such as the micro-Opener with #10/.06 taper) may help flare the orifice of the original pathway on the inner wall of the root canal coronal to the ledge (Fig. 20.24, *A*). The micro-Opener is available only in 16 mm and is short enough to provide tactile sensation and better control. However, in some cases a longer instrument is required to reach the ledge present in the radicular third of a long canal. In these instances, a hand instrument holder such as

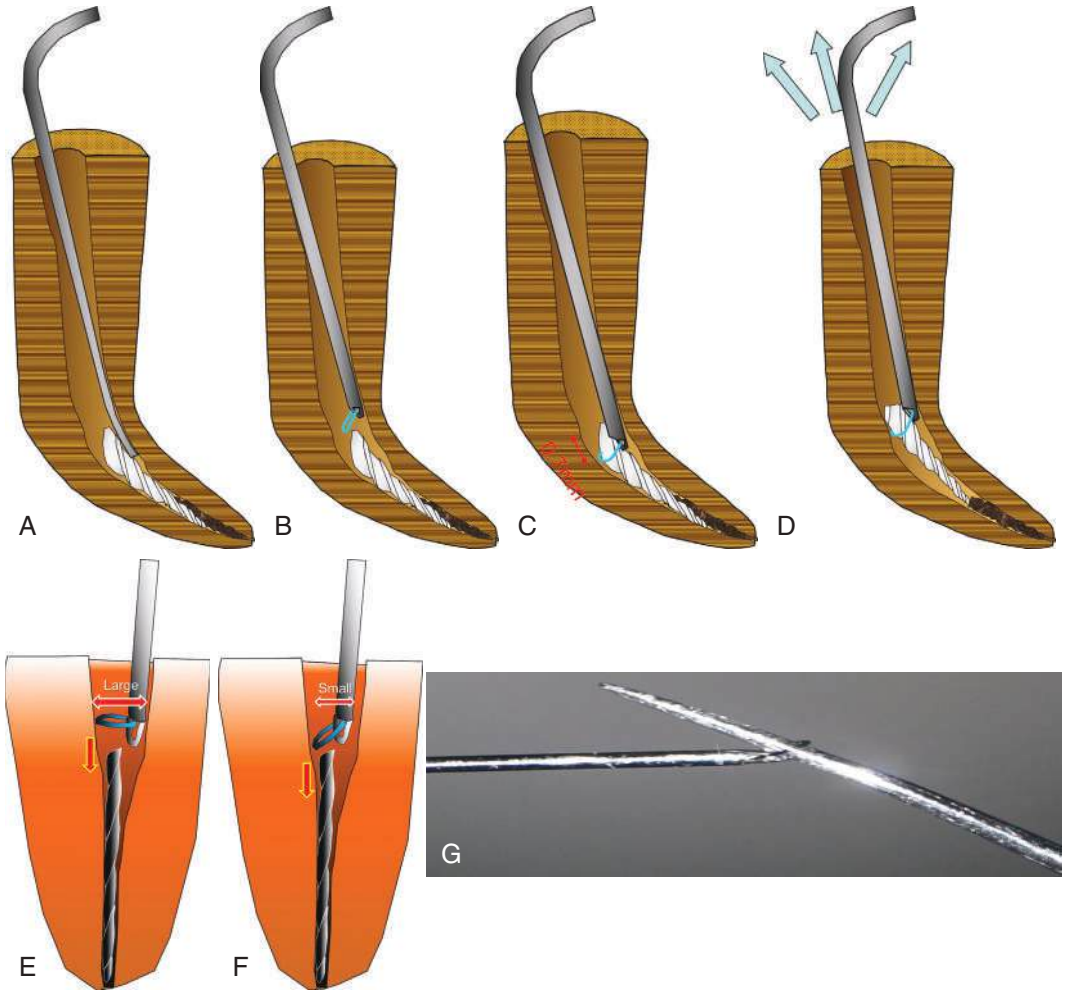


Fig. 20.21 **A,** Coronal portion of the separated file should be exposed at least 0.7 mm above the bottom of the space created around it. A #40 plugger should be placed in the space to assess the available space prior to the placement of the loop. **B,** Approaching the separated file with the loop bent to 45 degrees against it. **C,** Loop is pushed back to 90 degrees as it is placed over the separated file. Removal with the loop requires the diameters of the loop device (0.4 mm) and the separated instrument and a depth of at least 0.7 mm. **D,** Separated file is pulled out with the loop device in various directions. **E,** Loop prebent to 90 degrees occupying more room in the canal. **F,** Loop pre-bent to 45 degrees requiring minimum room in the canal. **G,** Loop size is adjusted using an endodontic explorer.

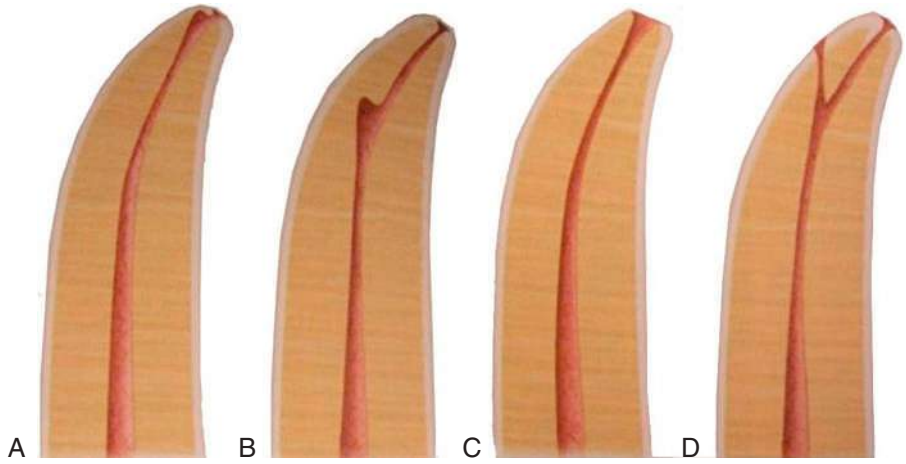


Fig. 20.22 Schematic diagrams showing the most common preparation errors. **A,** Radicular zip. **B,** Ledge. **C,** Radicular zip with perforation. **D,** Ledge with perforation.

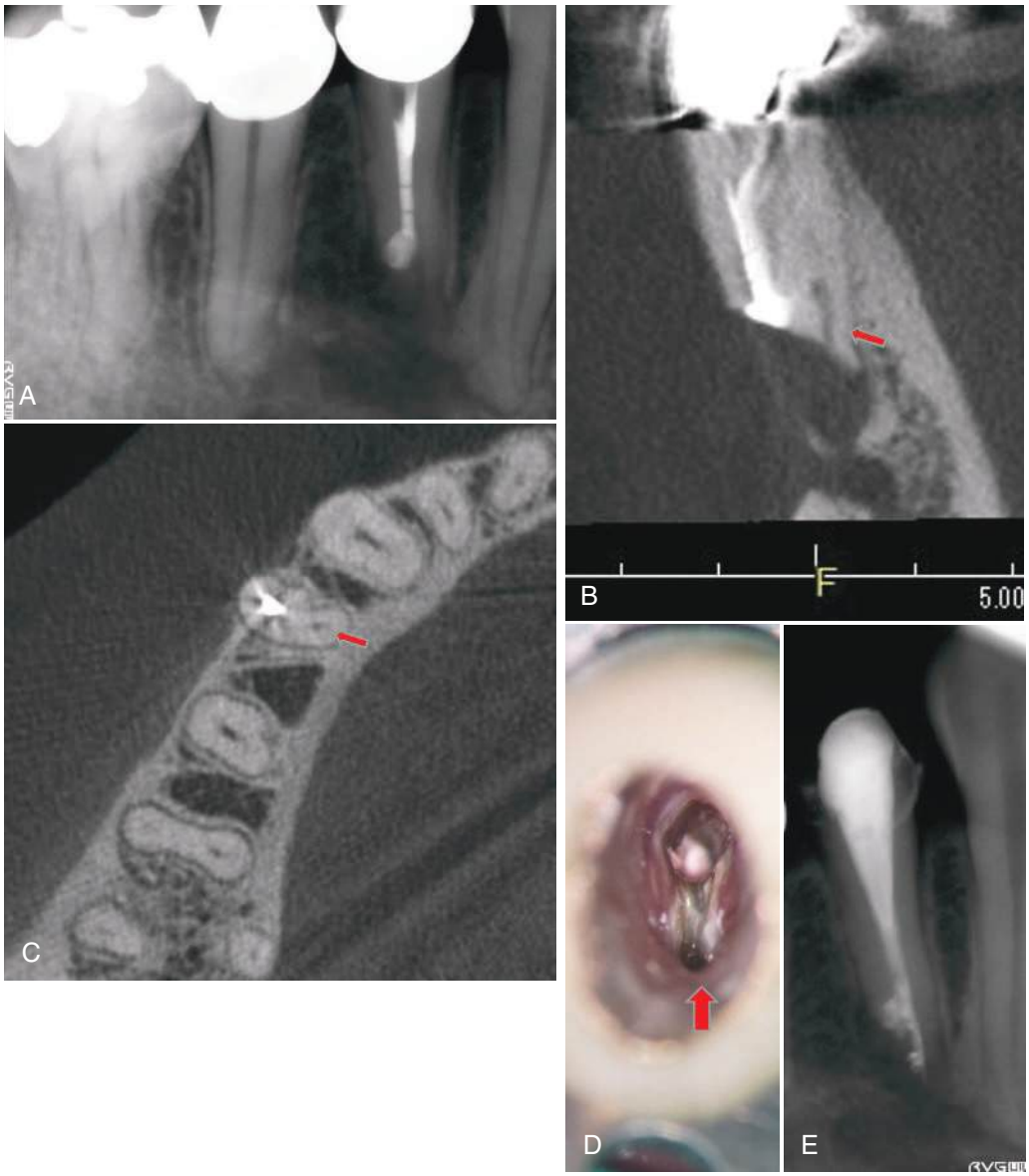


Fig. 20.23 **A,** Preoperative radiograph showing the failing outcome of endodontic surgery for the mandibular right first premolar with the root canal filling in one root canal. **B,** Cone-beam computed tomography (CBCT) image in coronal view showing a missed lingual canal *(red arrow)*. **C,** CBCT image in axial view showing the missed lingual canal lingual *(red arrow)* to the mesiodistal midline in relation to the buccal canal buccal to the midline with the same distance between the midline and the lingual canal. **D,** Magnification of the missed lingual canal *(red arrow)*. **E,** Postoperative radiograph showing root canal obturation in both the buccal and the lingual root canals.

an EndoHandle (Venta, Logan, UT) may help the clinician easily negotiate the canal radicular to the ledge. Any hand files can be attached to the handle for microscopic use. Other hand instruments are available specifically for this purpose (TFRK microexplorer instrument [TFRK-ME] or Gutta-Percha Removal Instrument [TFRK-GPR, DELabs] and ELES instrument [DenMat Lompoc, CA]) that are long enough to reach the ledge formed in the radicular third. Both instruments are made of stainless steel. The TFRK-ME has a 0.1 mm biconical tip with a .08 taper for the first 1-mm portion from the radicular end and a .06 taper for the next 10-mm portion, which gives the instrument more stability in push-pull motions than a .02 tapered K-file. The TFRK-ME is precurved to the original pathway in the radicular 2 to 3 mm and pecked with to explore the narrow original canal on the opposite side of the canal wall with the ledge (Fig. 20.25), while the ELES and the TFRK-GPR are used to reduce the ledge in push-pull motions as the 0.2 mm diameter tip portion of the ELES with a .06 taper is coated with fine diamond particles on the tip portion (Fig. 20.26) and the TFRK-GPR has a sharp conical tip. They can be used in sequence to widen the orifice of the original pathway, which results in reducing or removing the ledge. The maximum diameter of the TFRK-ME and the TFRK-GPR is 0.78 mm, which provides stability and maintains a high visibility under the DOM when it is precurved, whereas the maximum diameter of the ELES instrument is 0.68 mm, which provides more flexibility and more visibility, even around a curve, to facilitate the widening the orifice of the original pathway without worsening

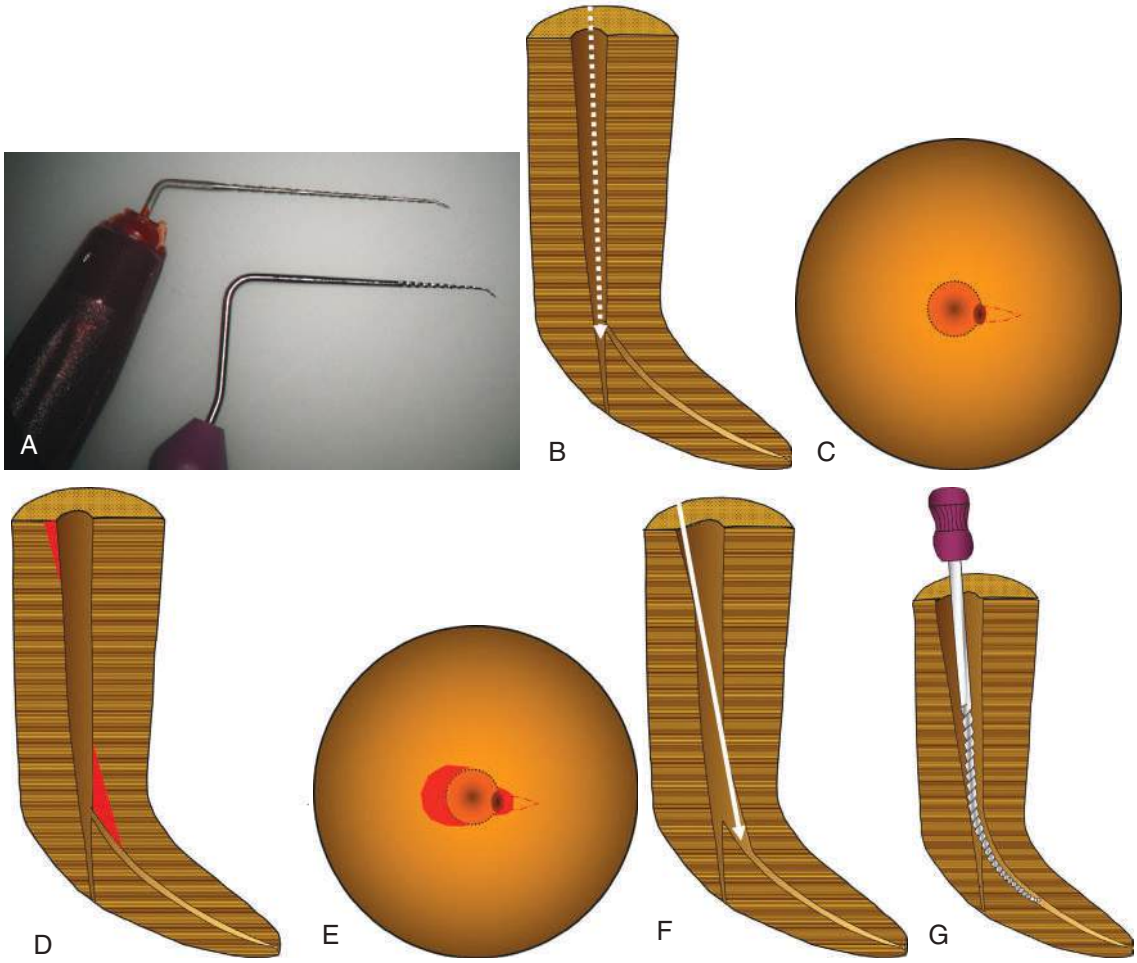


Fig. 20.24 A, Precurved microinstruments such as EndoHandle with a precurved #25 Hedstrom File (Venta Endo, Logan, UT) and micro-Opener (Dentsply Sirona) to explore a ledged canal for the original pathway and to reduce irregularities over the ledge. **B,** Ledge is formed with perforation in a radicular straight direction. **C,** Microscopic view showing the ledged canal in the center and a narrow access to the original pathway on the right side. **D,** The canal orifice as well as the access to the original pathway needs to be flared in an anticurvature direction *(red portions)*. **E,** Microscopic view showing red portions to be removed for easy access to the original pathway. **F,** Access to the original pathway was flared to facilitate file insertion. **G,** Bypassing file used in short push-pull strokes to create a distinct pathway to the original canal as the final glide path.



Fig. 20.25 A, Micro-hand instrument (TFRK-ME) to explore canals for the original pathway and bypass a ledged canal. **B,** Magnified image of a TFRK-ME.



Fig. 20.26 Micro-hand instrument with diamond coating *(ELES)* to bypass/eliminate/reduce a ledge and enlarge the orifice of the original pathway.

the ledge. Once the first TFRK-ME or any other micro-hand file has successfully bypassed the ledge, subsequently the ELES or the TFRK-GPR instrument or the same micro-instrument can be used in a push-pull motion to enlarge and reestablish the original pathway (Fig. 20.27), followed by hand instruments (such as ProTaper S1 and S2 hand instruments, DENTSPLY Sirona Tulsa Dental Specialties, Tulsa, OK) to both shape and reduce the ledge. S1 and S2 ProTaper Gold files (DENTSPLY Sirona Tulsa Dental Specialties) can also be precurved with ease and used as hand



Fig. 20.27 Schematic diagrams showing bypassing the ledge with ledge removal hand instruments. **A,** Ledge is created in a radicular straight direction around a curve. **B,** Original pathway is explored and opened up with the TFRK-ME in short push-pull strokes. **C,** Original pathway is widened with the ELES instrument. **D,** ELES instrument used in push-pull motions to reduce the ledge. **E,** Precurved TFRK-GPR is placed into the original canal and moved with short up and down strokes over the ledge. **F,** Ledge was reduced with the precurved TFRK-GPR. **G,** Precurved straight ultrasonic tip can also be used here to reduce the ledge instead of the ELES instrument. **H,** File with a biconical tip is brought into the canal to eliminate/reduce the ledge and negotiate the original canal. The primary cone of the biconical tip has no cutting tip. As it hits the curve or irregularities on the canal wall, it will slip alongside the curve and scoot over the irregularities. **I,** Secondary cone of the biconical tip removes the points at the transition angles and provides a smooth surface that guides the tip portion into the original pathway. **J,** Orifice of the original canal over the ledge needs to be enlarged for easy file insertion after the ledge has been bypassed successfully. **K,** Orifice of the original canal can be efficiently widened with a precurved instrument such as a ProTaper Gold S1 rotary file and a #15/.04 Vortex Blue rotary file used as a hand file, and the apical patency established. **L,** Original canal can be sequentially prepared with larger NiTi files. **M,** Postoperative canal showing the ledge is significantly reduced.

files in short pumping motions for the same purposes after the preliminary enlargement of the original pathway with those micro-hand instruments (Fig. 20.28).

As an alternative technique to bypass the ledge when the clinician wishes to use K files without using the DOM, precurving of a #8 or #10 hand file is a critical step. The small file should have a distinct curve at the tip. This file should be used initially to negotiate and explore the original pathway of the canal to the radicular foramen.[61,153,352] A rubber stopper with a directional indicator pointed in the direction of the original pathway should be set in the instrument.[153] A slight rotation of the file combined with a "pecking" motion can often help advance the instrument to the full working length of the canal.[153,352] If this technique is unsuccessful, the clinician should flare the canal coronal to the ledge in an anticurvature direction to obtain wider straight-line access to the original pathway of the canal, followed by a smaller K-file with the tip precurved to bypass the ledge (see Fig. 20.24).

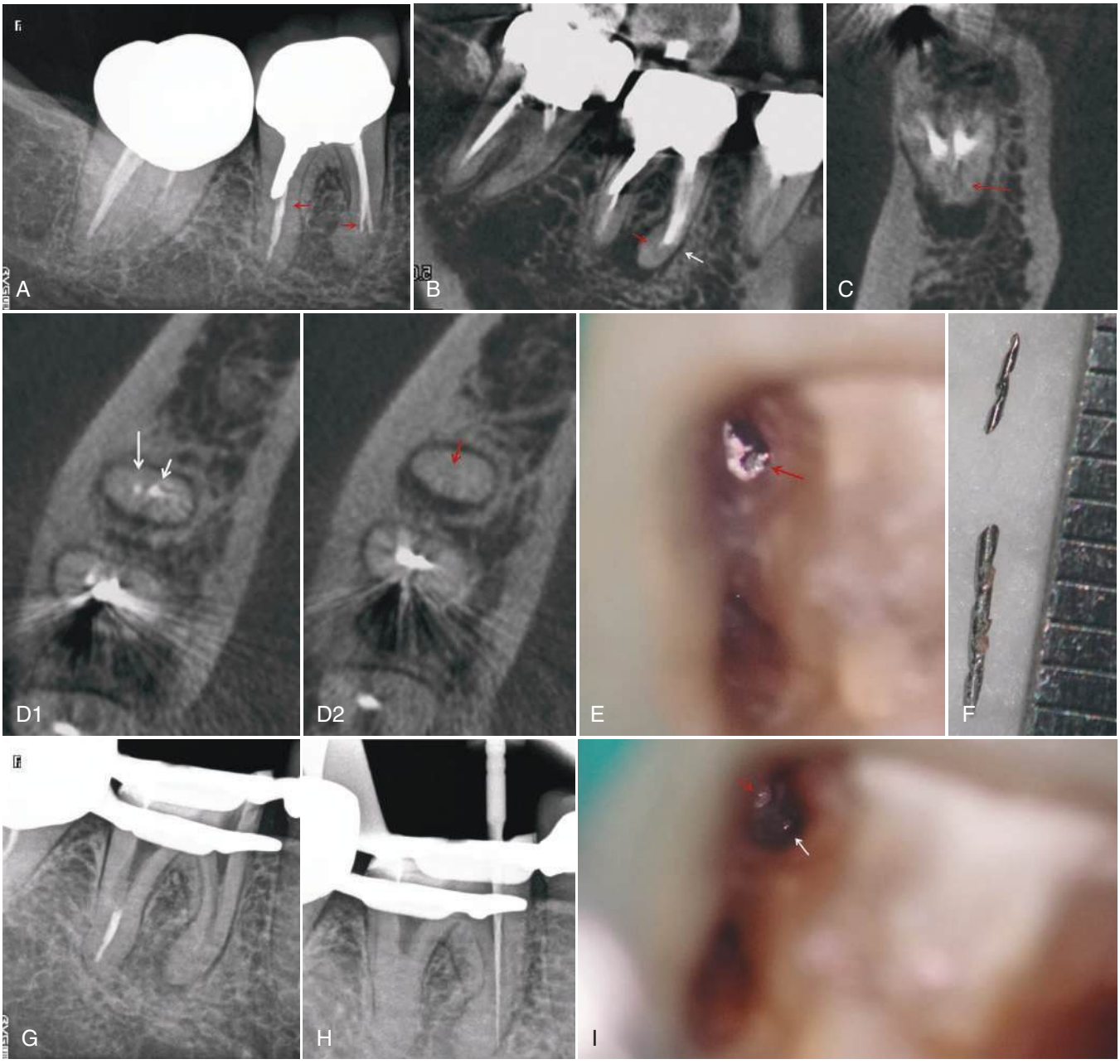


Fig. 20.28 **A,** Preoperative radiograph of a mandibular right first molar showing a periradicular lesion associated with the mesial root and two separated files in the mesial and the distal root *(red arrows)*. Note there are two separated files in the mesial and the distal roots in addition to the fact that the root filling in the mesial root deviates away from the center of the root. **B,** Cone-beam computed tomography (CBCT) image in sagittal view showing the original canal *(red arrow)* in the center of the root while the ledge *(white arrow)* turns away in the straight line in the mesial root. **C,** CBCT image in coronal view showing the MB and the ML canals merge into one canal *(red arrow)* in the apical third. **D1** and **D2,** Consecutive CBCT images in axial view showing the MB canal and the ML canal *(white arrows)* join into one oval canal *(red arrow)*. **E,** Magnification of the MB canal showing a separated file *(red arrow)*. **F,** Retrieved separated files from the mesial root and the distal root. **G,** Interoperative radiograph showing removal of the two separated files. **H,** Placement of a small file into the MB canal reveals a perforation on the outer wall in the straight line form from the orifice on the interoperative radiograph. **I,** Ledge *(red arrow)* and the original pathway *(white arrow)* were identified in the MB canal under the DOM based on the axial view of CBCT and the orifice of the original canal was enlarged with a precurved #10/.06 micro-opener with short push-pull strokes.

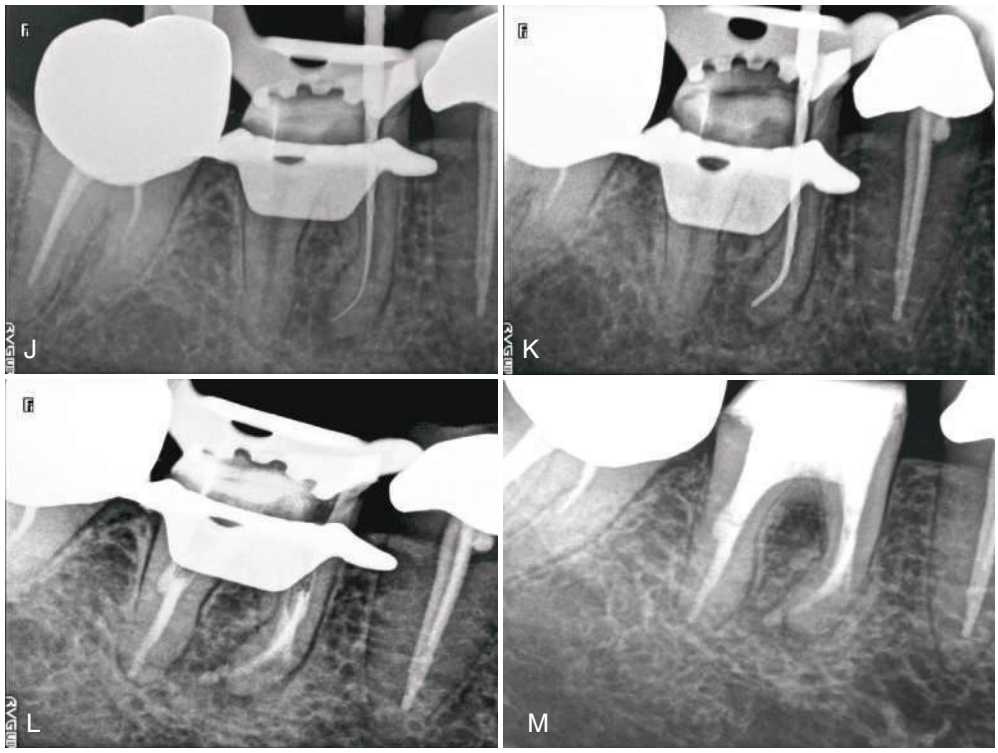


Fig. 20.28, cont'd J, Placement of a precurved small hand file (ProTaper S1) reveals it in the original canal reaching the working length on the interoperative radiograph. **K,** Placement of the final shaping file into the MB canal confirms it in the original canal reaching the working length without transportation on the interoperative radiograph. **L,** Postoperative radiograph shows root canal fillings with mineral trioxide aggregate (MTA) including the ledged canal with the perforation. **M,** Three-month postoperative radiograph shows a dissolving periapical lesion.

Using Ultrasonic Tips

Small diameter ultrasonic tips improve both cutting efficiency and visibility with the added benefit of reducing the need for sacrificing sound dentin in bypassing or removing the ledge.[192] The only drawback to the thinner tips could be that they are more prone to fracture during ultrasonic activation.[351] Therefore thinner ultrasonic tips should be activated in a pulsing and pecking fashion at the lowest possible power setting to prevent fracture, overextending the ledge, and heat generation as ultrasonics may get very aggressive in cutting.[282]

The use of longer ultrasonic tips may also facilitate bypassing or reducing the ledge in the radicular third as long as it is performed under the DOM. The entrance to the original canal just coronal to the ledge can be quickly enlarged with small diameter ultrasonic tips if it can be visualized under the DOM or felt with a sticky feeling from the micro-hand instruments (Fig. 20.29). First, an appropriate size of a small diameter ultrasonic tip (such as TFRK-S, DELabs and ET25, Satelec Corp, Merignac Cedex, France) is selected and precurved in the direction of the original canal. Then it is simply placed into the orifice of the original pathway in a radicular direction and activated in pecking and short push-pull motions at the same time, which will automatically enlarge the opening of the true canal in a few seconds so that files can consecutively be placed into the original canal for root canal preparation. Once the orifice of the original pathway gets wide enough for file insertion, negotiation files need to be placed into the canal to establish patency and to smooth out the irregularities on the canal wall that may have been created with ultrasonics, followed by root canal preparation with NiTi rotary or hand files (Fig. 20.30).

Using Rotary Instruments

Before using rotary instruments to bypass the ledge, the canal should be enlarged to the area coronal to the ledge for better visualization of the original pathway and then the orifice of the original pathway must be located and opened. Precurved smaller diameter rotary files with smaller tapers such as PathFiles (Dentsply Sirona, Ballaigues, Switzerland) and Race 10 (FKG Dentaire, La Chaux-de-Fond, Switzerland) are also beneficial for negotiating beyond a ledge. First, the radicular 2 to 3 mm of the file is precurved to the original pathway based on the coronal or the sagittal view of CBCT and is rotated at 100 rpm or with 90° CW/30° to 60° CCW reciprocating motions at 150 rpm in short up-and-down motions as it is brought slowly into the canal and moved radicularly (Fig. 20.31). If a ledge or sharp radicular curvature is encountered, the instrument is withdrawn about 1 mm and immediately reinserted. This allows the precurved tip to move to a different orientation within the canal as it is reinserted. This may require one or two withdrawals and reinsertions to negotiating the original canal. This procedure is followed in sequence by the next larger rotary files to preliminarily enlarge the original pathway and reduce the ledge prior to the conventional root canal preparation with NiTi rotary instruments.

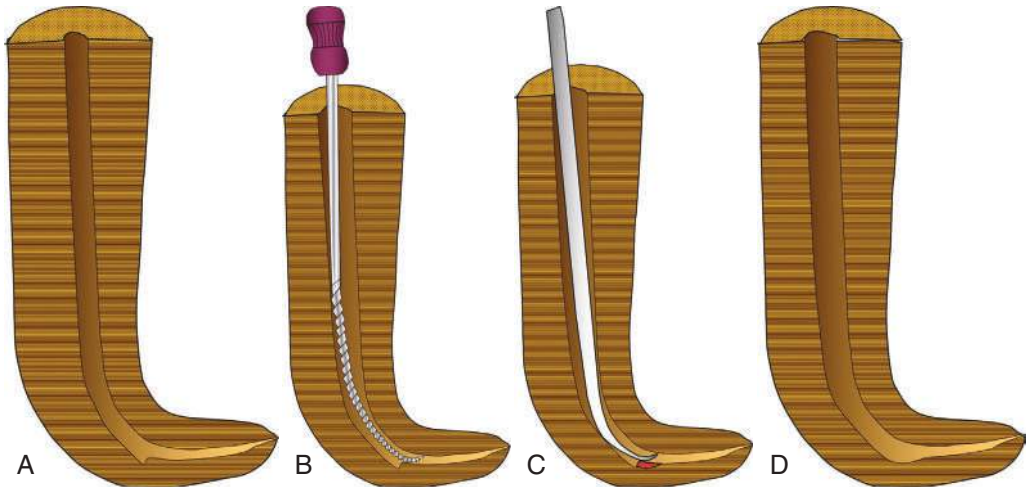


Fig. 20.29 Nonvisible ledge removal with ultrasonics. **A,** Ledge formed beyond the curve. Cone-beam computed tomography (CBCT) is taken to locate both the ledge and the original pathway. **B,** Canal coronal to the ledge is flared in an anticurvature direction with NiTi rotary files in brushing motions to possibly visualize the ledge and the original pathway. Original pathway to the canal terminus is explored with a #10 precurved K-file or the precurved TFRK-ME for a sticky feeling and then radiographs are taken with it inserted in the canal to see if it is in the original canal beyond the ledge. **C,** Straight ultrasonic tip is precurved the same way as the hand instrument used and placed into the original canal to eliminate/reduce the ledge and to open up the entrance to the original pathway. The red portion needs to be smoothed out to facilitate the insertion of the next larger file into the original pathway. **D,** Postoperative canal showing the ledge is significantly reduced.

POTENTIAL COMPLICATIONS OF REMOVAL OR BYPASSING A LEDGE

During ledge removal or bypassing a ledge, complications can occur, including root fracture, root perforation, worsening the ledge, and instrument separation. Since a ledge is likely to form around a curve, attempts to bypass or eliminate it may often involve dentin sacrifice, which may eventually compromise the integrity of the tooth and increase the risk of perforation or root fracture. Care should be taken not to remove excessive root structure or overstress the instruments during the ledge management. If a ledge cannot be bypassed with any of the techniques described earlier, and if signs or symptoms of the periradicular lesion persist, then the treatment options are limited to either the use of GentleWave, which applies advanced fluid dynamics, acoustics, and tissue dissolution chemistry to clean and disinfect the entire root canal system, regardless of any anatomic complexities (including the original pathway beyond the ledge), leading to success in eliminating infection from the canal system without shaping them where traditional instruments cannot access[301,302,349] or surgical intervention such as periradicular surgery or intentional replantation.[153]

PREVENTION OF LEDGE FORMATION

Glide path preparation helps clinicians prevent mishaps such as ledge formation and canal transportation as it guarantees that the root canal diameter is sufficiently large to receive the first shaping instrument.[27,234] Precurving the instruments and not forcing them into the canal is one of the most important considerations in the prevention of ledge formation.[153]

NiTi reciprocating instrumentation shows a lower frequency of procedural errors such as ledge formation and instrument separation even in the hands of novice clinicians, compared to K-file/NiTi rotary instrumentation.[73,128,202,283,328] The use of NiTi files and instruments with noncutting tips helps to maintain original root canal curvatures.[153] The concept of use of these instruments is that the rounded tip does not cut into the wall but will slip alongside it.[152,153] In addition to the tip design, flexibility of an instrument also plays an important role in keeping the instrument centered in the canal, with NiTi instruments favored over stainless steel instruments.[143,150,223] The use of novel NiTi alloy instruments with new heat treatment methods such as the Controlled Memory alloy technology (Coltene/Whaledent AG) and the GOLD alloy technology (Dentsply Maillefer, Ballaigues, Switzerland) show less canal transportation and more canal centering ability than conventional NiTi alloy instruments.[135,227,244,348]

PROGNOSIS

A ledge in a canal is analogous to any intracanal obstruction that hinders the cleaning and shaping of the root canal to the radicular terminus. The presence of a nonnegotiated ledge may have a similar prognosis as a canal with a retained separated instrument. If the ledge can be bypassed or eliminated, and if the canal beyond the ledge can be cleaned without excessive enlargement or perforation, then the prognosis should not be that greatly reduced.[338,339] Therefore the prognosis in general for such a case depends not only on the efficacy of the disinfection process but also on the host response to remaining debris or biofilms (bacteria) that are traditionally not removed with conventional root canal therapy (Fig. 20.32).[74]

Radicular Extrusion of Root Canal Filling Materials

The extrusion of root filling materials beyond the radicular foramen or the perforation site may occur either when the canal is obturated or during the removal of obturation materials. Radicularly extruded materials such as obturating

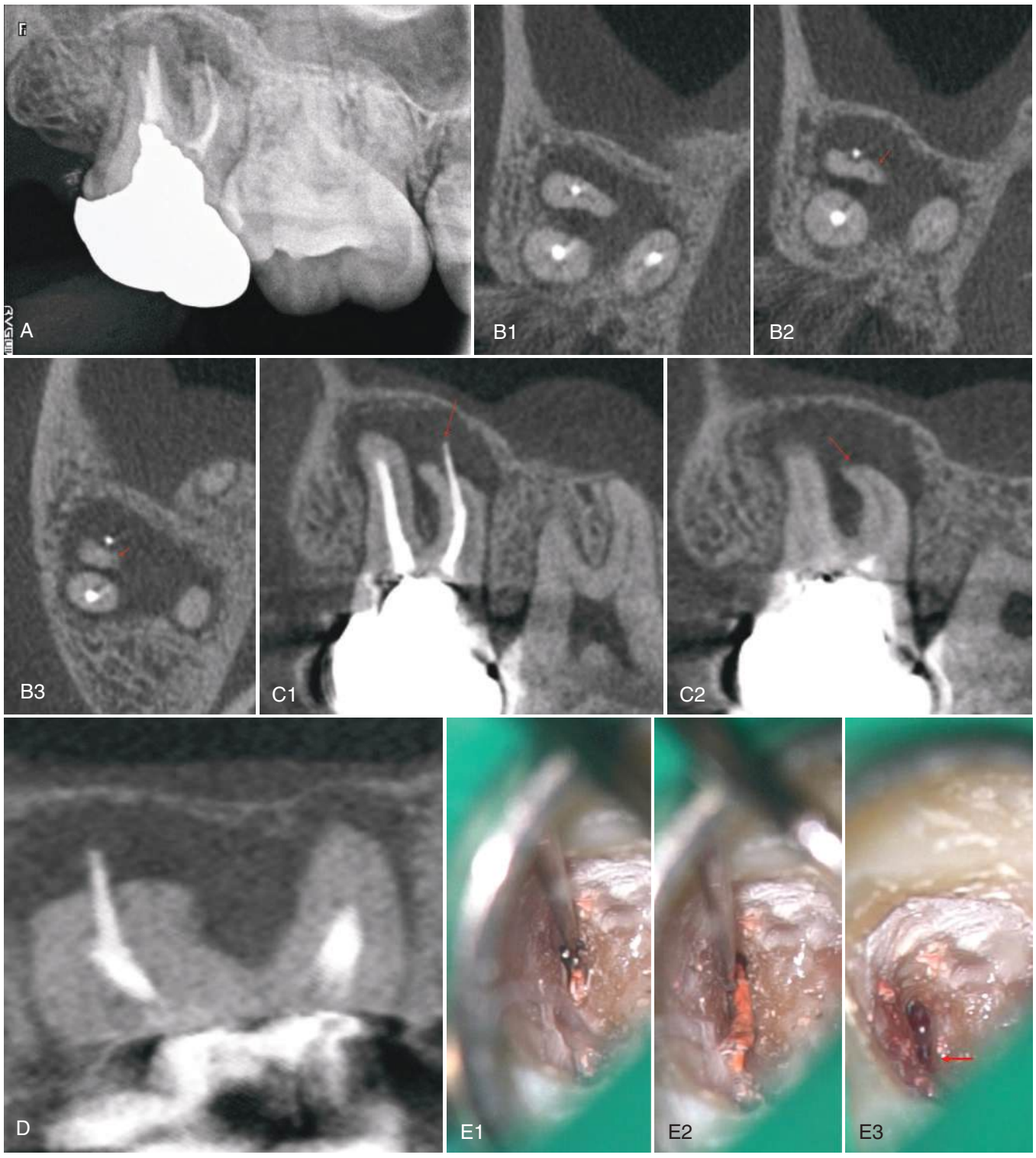


Fig. 20.30 A, Preoperative radiograph of a maxillary right second molar showing periradicular lesions. Note the root filling in MB is overextended beyond the radicular foramen. **B,** A series of cone-beam computed tomography (CBCT) images in axial view from the apical third to the radicular foramen (**B1** to **B3**) reveals an oval MB canal turning round toward the radicular foramen and the original pathway *(red arrows)* curves in a distopalatal direction with the root filling turning away from the original canal (off-center). **C1** and **C2,** A pair of CBCT images in sagittal view clearly show both the overextended root filling (*red arrow* in **C1**) and the presence of the distally curved original canal (*red arrow* in **C2**). **D,** CBCT image in coronal view also shows a large lesion associated with the MB canal, with the root filling extruded beyond the radicular foramen (perforation). **E1 to E3,** A series of microscopic images shows the overextended root fillings being removed with the TFRK-GPR (**E1** and **E2**) and the original canal (**E3,** *red arrow*) was located in the MB canal and enlarged with ultrasonics.

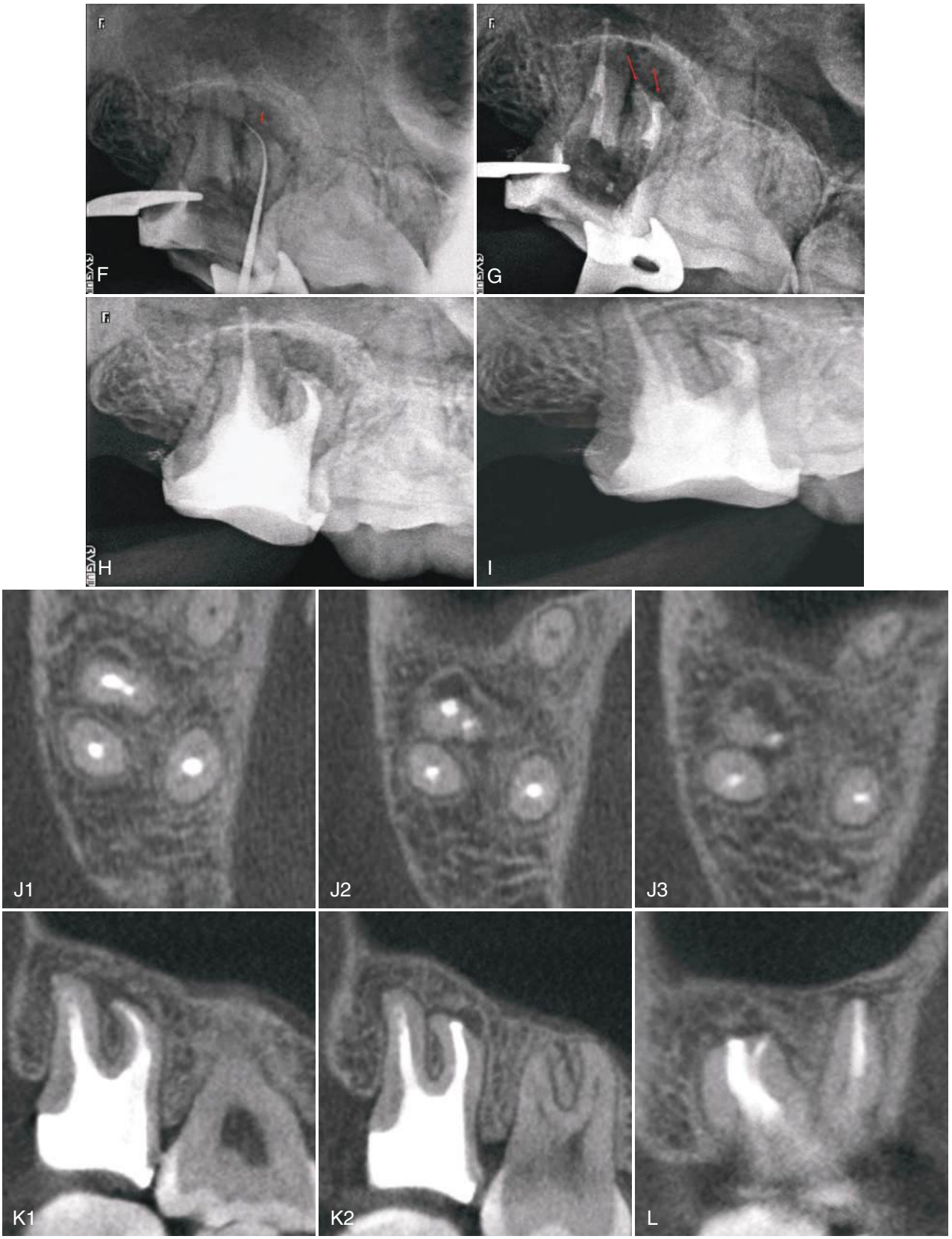


Fig. 20.30, cont'd F, Original canal was negotiated and then the apical patency was established with a precurved micro-opener followed by a precurved ProTaper S1. **G,** Postoperative radiograph showing the completed root fillings including the original canal and the ledged canal with perforation *(red arrows)*. **H,** Three-month follow-up showing healing in progress. **I,** Six-month follow-up showing nice periapical healing. **J,** A series of 6-month postoperative CBCT images in axial view from the apical third to the radicular foramen (**J1** to **J3**) reveals root fillings in both the original canal and the ledged canal with nice periapical healing. **K,** A pair of 6-month postoperative CBCT images in sagittal view (**K1** and **K2**) shows nice periapical healing compared to the preoperative CBCT images. **L,** Six-month CBCT image in coronal view also shows nice periapical healing.

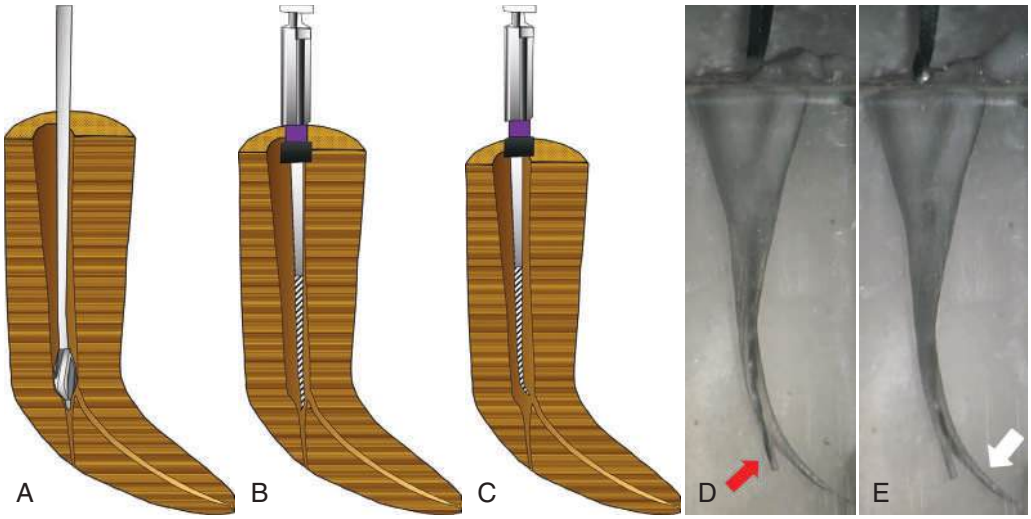


Fig. 20.31 **A,** Canal coronal to the ledge is enlarged with a #3 GG bur. **B,** Small straight rotary file (#10/.02 or #10/.04) rotating at 300 RPM or faster will not get into the original pathway. **C,** Small precurved rotary file rotating at 100 RPM or 150 RPM with 90-degree clockwise and 30-degree counterclockwise reciprocating motions will most likely get into the original pathway. **D,** Small precurved rotary file rotating at 300 RPM ended up going into the ledged canal *(red arrow)*. **E,** Small precurved rotary file rotating at 100 RPM was able to get into the original canal *(white arrow)*.

materials, necrotic pulp tissue, bacteria, medicaments, and irrigants have been associated with posttreatment inflammation and flare-up or the lack of radicular healing.[213,295,305,335] In vivo histologic studies show the most favorable histologic outcome when the root filling material remains within the radicular constriction. When gutta-percha is extruded into the periradicular tissue, there can be a severe inflammatory reaction.[265] However, other types of materials, such as ProRoot mineral trioxide aggregate (MTA) (DENTSPLY Sirona Tulsa Dental Specialties) and EndoSequence BC Sealer (Brasseler, Savannah, GA), are more biocompatible[1,57,111,117,167] and do not seem to produce much periradicular inflammation if extruded beyond the root of the tooth.[93,312]

Studies show that almost all instrumentation techniques or types of NiTi instruments promote the radicular extrusion of debris including root filling materials.[7,144,329,336] Furthermore, gutta-percha and Resilon filling materials obtained from teeth associated with refractory radicular periodontitis have indicated the presence of bacterial biofilm and *E. faecalis.*[122,214,223,296] Another study stressed the possibility of gutta-percha–centered *E. faecalis* biofilm as well as the role of tissue fluids such as saliva and serum in the biofilm formation.[108] *E. faecalis* adheres well to gutta-percha with and without canal sealers, with the biofilm formation more effective when it is in contact with serum in periradicular tissues. Because bacterial biofilm is considered the primary cause of chronic and recurrent endodontic infections,[341] it is crucial to keep gutta-percha filling materials within the canal system as much as possible to both prevent potential periradicular periodontitis and obtain the optimal results.[26,83,289]

Uncontrolled obturation can cause the extrusion of root canal filling materials into the maxillary sinus[86] or into the territory of the inferior alveolar nerve (IAN).[246] The systematic review shows most of the nerve injury cases with lateral condensation fully recovered over time, compared with none of the cases with vertical condensation that recovered.[273] Serious complications such as maxillary sinusitis, aspergillosis infection, paresthesia, dysesthesia, and similar neural complications may occur after extruding the root filling materials including a core carrier of an obturator into such sensitive areas.[30,173,203]

Endodontic-related paresthesia can be caused mechanically, thermally, chemically, or a combination of these via the passage of endodontic materials into the vicinity of the major nerves such as the IAN or its branches.[3,249] When the IAN is injured, the pain is not only located in the immediate periradicular area but may radiate and refer to the surrounding or distant structures.[222] Inflammation caused by mechanical irritation is likely to persist until the extruded materials are removed, and it may become permanent secondary to the mechanical, thermal, or chemical insult.[136] These extruded materials include root canal irrigants, sealers, and paraformaldehyde-containing pastes.[350] Most extrusion cases with resin-based sealers showed full recovery from nerve injuries (62%), but only 27% of those with paraformaldehyde based sealers showed full recovery.[273] The use of paraformaldehyde-containing pastes is not recommended for endodontic obturation[210,249] because paraformaldehyde is a polymeric hydrate of formaldehyde that releases formaldehyde gas and may cause permanent damage to the nerve when it is in contact with water.[114] On the other hand, extrusion of endodontic materials, even into the inferior alveolar canal, will not always result in altered sensation.[246] The possible explanation for this is that some extruded root canal sealants demonstrate their neurotoxic properties only when they come into direct contact with the individual fascicles through the pathway created by over-instrumentation, whereas they are less harmful to the sensory nerves when they are outside the epineurium.

Significantly higher or lower pH materials than those of body fluids (around 7.35) are also likely to cause cellular necrosis of tissues when they are in direct contact with these materials. The clinician must also consider the pH of some of the routinely used endodontic and related dental materials. It has been reported that an alkaline pH of root canal sealers could neutralize the lactic acid from osteoclasts and

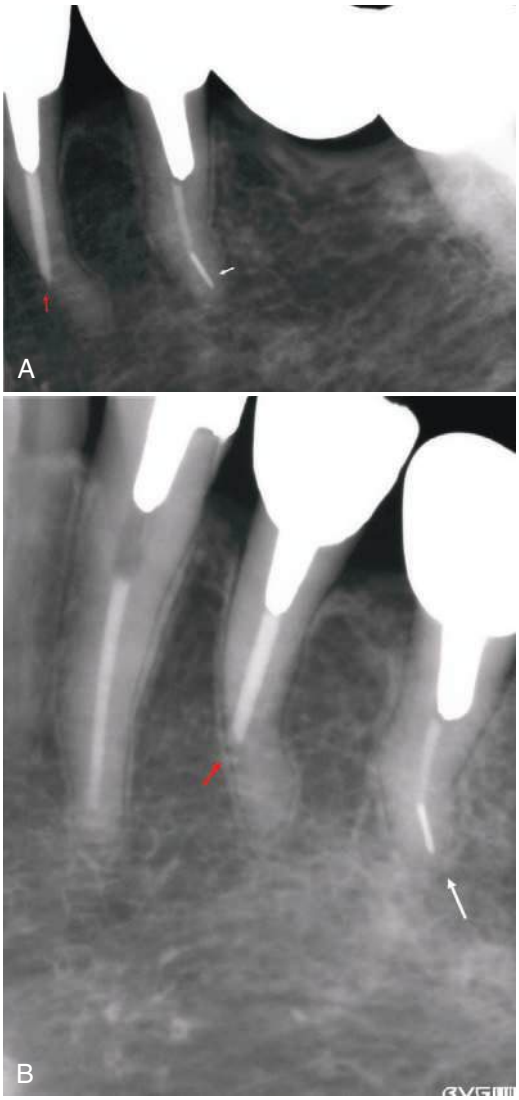


Fig. 20.32 A, Radiograph taken in 2010 for evaluation of the ledge *(red arrow)* in the mandibular left first premolar and the separated file *(white arrow)* in the mandibular left second premolar showing no radiographic periapical pathosis and both of the teeth were asymptomatic. **B,** Radiograph taken in 2019 for evaluation of those teeth showing the same without any sign of periapical pathosis *(red and white arrows)*.

prevent dissolution of mineralized components of teeth; therefore root canal sealers, especially calcium silicate–based cements or sealers, can contribute to hard tissue formation by activating alkaline phosphatase.[321] Commonly used endodontic medicaments are as follows[47,105,245,258,303]:

Formocresol: pH 12.45 +/− 0.02
NaOCl: pH 11 to 12
Calcium hydroxide (Calyxl): pH 10 to 14
ProRoot MTA: pH 11.7 to 7.12 at 3 hours to 28 days
Endosequence BC Sealer: pH 10.31 to 11.16 at 3 hours to 240 hours
MTA Fillapex: pH 9.68 to 7.76 at 3 hours to 168 hours
AH plus: pH 7.81 to 7.17 at 3 hours to 240 hours
Sealapex: pH 9.72 to 9.63 at 3 hours to 24 hours
Antibiotic-corticosteroid paste (Ledermix): pH 8.13
Pulp Canal Sealer: 8.07 to 7.55 at 3 hours to 24 hours
Eugenol: pH 4.34 +/−0.05
Iodoform paste: pH 2.90 +/−0.02

Neural damage from those higher or lower pH materials may be permanent and can cause neuropathic pain (i.e., dysesthesia) or anesthesia.

Orbital pain and headache may occur secondary to the local compression of overextended material into the maxillary sinus.[365] Sinus infection caused by zinc-oxide-eugenol–based filling materials extruded into the maxillary sinus is often associated with aspergillus growth and biofilms (Fig. 20.33).[21,22,112,175,274] Although aspergillosis of the paranasal sinuses is a relatively rare finding in nonimmunocompromised patients, it is known as an opportunistic infection.[166] Many studies have demonstrated that maxillary sinus aspergillosis is generally caused by extruded root canal filling material containing zinc oxide and formaldehyde.[21,22,175,188,319] However, even those extruded materials that are not considered toxic[315] may cause an additional foreign body reaction.

The systematic review shows nerve injuries caused by overextruded endodontic materials were reported to be 72% for the first or the second mandibular molars followed by the second mandibular premolars (19%), and none in maxillary teeth.[273]

CAUSES OF EXTRUSION OF OBTURATION MATERIALS BEYOND THE RADICULAR FORAMEN

The extrusion of obturation material that was thought to be the cause of endodontic failure is reported to be as low as 3%,[311] with the majority of causes thought to be more from coronal leakage. However, several researchers have seen a higher frequency of failure and unhealed periradicular lesions in cases where the root filling material had extruded beyond the radicular foramen without filling small anatomical spaces within the root canal system and harboring biofilms on the surface of the extruded material.[26,30,40,239,296] Overextension of the root canal filling materials occurs mainly for lack of material control.[43,126] The degree of extrusion of obturation materials into periradicular tissues depends on the techniques used for root canal preparation and obturation,[266] which include accidental over-instrumentation and incorrect canal working length measurement. Caution should also be taken when obturating a canal with open apices and larger minor foramen diameters, which are often found in cases with radicular inflammatory resorption. Warm vertical compaction techniques produce a homogeneous obturation that adapts well to the canal walls[290,362] but may result in the extrusion of gutta-percha filling into the periradicular tissues and may damage them mechanically and thermally.[43] Carrier-based obturation was also found to have more radicular extrusion of gutta-percha when compared with warm vertical compaction.[184] The type of canal shaping may also be a contributing factor. The canal shape created with the Profile GT system is more prone to extrusion of gutta-percha when obturated with the carrier-based technique than the canal shape created using Profile 0.06 instruments. The Profile GT system created a more parallel preparation coronally and a tapered preparation radicularly, whereas the Profile 0.06 system created a preparation with uniform taper from the apex to its most coronal extent.[266] Therefore possibly the more continuously tapered the canal shape is, the less obturation material is extruded beyond the radicular foramen.

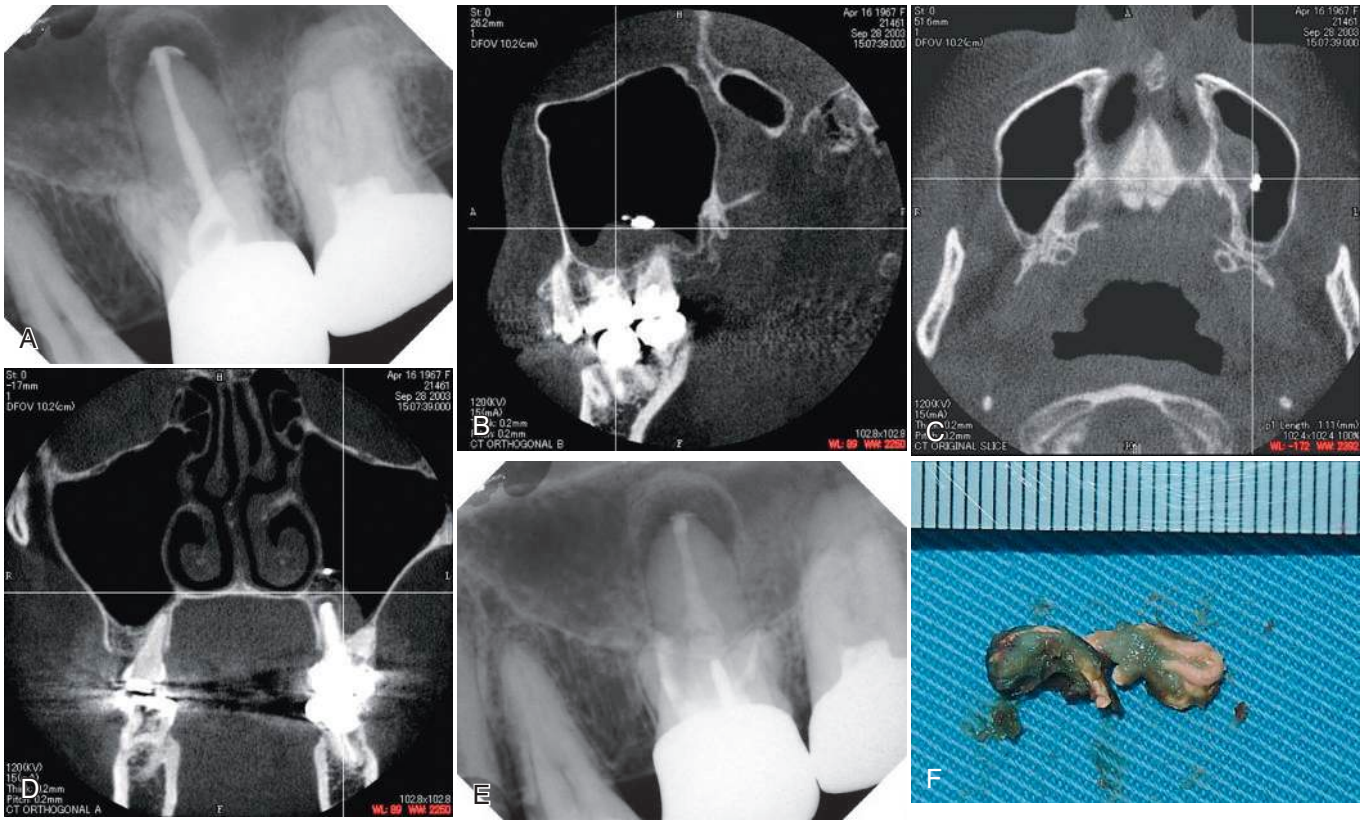


Fig. 20.33 Maxillary sinus aspergillosis. **A,** Preoperative radiograph showing a lesion associated with the palatal root, from which the extrusion of the root filling materials occurred in the maxillary sinus. **B,** Cone-beam computed tomography (CBCT) image in sagittal view showing the relation between mucositis on the maxillary sinus floor and the gutta-percha filling materials. **C,** CBCT image in axial view showing the relation between mucositis on the mesial wall of the maxillary sinus and the extruded root filling materials. **D,** CBCT image in coronal view showing the relation between the palatal root and the extruded root filling materials. **E,** Postoperative radiograph of orthograde root canal retreatment on the maxillary left first molar showing the mesiobuccal, distobuccal, and palatal root canals were adequately filled without overextension of the filling materials. **F,** Aspergillus was found on the filling materials retrieved from the maxillary sinus.

When endodontic retreatment is performed, irritants in the form of root filling materials can be extruded into the periradicular tissues.[7,144,329] Techniques involving push-pull filing strokes usually create a greater mass of debris than those involving some sort of rotational motions.[5,38,94,165,205,254]

MANAGEMENT OF OBTURATION MATERIALS EXTRUDED BEYOND THE RADICULAR FORAMEN

Nonsurgical Management of Extrusion of Obturation Materials

Methods to remove the gutta-percha filling material from the root canal include the use of heat-carrying instruments, ultrasonics, solvents, and rotary or reciprocating or hand files. Mechanized methods to remove filling materials are faster but may possibly be no different from manual methods in terms of the amount of remaining material in the root canal.[129,164,269,270,327] Heat-carrying instrument may damage the periradicular tissues if the removal attempts extend beyond the radicular foramen. The complete removal of gutta-percha from the root canal system has proven difficult.[129,327,368] Poorly compacted gutta-percha obturation can easily be removed in large masses of gutta-percha root fillings with micro-hand instruments under the DOM or Hedström files in combination with solvents such as chloroform,[59,154] although the extruded portion of root canal fillings will often remain in the periradicular tissue (Fig. 20.34).[59] The relative ease of gutta-percha removal suggests a failure of previous endodontic treatment, possibly related to poor obturation compaction or canal shaping, which resulted in both radicular extrusion and failure of the radicular seal. Comparatively, well-filled canals with radicular extrusion do not appear to have a significant failure rate.[15,154] On the other hand, the extrusion of the MTA root filling material into the periradicular tissues does not appear to negatively affect the periradicular healing due to the high biocompatibility and antibacterial properties that the calcium silicate–based cement possesses,[196,326] compared to gutta-percha filling materials (Fig. 20.35). Hence the MTA root filling extruded into periapical tissues does not have to be removed.

When gutta-percha is overextended into the periradicular space, removal can be attempted by inserting a new Hedström file, usually a #15 or #20, into the radicular obturation, using a gentle clockwise rotation to a depth of 0.5 to 1 mm beyond the radicular constriction. The file is then slowly and firmly withdrawn with no rotation, thereby engaging and removing the overextended material.[207] This technique works frequently, but care must be taken not to force the instrument, or further extrusion of the gutta-percha or

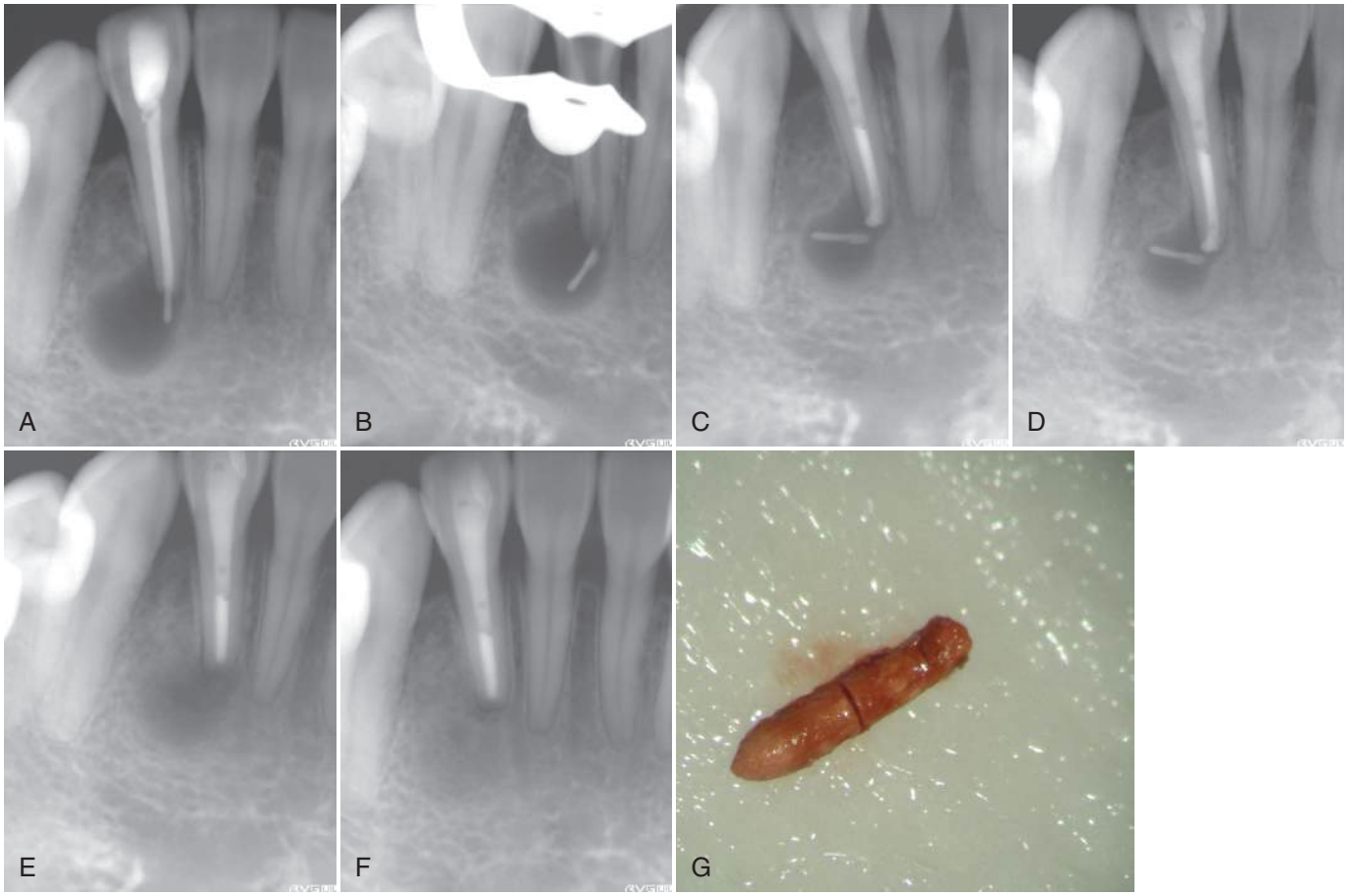


Fig. 20.34 Root filling materials extruded beyond the radicular foramen. **A,** Preoperative radiograph showing overextension of the root filling material with a large lesion associated with the root. **B,** Root filling in the canal was mechanically removed, but the extruded portion of it remained in the periradicular tissues. **C,** Mineral trioxide aggregate (MTA) was filled in the canal with the extruded portion remaining in the lesion. **D,** Six-month postoperative radiograph showing the reduced size of the lesion, but the lesion surrounding the extruded portion stayed the same in size for the next 6 months, indicating infection of the extruded gutta-percha point. The extruded portion was surgically removed from the lesion at this point. **E,** Three-month follow-up radiograph after the surgery showing the resolving lesion. **F,** Twelve-month follow-up radiograph after the surgery showing a more periradicular healing. **G,** Gutta-percha point retrieved from the lesion.

separation of the file may result. The overextended radicular portion of the gutta-percha should not be softened with heat or solvent, because this application can decrease the likelihood of the Hedström file engaging the obturation, hindering its retrieval.[318] In addition, care should be taken not to overextend the file into the area of the IAN canal, which could cause irreversible damage to the associated nerves.

The use of rotary NiTi files that are specifically designed for gutta-percha removal with active tips and more aggressive cutting may not be superior to conventional files.[348,349] Hand instrumentation may be associated with longer times to remove root filling materials than rotary/reciprocation systems. However, no iatrogenic errors were reported with hand instrumentation.[269,270,271] Hand instruments used for the removal of root fillings include H and K-files, Micro-debriders (DENTSPLY Maillefer, Ballaigues, Switzerland), EGPR-L/R/U/D (Den-Mat Holdings LLC, Lompoc, CA), and the TFRK gutta-percha removal instrument (TFRK-GPR, DELabs). Single-file systems such as the self-adjusting file (SAF) (ReDent-Nova, Ra'anana, Israel), Hyflex EDM "one file" (EDM) (Coltene/Whaledent, Alstatten, Switzerland), WaveOne Gold (WOG) (Dentsply Sirona, Ballaigues, Switzerland), RECIPROC blue (VDW, Munich, Germany), and XP Shaper (XP) (FKG, La Chaux de Fonds, Switzerland) have improved mechanical properties that can potentially make them suitable for gutta-percha removal. The XP is a minimal taper file that can expand beyond its core size to adapt to the anatomy of the root canal space and may be more efficient in removing root filling materials than other single-file systems as it provides more space for the debris to be displaced.[16,304] The XP has been reported to provide a shorter operation time and showed the optimum results without any added untoward events when operated at a higher speed (3000 rpm) when compared to other single-file systems.[16] A full rotation motion may also be a more suitable cutting motion in root filling removal than a reciprocating motion.[16] Rotary retreatment files may be more advantageous in the amount of debris extrusion when compared with manual techniques using Hedström files with chloroform,[124,337] although all instrumentation techniques resulted in some radicular extrusion.[146,310] The literature shows more debris extrusion produced by instruments used with a reciprocating movement than instruments used with a continuous rotary movement.[44,45,51,336,340] Likewise, the rotary XP shaper instrument has been reported to extrude a significantly smaller amount of debris compared with reciprocating Reciproc Blue

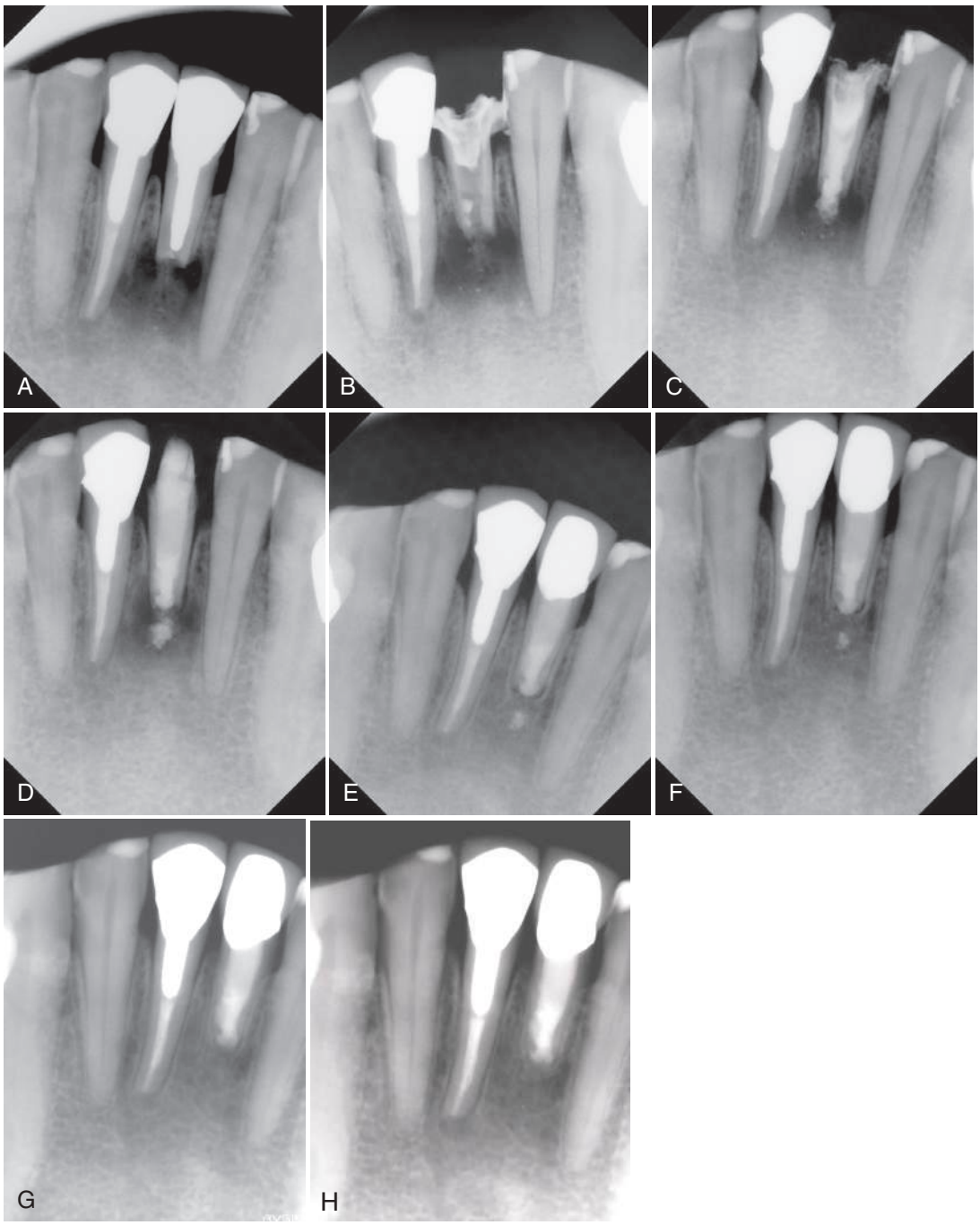


Fig. 20.35 **A,** Preoperative radiograph showing a large periradicular lesion associated with the root of the mandibular left incisor. **B,** Interoperative radiograph showing the cast post and the root filling in the canal was mechanically removed. Calcium hydroxide was placed as an intracanal dressing on the first appointment. **C,** Postoperative radiograph on the second appointment showing the canal was obturated with mineral trioxide aggregate (MTA), which resulted in overextension of the material into the periradicular tissues. **D,** Six-month postoperative radiograph showing the significantly reduced size of the lesion. **E,** One-year postoperative radiograph showing a nice periradicular healing with hard tissues formed on the radicular end of the MTA filling, resulting in a longer root aside from a residual of the MTA root filling extruded into the periradicular tissues. **F,** Three-year postoperative radiograph showing the tooth has maintained healthy periradicular tissues with the extruded MTA filling reduced in size. **G,** Twelve-year postoperative radiograph showing the tooth has maintained healthy periradicular tissues with the extruded MTA filling completely dissolved in the bone. **H,** Sixteen-year postoperative radiograph showing the tooth has maintained healthy periradicular tissues with healthy lamina dura surrounding the root.

instruments because of its expanding snake-like structure, which may be ideal to mechanically remove root fillings extruded into periradicular tissues (Fig. 20.36). Furthermore, a combination of both rotary instruments for initial quick removal of gutta-percha and hand instruments to refine and complete the cleaning of the canal, especially in the radicular third of the root, shows the better protocol for obtaining clean canal walls during endodontic retreatment.[28,151] It has been reported that the use of a solvent after mechanical retreatment procedures may enhance root filling removal.[50] The use of a solvent, such as chloroform, may help to dissolve the gutta-percha root fillings remaining in

both the periradicular tissues and the root canal system after mechanical removal attempts using a micro-hand instrument or an efficient rotary single-file system such as the XP (Fig. 20.37 *D*–*G*). In addition, chloroform has been reported to be capable of reducing the intracanal levels of *Enterococcus faecalis*.[79,209] However, when chloroform is used during the early stages of gutta-percha removal, more filling material will remain in the canal and some of it may be extruded beyond the radicular foramen.[146,198] Although the carcinogenicity of chloroform in humans is suspect, chloroform is regarded as a safe and effective endodontic solvent when used carefully.[56,204]

Unlike rotary instruments, the EGPR-L/R/U/D hand instruments and the GPR instrument are 30 mm in length, allowing them to remove obturation materials not only from the radicular third of the canal but also from the periradicular tissues through the radicular foramen. The micro-debriders are designed to be used in conjunction with a DOM. They have a Hedström cutting configuration with a standard .02 taper in sizes 20 and 30 with 16 mm of cutting flutes, which may not be long enough to reach the gutta-percha filling material beyond the radicular foramen. Another option to reach the root fillings extruded into the radicular tissues is to use an Endo Handle (Vento Endo, Logan, UT) which is designed to be used under the DOM with a pre-bent 30-mm hand file attached to the locking chamber.

First, a small hole must be created between the canal wall and the gutta-percha filling with a small diameter ultrasonic tip to facilitate the root filling removal with the hand gutta-percha removal instrument placed into the hole. It is ideal to remove the root filling in one large piece from the periradicular tissues. Care should be taken not to push the



Fig. 20.36 **A,** Preoperative radiograph of the mandibular right first molar showing extruded root filling materials into periapical tissues *(red arrows)*. **B1** and **B2,** Cone-beam computed tomography (CBCT) images in axial view showing the perforation site from the mesiobuccal canal into the furcation *(red arrow)*. **C,** CBCT image in sagittal view showing extruded root fillings from the distal root *(red arrow)* and the perforation *(white arrow)*. **D1** and **D2,** CBCT images in coronal view showing the mesial root without extruded root fillings *(red arrow)* (**D1**) and the distal root with extruded root fillings *(red arrow)* (**D2**). **E,** Magnification of the MB canal with perforation *(red arrow)*. **F,** Interoperative radiograph showing removal of the intracanal root fillings.



Fig. 20.36, cont'd G, Magnification of the MB canal showing removal of the extruded root fillings from the perforation with an XP-Endo Shaper. **H,** Magnification of the distal canal showing removal of the extruded root fillings from the distal root end with an XP-Endo Shaper. **I,** Magnification of the root canals after the removal of extruded root fillings *(white arrows)*. **J,** Postoperative radiograph showing all the root canals including the perforation were successfully filled with mineral trioxide aggregate (MTA). **K,** Three-month postoperative radiograph showing reduced periapical lesions. **L,** Ten-month postoperative radiograph showing nice periapical healing. **M,** Ten-month postoperative CBCT image in sagittal view showing furcal healing. **N,** Ten-month postoperative CBCT image in sagittal view showing nice periapical healing around the perforation site *(white arrow)*. **O,** Ten-month postoperative CBCT image in coronal view showing nice periapical healing associated with the mesial root *(white arrow)*. **P,** Ten-month postoperative CBCT image in coronal view showing nice periapical healing associated with the distal root *(white arrow)*. **Q** and **R,** Ten-month postoperative CBCT image in axial view showing periapical healing associated with the perforation site *(white arrows)*.

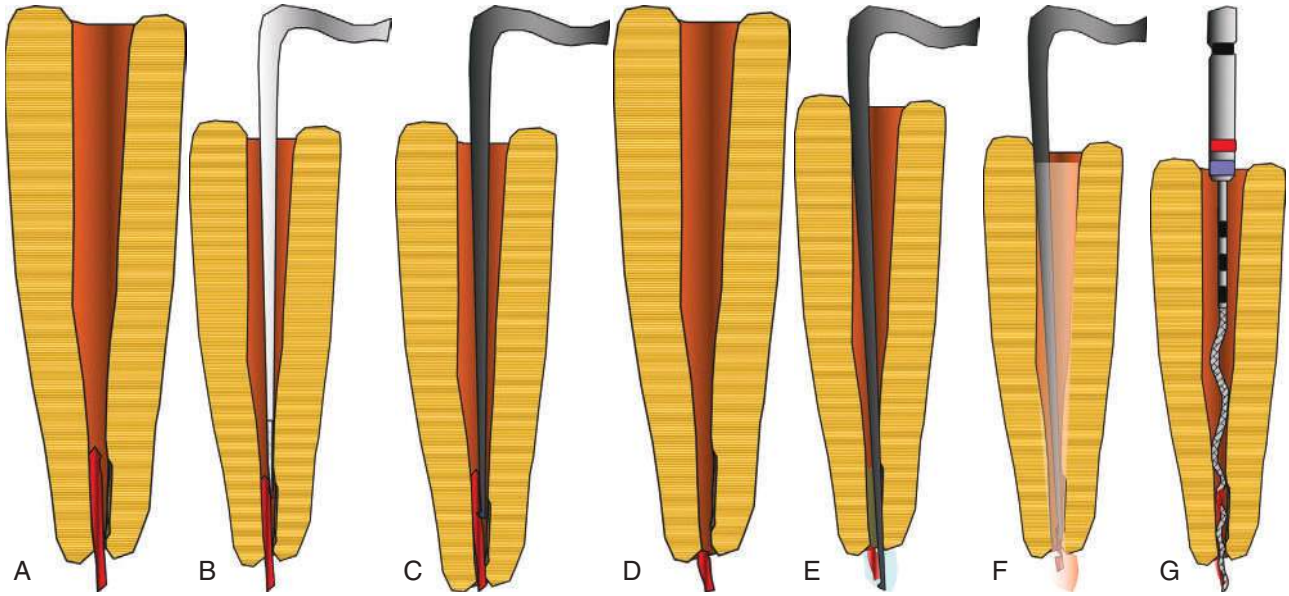


Fig. 20.37 Diagrams showing removal of gutta-percha filling materials extruded beyond the radicular foramen. **A,** Solid gutta-percha filling extruded beyond the radicular foramen. **B,** Small space is created between the canal wall and the gutta-percha filling with a thin ultrasonic tip (such as TFRK-6/12 and TFRK-S ultrasonic tips). **C,** Gutta-percha removal hand instrument, such as the TFRK-GPR and the EGPR, is brought into the space with the tiny projection pressed against the gutta-percha filling to pull it out. **D,** Extruded gutta-percha root filling portion separated from the coronal portion within the canal. **E,** Chloroform or some other solvent is filled in the canal to agitate it beyond the widened radicular foramen with a gutta-percha removal hand instrument or expandable rotary files to dissolve the extruded portion. **F,** Gutta-percha filling material is diluted, dissolved with the hand instrument in push-and-pull motions, and is irrigated out with a minimally invasive irrigant such as saline. **G,** Extruded gutta-percha filling material can also be braided out with those expandable rotary files with caution not to break them or not to damage the inferior alveolar nerve if close to it.

root filling materials in a radicular direction with those micro-instruments especially for a canal with large radicular foramen. Then the gutta-percha filling is scraped and taken out from the canal wall with the micro-instrument.

The EGPR instrument has a tiny projection on one side of the tip for scraping and engaging the gutta-percha filling left on the canal wall. The overall diameter of the tip is 0.3 mm and the length of the projection is 0.1 mm. This hand instrument works like a scraper on the canal wall. When the radicular portion of the EGPR comes down to the bottom of the hole created, the projection is directed to the gutta-percha filling so that it can be hooked on the projection pressed against it. Then the gutta-percha filling should come apart in large pieces from the canal wall (see Fig. 20.37, *A–C*).

On the other hand, the TFRK-GPR instrument has a tiny conical harpoon-shaped tip that is sharp and small enough to sneak into a narrow space on the canal wall and hook the gutta-percha on the rim on the return stroke. The tip portion of this instrument has a .40 taper with the minimum tip diameter of 0.1 mm and the maximum diameter of 0.3 mm with a 0.2-mm diameter shank supporting the tip portion. The 0.1-mm gap between the 0.3-mm and the 0.2-mm diameters works practically as a tiny circular projection over the shank portion that can hook a mass of gutta-percha fillings. The conical tip with the maximum diameter of the shank portion being 0.6 mm makes it possible to place the instrument into the canal beyond the curve just like a Hedström file. The gutta-percha filling can also be hooked by pressing the projecting rim against it, and then should be removed in large pieces from the canal wall.

When a small amount of gutta-percha filling remains on the inside curve or in the radicular third of the canal after the mechanical removal attempts, chloroform should be filled in the canal and the GPR instrument can be used to agitate it in short push-and-pull motions in the canal, or even outside the canal if root filling is extruded beyond the radicular foramen, to actively dissolve the residual root filling. The mean percent total values of residual material ranged from 43.9% to 0.02%, with the vast majority of studies reporting values less than 10%.[275]

MANAGEMENT OF TISSUE DAMAGE CAUSED BY EXTRUSION OF ROOT CANAL FILLING MATERIALS

When root canal filling material extends into the periradicular spaces, tissue damage may occur. The majority of these cases are managed by nonsurgical methods such as analgesics, cold packs, corticosteroids, and antibiotics.[90] Dexamethasone is a corticosteroid that has been widely used in dentistry, and it seems to decrease periradicular inflammation caused by a foreign body[210] with minimal side effects.[65] If the tissue damage is extended to the IAN or maxillary sinus, then a different approach may be necessary.

Sinus Perforation

There is a close proximity of maxillary premolars and molars to the sinus, and the apices of these teeth can be less than 2 mm from the floor of the sinus.[78] Pathologic destruction of the sinus floor may predispose a maxillary sinus communication,[294] producing an oroantral communication,[98,190,278,355] also known as an OAC. In general,

exposure of the sinus lumen is well tolerated,[155,238] with no overt repercussions on healing. One investigator[355] found no significant difference in the healing of the mucosa in patients with and without intraoperative perforations and regeneration of the membrane, complete with cilia, in approximately 5 months following complete removal.[25] However, introduction of foreign materials during the access or apical resection/manipulation can initiate thickening of the sinus mucosa with symptoms of maxillary sinusitis.[84]

Conventional radiographs appear to be of limited value in predicting the potential for OAC, with the position of the root tip, periapical lesion, or outline of the lesion not statistically significant in that respect.[224] In contrast, a CBCT survey of the intended surgical field is much more informative, providing accurate measurements of the osseous thickness, distance to the sinus floor, and presence of pathosis.[33]

Maxillary sinus complications secondary to the extrusion of root canal filling material need to be managed differently. Approximately 10% of all patients with chronic sinusitis are found to have an aspergilloma.[120,193] It has been suggested that root canal treated teeth with overextension of the root canal sealer into the maxillary sinus might be the main cause for aspergillosis in healthy patients.[21,22] Surgical intervention is usually the most predictable method for removing extruded root canal filling material and allowing for adequate healing secondary to IAN or sinus complications.[333]

Inferior Alveolar Nerve Injury

Sensory nerves are extremely sensitive to physical, radiation, chemical, thermal and ischemic damage. Despite their capacity to heal, if the injury causes sufficient damage, then persistent posttraumatic sensory neuropathy will result. The presenting symptoms of sensory neuropathy include; a neuropathic area within the dermatome where the surgery or injury took place,[259] allodynia (pain to touch or cold), hyperpathia (continued pain or altered sensation after stimulus removed), hyperalgesia (increased pain to a painful stimulus for example exaggerated response to sharp blunt discrimination test despite ongoing numbness). Additional clinical symptoms may include ongoing, spontaneous and/ or elicited altered sensation (paraesthesia or dysesthesia), a tingling or formication (ants crawling on the skin), burning or shooting pain or electric shock. The sensory deficit to mechanical function indicates damage to larger myelinated fibers, whereas sensor deficit to noxious and thermal stimuli indicates damage to small diameter afferent fibers of the spinothalamic tract.[256]

Clinicians must use appropriate medical language and not refer to sensory neuropathy as 'paraesthesia' which is a clinical sign and not a diagnosis. Generally, neurologists refer to either hypoesthetic (predominantly anesthetic [numb] with reduced mechanosensory function) or hyperesthetic (ongoing, spontaneous and/or elicited pain or dysesthesia) sensory neuropathies. Posttraumatic neuropathic pain (PTNP) is a recognized neurologic diagnosis (ICOP).[154a]

Iatrogenic trigeminal nerve injuries remain significant and complex clinical problems. The impact on the patient of posttraumatic sensory neuropathy can be functionally and psychologically severe, particularly when painful and iatrogenic. The trigeminal nerve underpins our very existence, providing sensory protection to our eyes, nose, mouth and brain. Thus the affective and emotional reaction to the threat or experience of trigeminal pain is thus proportionally immense. Altered sensation and pain in the orofacial region may interfere with speaking, eating, kissing, shaving, applying make-up, tooth brushing and drinking—in fact, just about every social interaction we take for granted. Thus these injuries have a significant negative effect on the patient's quality of life and the iatrogenesis of these injuries lead to significant psychological effects.[308] The management of patients with posttraumatic sensory neuropathy is complex and 'curative' treatment is unavailable, thus prevention of these injuries is paramount. Trigeminal nerve injury is the most challenging consequence of contemporary dental surgical procedures with significant medical-legal repercussions (see also Chapter 27).[46] An approximate estimation places the incidence if IAN injury at 5%, with approximately one third of these patients afflicted with neuropathies, including neuropathic pain.[262] Common causes of injury to the IAN are listed in Box 20.2.[258]

The highest incidence of reported damage to the inferior alveolar and lingual nerves is associated with third molar extractions.[18,140,141] Local anesthesia was the second most common cause of nerve injury, and its exact mechanism can be confusing. It may be physical (needle damage, epineural/perineural hemorrhage) or chemical (local anesthetic contents). Reports have implicated certain local anesthetics of high concentrations (prilocaine 4% and articaine 4%) with IAN neuropathies (see also Chapter 6).[17,131,142,248] A relatively small percentage of IAN injury cases (8%) are associated with an endodontic procedure.[261]

IAN damage has been suggested to occur in up to 1% of mandibular premolars that receive root canal treatment.[87] In one retrospective study examining cases of paraesthesia related to endodontic treatment of mandibular premolars, the incidence was 0.96% (8/832).[172] In a survey of 2338 patients, 7% sustained chronic neuropathic pain after a single endodontic procedure.[170] Most reports of endodontic-related nerve injuries are case reports with very little analysis of the clinical impact for the patient, risk factors, or management issues. To date, there are four case series reports on inferior alveolar nerve injuries (IANIs) related to root canal treatment (RCT) including 61 cases,[246] eight cases,[172] four cases,[292] and one case series reported on the treatment of 11 cases.[121] These case series predominantly

Box 20.2 Common Causes of Damage to the Inferior Alveolar Nerve

1. Local anesthetic injections
2. Third molar surgery
3. Implants
4. Endodontics (extrusion of the materials used in endodontic treatment, overinstrumentation, elevated temperatures)
5. Ablative surgery
6. Trauma
7. Orthognathic surgery

focus on the outcomes surgical exploration and irrigation of damage nerves.

The mechanisms of sensory neuropathy are complex, and they are likely not only due to mechanical damage including perforation or compression. Pogrel[247] concluded that the severity of the injury was proportional to the process by which the injury was created, and these processes included scalpel blade wounding/transection, crushing, and stretching damages. These would be analogous to the dissection and reflection of the flap, improper retraction location/application, and postoperative edema. However, endodontics is complicated with use of chemicals and fillers that provide additional risk of nerve injury. It is likely that chemical (pH of dental materials), ischemia, hemorrhage (the iron in hemoglobin irritates nerve tissue), and thermal factors may compound the nerve injury resulting in persistent neuropathy. Compression of peripheral sensory nerves over 6 hours can evoke nerve fiber atrophy.[298] Ischemia alone without direct nerve damage will cause sufficient neural inflammation and damage to cause permanent nerve injury within only 6 hours.[230] The IAN may be damaged when an increase in temperature proximal to the IAN is greater than 10°C.[87] The type of sealer cement used may influence the symptoms of IAN damage.[216]

The three primary factors responsible for IAN damage during endodontic treatment are as follows:

1. Chemical toxicity and mechanical pressure produced by leakage of sealers into areas close to the mandibular canal are other potential mechanisms for nerve injury related to endodontic procedures, with the IAN and mental nerve being most affected. Cytotoxic materials employed either in the preparation (e.g., irrigation liquids, intracanal medicaments)[2,109] or the obturation of the canal space[277] as the materials most commonly associated with these complications are those containing paraformaldehyde.[3] In addition, most endodontic materials have very high pH levels, resulting in immediate nerve damage likely to be permanent (Fig. 20.38).[37]
2. Mechanical factors: repeated insertions of endodontic instruments through the apex into the IAN and surrounding tissues[70]
3. Thermal factors: inappropriate/prolonged heat application[89]

RISK ASSESSMENT FOR THE PREVENTION OF ENDODONTIC NERVE INJURY

Preoperative assessment is crucial in prevention of nerve injury related to endodontics (Box 20.3). Prevention is the key to successful treatment. Damage, either directly to the IAN or to the vessel(s) that nourish it, will result in sensory events that may be transient or permanent. Pretreatment radiographic assessment using periapical images is mandatory; in critical cases, CBCT may be more appropriate and precise in establishing proximity (Fig. 20.39). Assessment should include the following:

- Proximity of the apex of the tooth to the IDC. Lower molar teeth where root apices are is close proximity to the Inferior dental canal and or mental foramen. Mandibular second molars are most commonly associated with

Box 20.3 Key Points for Prevention of Nerve Injury Related to Endodontics

- Aways assess the proximity of the tooth apex to the inferior alveolar nerve (IAN).
- Always assess additional risk factors in root morphology and bone factors contributing to leakage of chemicals adjacent to the IAN.
- Take all the steps to minimize the apical breach during endodontic treatment if any of the previous risk factors exists. This will allow leakage of high pH chemicals into the IAN, rendering permanent chemical damage to the IAN.
- Consider immediate extraction of the tooth and generous lavage of the socket along with adjunctive medical treatment (described in this chapter) if the patient develops sensory neuropathy within 3 days of endodontic treatment.

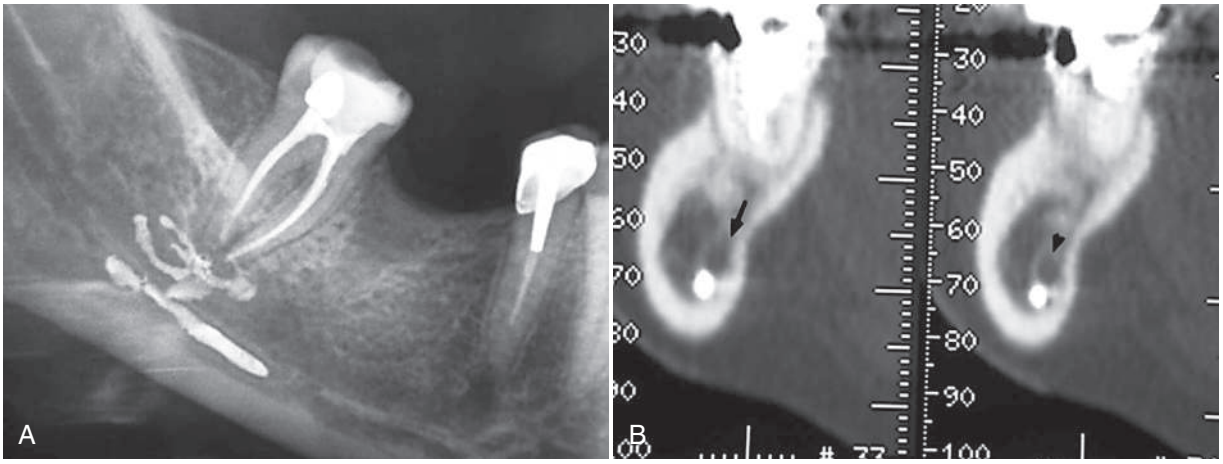


Fig. 20.38 A, Panoramic radiograph demonstrating significant overfilling and extension in the radicular vacuoles and overlying the mandibular canal. **B,** A coronal CT section demonstrating that the extruded material is lateral and below the mandibular bundle *(arrows)*. The patient's sensory complications were temporary. (From Tilotta-Yasukawa F, Millot S, El Haddioui A, et al: Labiomandibular paresthesia caused by endodontic treatment: an anatomic and clinical study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102:e47, 2006.)

this population, but cases involving treatment of mandibular first molars and premolars have also been reported.[172] Careful consideration should also focus on the distance between the root apices and the mandibular canal. For mandibular first molars, this distance varies between 1 and 4 mm, and for mandibular second and third molars, the distance can be less than 1 mm.[334] With mandibular premolars, the proximity of a mental foramen should always be considered.[219] In addition, a study using CBCT has shown that the mesiobuccal root apex of maxillary second molars is frequently in very close proximity with the sinus floor (see Fig. 20.39).[228]

- Tilotta-Yasukawa and colleagues (2006) determined the proximity of the apex of the premolars and molars in relation to the mandibular canal, as well as the relationship between the IAN and its corresponding artery, with the goal of understanding how endodontic filling material spreads through the bone to penetrate the mandibular canal. They observed that the distance between the dental apex and the mandibular canal was more variable (and generally greater) for the first molar than for the second and third molars (1 to 4 mm vs. <1 mm [35 cases in 40 mandibles examined]). The presence of bone between the apices and the IAN canal does not necessarily protect against injury to the nerve. Research has provided an enlightening explanation to this puzzle (Figs. 20.40–20.42).[334] Using fresh cadaver specimens (mandibles), researchers examined the true anatomic relationship among the root apices, IAN, and the artery. They drew the following conclusions:
  - There was never any compact layer of cortical bone surrounding the mandibular nerve sheath. In some cases, a slightly denser cancellous bone was found, but it was replete with multiple perforations that would not afford significant resistance to penetration.[137]
  - The proximity of the IAN canal to the apices of the second molars was < 1 mm, and it was more variable (1 to 4 mm) with first molars. This could explain the higher percentages of paresthesias reported with these teeth.

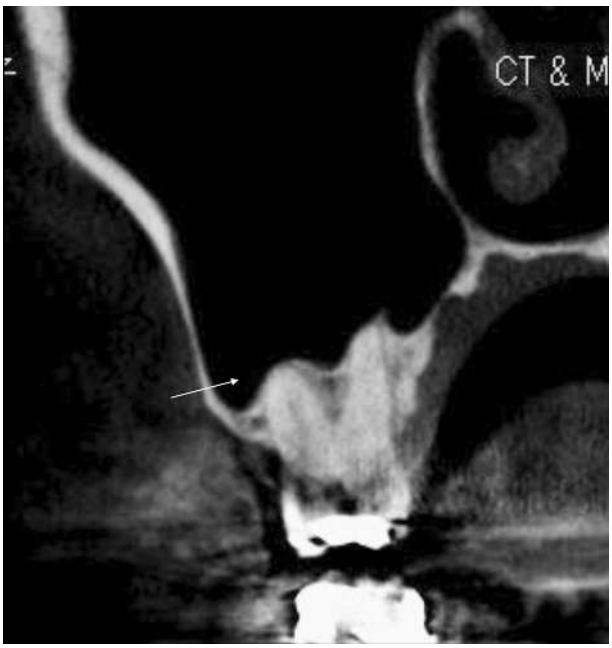


Fig. 20.39 Cone-beam computed tomography (CBCT) image in coronal view showing that the mesiobuccal root apex of a maxillary second molar is in close proximity with the sinus floor.

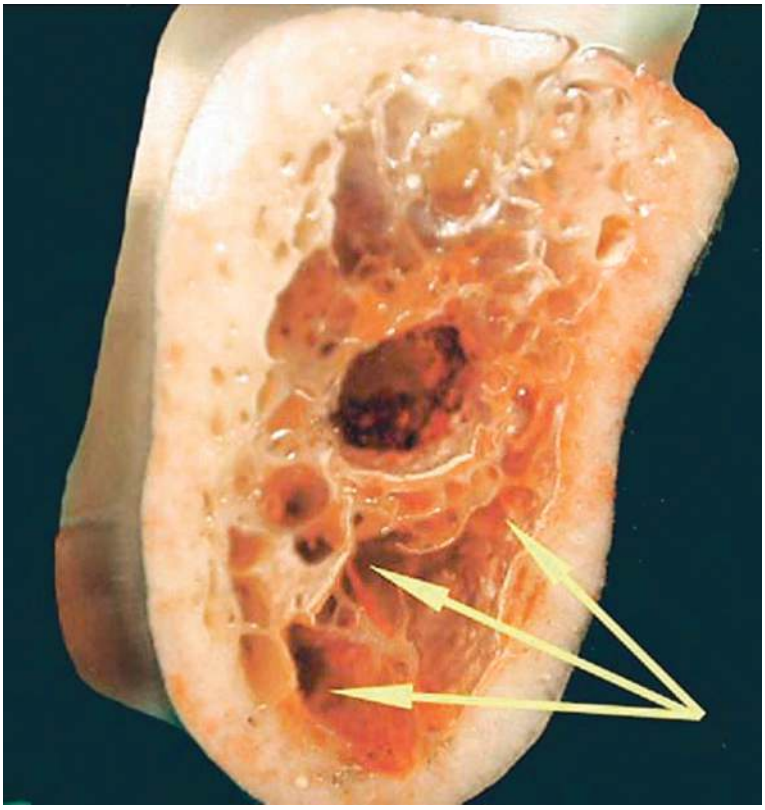


Fig. 20.40 Cross section of mandible in the molar region showing the presence of bone vacuoles *(yellow arrows)*. (From Tilotta-Yasukawa F, Millot S, El Haddioui A, et al: Labiomandibular paresthesia caused by endodontic treatment: an anatomic and clinical study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102:e47, 2006.)

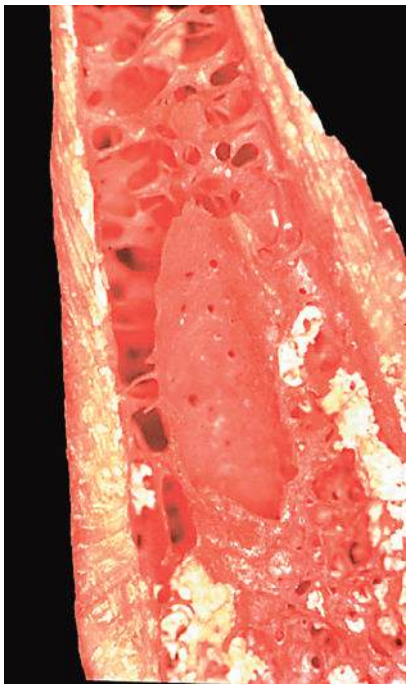


Fig. 20.41 Longitudinal mandible cross section in the molar region showing the spongy bone cover of the neurovascular bundle. Note the multiple fenestrations and honeycombed trabeculations. (From Tilotta-Yasukawa F, Millot S, El Haddioui A, et al: Labiomandibular paresthesia caused by endodontic treatment: an anatomic and clinical study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102:e47, 2006.)

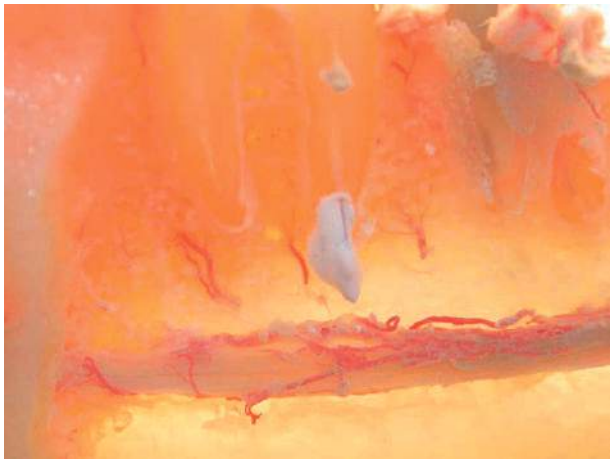


Fig. 20.42 Overfill of the distal root under direct observation. Note the proximity of the extruded material to the artery and nerve bundle (pedicle). (From Tilotta-Yasukawa F, Millot S, El Haddioui A, et al: Labiomandibular paresthesia caused by endodontic treatment: an anatomic and clinical study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102:e47, 2006.)

- The mandibular artery ran along the superomedial side of the IAN up to the first molar, and then crossed on the upper side until the mental foramen. Perforation of this vessel, with the concomitant hemorrhage exerting pressure on the pedicle, could feasibly produce a paresthetic event via the resulting ischemia.[71]
- Only with root apices in direct contact with the IAN canal were extrusions able to penetrate the nerve. If cancellous bone was interspersed between the apices and pedicle, the spread of the extrusion through the trabeculations remained at the periphery of the sheath.
  - The low density of trabeculations and presence of numerous vacuoles provided an avenue for the diffusion of extruded irrigation fluids and obturation materials toward the IAN bundle. In a tooth with apical resorption, the foramen is frequently enlarged[363]; these two factors in concert could explain the occurrence of sensory symptoms in the absence of clinical or radiographic signs of overexpression or close proximity to the mandibular nerve.
- Predisposing radiological factors that may indicate a higher risk of an adverse incident during root canal treatment or cause an extrusion of endodontic filler/hypochlorite accident
  - Resorption defects where extent is not identified, such as internal/external communicating with root canal and external surface of the root
  - Suspicion of a perforation communicating with the external root surface
  - Root fracture where there could be a potential communication of the root canal with external root surface
  - Sclerosed root canal
  - Dens invaginatus
  - Presence and the diameter of bone lesions of endodontic origin may also influence the occurrence of paraesthesia, especially when associated with the premolars and lower molars.[176]

OPERATIVE TECHNIQUES TO MINIMIZE NERVE INJURIES

Surgical endodontic procedures need to be designed to minimize trauma to nerves including the IAN, mental nerve, and mental foramen. Although the mental foramen is most often located apically to the second premolar,[217,241,242,243] it can vary in anterior-posterior locations,[162,220] coronal-apical locations,[95] and number.

Careful presurgical assessment via radiographs and CBCT provides critical information on the size, location, and spatial relation of the planned surgery to nearby trigeminal nerves.[32] Before exiting the mandibular canal, the nerve often loops anteriorly, then superior and posteriorly in the apical portion of the premolar region.[119] If an apical procedure is planned in an area commensurate with this anatomic variant, then CBCT is warranted and strongly advised.

Controlled reflection of the tissue overlying the suspected location of the foramen will diminish the incidence of transection. Cautious reflection of the flap will reveal evidence of tissue "funneling" at its base (Fig. 20.43); this is the telltale sign of the exiting nerve bundle.[221] This exiting tissue, upon discovery, can be marked with a dot of gentian violet or methylene blue dye. By highlighting it in this fashion, the surgeon can avoid unintentional impingement with a retractor. The tissues should be minimally distended and the underside of the flap kept well hydrated.

Prevention of over instrumentation: Every attempt should be undertaken to confine the instrumentation and irrigation processes to within the canal space; damage from extrusions can be chemical or mechanical in origin,[292] and

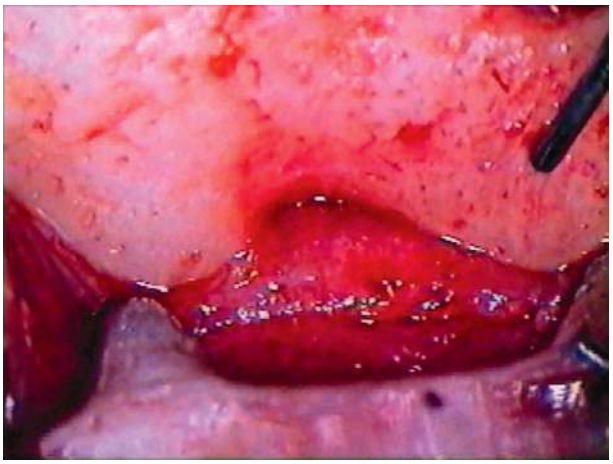


Fig. 20.43 Full-thickness reflection of the flap in the region of the premolar root apices reveals the mental nerve exiting the foramen; note how the tissue "funnels" from the reflected flap to the foramen. (From Tilotta-Yasukawa F, Millot S, El Haddioui A, et al: Labiomandibular paresthesia caused by endodontic treatment: an anatomic and clinical study. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 102:e47, 2006.)

toxic products can easily diffuse through the trabecular spaces and vacuoles surrounding the nerve bundle. This latency in diffusion, and the potential onset of paresthetic symptoms, dictates close monitoring for 72 hours postoperatively in cases where extrusion is apparent.[246]

Prevention of over filling with beyond apex extrusion of filler or treatment chemicals: Any tooth requiring endodontic therapy, which is in close proximity to the IAN canal, should require special attention, including: NaOCl (as irrigation) and calcium hydroxide (as sealant; calcium hydroxide medicaments breach the canal roof, precipitating a vascular bleed resulting in hemoglobin irritation of the nerve due to the iron content).[31,87]

Prevention of thermal injury: Imprudent heat application during thermoplastic compaction of gutta-percha may also lead to IAN injury.[31]

The clinician must be aware and record any events that may indicate operative nerve injury, including extreme pain during LA IDB, canal instrumentation, irrigation, medication, or filling; and sudden and profuse hemorrhage arising from the apex of the tooth

Appropriate postoperative periapical radiographs must be taken, in accordance with endodontic clinical guidance, to check for any extrusion of dressing or filling materials into the inferior dental canal or around the mental foramen.

Home check should be undertaken, particularly for high-risk cases, to assess signs of persistent or new neuropathy.

MANAGEMENT OF NERVE INJURIES

The priority in management of patients with iatrogenic nerve injuries is for them to understand what has happened, the likely prognosis and possible support, and medical or surgical interventions. Management should be holistic and of the patient with the nerve injury and not just the neuropathy itself.[261] Pain, functional problems, and psychological impact should be assessed using questionnaires.[82] Grötz and colleagues[121] reported on 11 patients with endodontic-associated neuropathy and their management, reporting that the neurologic findings were dominated by hypoaesthesia and dysesthesia, with 50% of patients reporting pain. Initial x-rays showed root filling material in the area of the mandibular canal. Nine cases were treated with apicectomy and decompression of the nerve: in two cases, extraction of the tooth was necessary; only one patient reported persistent pain after surgery.

It is important to recognize that IANI can occur due to local anesthetic block injections and the clinician can often discriminate between endodontic and local anesthesia-caused nerve injuries by careful questioning and careful clinical neurologic assessment. Mapping of the neuropathic area will discriminate between inferior dental block (IDB) and endodontic nerve injury (Figs. 20.44 and 20.45).[49]

Diagnosis is based upon the signs and symptoms described earlier in this section. Posttraumatic sensory neuropathy is a

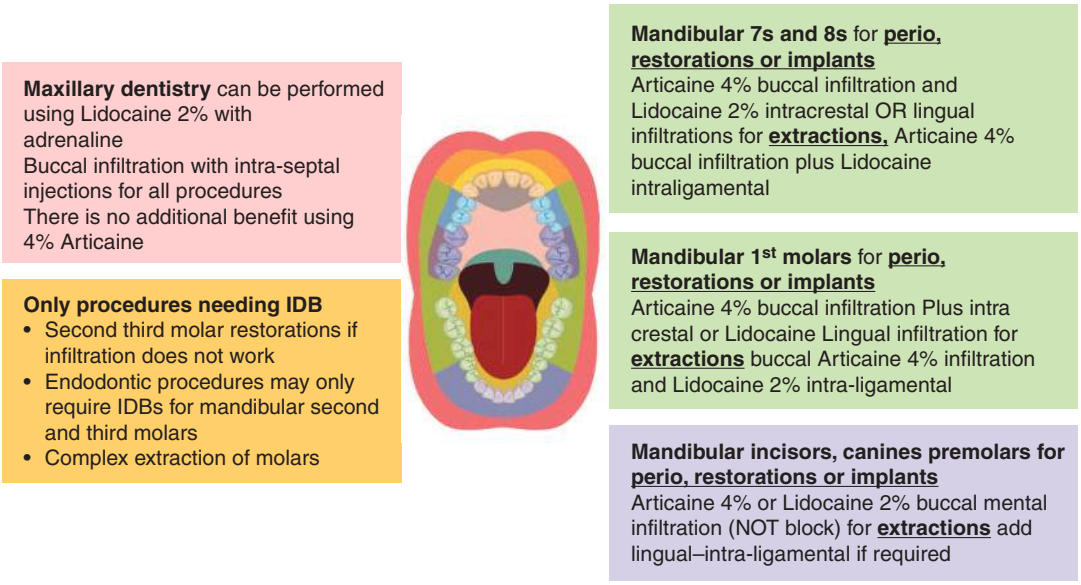


Fig. 20.44 Infiltration dentistry is dependent upon the site and procedure.

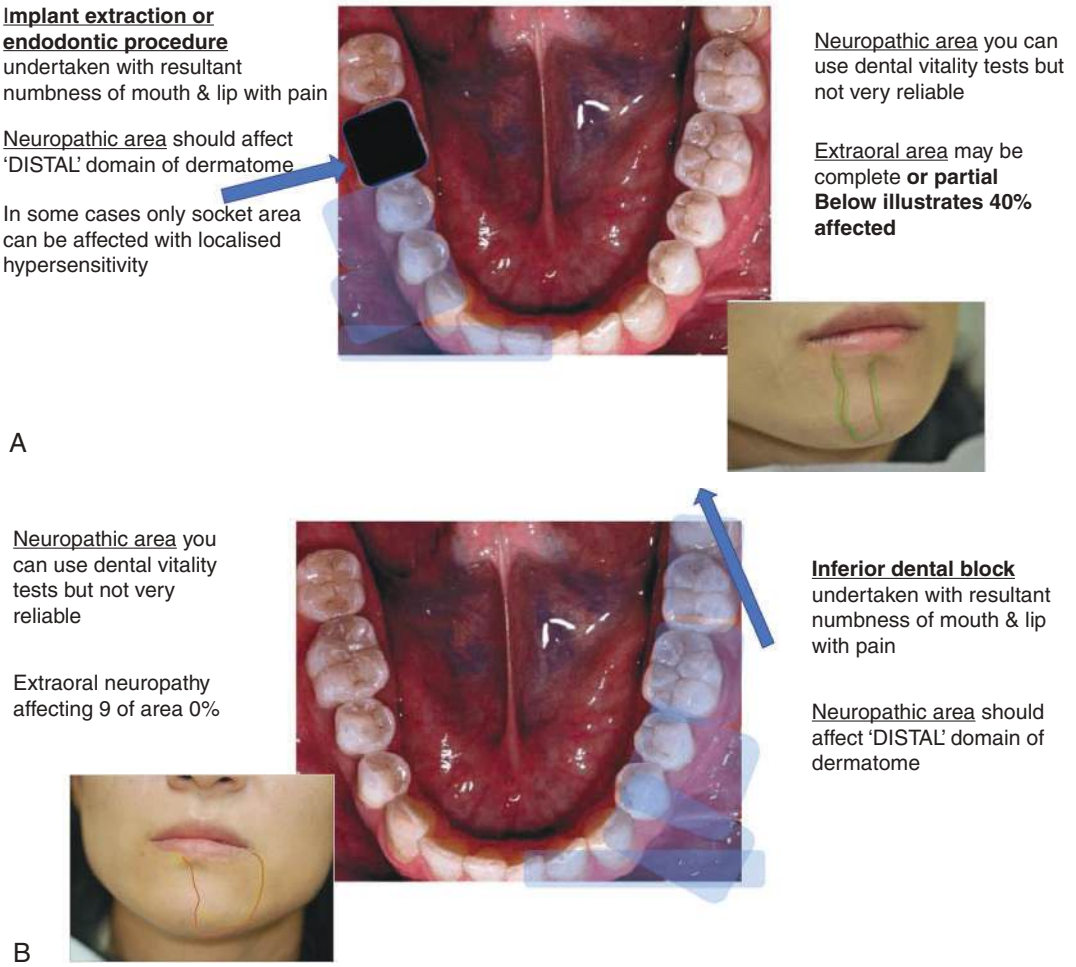


Fig. 20.45 A, Nerve injury proximal to endodontically treated tooth. Cause = endo. **B,** Inferior dental block nerve injury initially will cause neuropathy of the whole dermatome.

clinical diagnosis and knowledge of the preoperative risk along with the posttreatment radiographs should be sufficient in confirming the diagnosis. Further radiographic assessment, especially CBCT, may be useful in revealing the extent of location of extruded materials in relation to the mandibular canal or the maxillary sinus) but is unlikely to improve treatment outcome.[31,103,113] Chemical nerve injury may not be obvious radiographically: If the patient is suffering from neuropathy after the local anesthetic has worn off and the postoperative radiographs confirm that there is no radiopaque material in the canal, chemical nerve injury may be presumed

Transient paresthesia may occur in cases where a protracted surgical procedure, inadvertent stretching, or excessive postoperative edema has occurred. Appropriate reevaluation appointments include neurosensory testing and mapping the outline of the sensory deficit. Return of sensation within the first few weeks indicates a neuropraxia and may indicate prognosis but cannot predict full recovery. It is not possible to distinguish between temporary or permanent nerve injuries early postoperatively, thus treatment must be based upon the patient's symptom severity, the extent of neuropathy, and the evidence of injury (recent treatment and radiographic overfill). Timing of intervention is paramount and ideally high-risk patients should be followed up for 24 to 60 hours post endo to check for onset of symptoms of neuropathy.

Assessment

The patient must be assessed for pain, functional limitations, and psychological impact. Further confirmation of the neuropathic area and mechanosensory testing is important to establish the extent and symptoms of the neuropathy. The patient must be assessed holistically, including their history of the event and if it is related to the initiation of pain.[261] Ensure that the pain history excludes preexisting neuropathic pain, including the following:

- Severe pain during procedure (funny bone pain)
- High-level postsurgical pain (indicative of nerve injury)
- Ongoing pain, altered sensation, and/or numbness
- Functional problems
- Psychological issues. Important questions about the mechanism and duration of the neuropathy will drive the timing and type of management.[260]

Necessary investigations include:

- Radiological: LCPA; CBCT necessary posttrauma
- Neurosensory to confirm that the presence of a neuropathy and distribution correlates with potential nerve injury

- Diagnostic local anesthesia blocks may be useful in evaluating the potential of some peripheral pain management strategies when medical management is unsuccessful.

Recent developments of medical resonance neurography may enhance the ability of the clinician to assess the trigeminal nerve injury early postoperatively, allowing for earlier surgical intervention, if required.[375]

IMMEDIATE INTERVENTION

If nerve section or severe injury is known or suspect, likely due to scalpel or drill, then immediate repair must be undertaken. If the transected ends are closely approximated following the injury, either naturally or surgically, proliferating axons will attempt to bridge the gap and, if they find a perineural tube to populate, gradually restore function. However, in cases where the gap is beyond a critical length, a neuroma may develop at the proximal end. Neuromas consist of dense fibrous tissue intermingled with nerve fascicles, with a disproportionate proliferation of unmyelinated fibers. Treatment, if initiated within 3 months of the initial event, often entails resection of the neuroma and repair by tension-free anastomosis or nerve grafting.

EARLY INTERVENTION

If the patient exhibits neuropathic signs during instrumentation (severe neuralgia) or after the local anesthesia has worn off, posttreatment radiographs should be reevaluated to observe the possibility of any obturation material extending into the IAN canal. Devine et al.[72] reported that there may be delayed development of sensory neuropathy, usually 2 to 3 days, likely due to chemical leakage through the treated tooth apex, resulting in chemical nerve injury which will not be apparent on a posttrreatment radiograph. Three months after the IAN injury, permanent central and peripheral changes occur within the nervous system subsequent to injury, which are unlikely to respond to surgical treatment intervention.[367]

There is a strong correlation between duration, origin, significance of the injury, and the prognosis for resolution of the paresthesia. The longer the irritation persists, either chemical of mechanical, the more fibers are involved in a degenerative process and the more likely the paresthesia will become permanent.[284] If injury to the nerve is demonstrated by postoperative neuropathy and posttreatment radiographs (not for chemical leakage) confirm the presence of a significant extrusion or overinstrumentation into the IAN canal, management entails the surgical removal of the offending material as soon as possible if the patient is symptomatic.[246,258,262]

Management should be performed by immediate removal of endodontic materials within 24 to 48 hours via periapical surgery, tooth extraction, or surgical debridement, whichever will be the most effective option with the least potential damage to the IAN.[75,178,246,258] If the patient wishes to retain the treated tooth and overfill is present in the inferior dental canal (IDC), then immediate mandibular buccal corticotomy under general anesthesia may be advised, with removal of overfill and hours of neural lavage with saline.[245a]

However, in spite of the best efforts to eliminate the causative agent, the symptoms may persist because of the associated chemical damage to the nerve. If sensory improvement has not been evident after 2 months, with or without repair during interventional surgery, then the prognosis for recovery is poor, and steps should be taken to alleviate the patient's concerns through counseling and systemic medications for chronic pain.[261]

Counseling and medical therapeutic management are indicated within 48 hours for injuries caused by endodontic treatment or local anesthetic injections.[99]

WATCH AND WAIT?

If the patient presents with a minimal neuropathic area which is tolerable and hyposensitive (nonpainful) then intervention may not be justified. Decisions about intervention have to be based on the patient's understanding of the risks and benefits. Although the prognosis in the case of the IAN damage caused by resorbable or inert material is favorable, as it is dissolved over time, the toxicity of the material, whether chemical or pH, may cause permanent damage if left in place.[222] In addition, pressure from the material on the IAN may be caused by a hematoma or inflammatory, which can also contribute to temporary or permanent IAN damage.[90,222] The prognosis of the IAN damage depends on the severity of the direct injury to the nerve, the toxicity of the irritant, the amount and speed of its resorption, and how rapidly the causative agent is removed.[222,226,246] Because the paresthesia can be of both a mechanical and a chemical nature, removal of the excess material might not be sufficient if the nerve fibers have already undergone degeneration from a chemical irritant. Nevertheless, paresthesia caused by an instant irritation of the nerve, like a minor overinstrumentation or swelling from infection or minor inflammation, usually subsides within days. If no signs of healing are observed within 3 to 6 months, the chances of healing are considered much lower, although normal sensory function might still return after this time.[113,334]

If the opportunity for immediate repair or early decompression has been missed, then the patient must be treated therapeutically, which may include medical and psychological interventions.

Psychological interventions may include counseling, mindfulness, acceptance, and commitment as well as cognitive behavioral therapies for patients with nerve injuries, which is very effective (there is limited evidence for success of this treatment for endodontic-related IANIs, but evidence does support psychological therapies for chronic pain and IANIs).[261]

Early medical management to minimize acute surgical neural inflammation using protocols undertaken for other acute sensory nerve injuries (Gatot, Tovi, 1986; Grötz et al., 1998). Steroids may maximize recovery of peripheral nerve injury (evidenced in spinal nerve injuries) using prednisolone (15 mg for 5 days, 10 mg for 5 days, and 5 mg for 5 days) and high-dose nonsteroidal antiinflammatory drugs (NSAIDs) (600 mg ibuprofen) as long as the patient has no contraindications to NSAIDs (gastric or peptic ulceration or allergy to NSAIDs) and a timely referral is made to an appropriately trained oral or maxillofacial microsurgeon. If the opportunity for early repair is missed, long-term therapeutic pain management will be necessary.[197a] Vitamin B complex and magnesium supplements may have some beneficial effects for neural recovery. Antibiotics are not recommended for nerve injuries unless infection related

neuropathy is suspected but, in these cases, the tooth should be endodontically treated or extracted.

Regarding late medical management of chronic pain associated with endodontic treatment. Oshima[225] reported that 16 out of 271 patients presenting with chronic orofacial pain were diagnosed with chronic neuropathic tooth pain subsequent to endodontic retreatment. Most of these patients were treated for maxillary teeth. Seventy percent of the patients responded to tricyclic antidepressant therapy, which highlights the importance of establishing whether the patient has neuropathic pain. In Renton and Yilmaz's study,[261] all the patients presented too late for surgical decompression or it was not indicated. Thus two patients were managed with oxcarbazepine for neuralgic pain elicited with touch or cold and with topical clonazepam intraorally to manage the severe gingival discomfort. Two patients were prescribed topical 5% lidocaine patches (12 hours on nocte and 12 hours off daily) for debilitating mechanical allodynia in the extraoral dermatome of the IAN, causing pain and functional problems. This is a treatment used successfully for patients with chronic orofacial pain, particularly those with mechanical or cold allodynia of the face.

Recommendations for the treatment of neuropathic pain are reported; Medical guidance is provided by Alonso-Ezpeleta et al.[9] Surgical intervention using direct repair or allograft cadaveric grafts has limited impact on trigeminal neuropathic pain,[376] and other interventions for neuropathic pain have been summarized.[77] Overall, the management of neuropathic pain remains unsatisfactory in that the side effects of medications are significant. Recent reports of Botoxin A and Lidocaine patches may improve the patients' quality of life with a low evidence base, but most importantly, acknowledgement and holistic support for the patient with iatrogenic nerve injuries must not be underestimated, and the prevention of these nerve injuries is paramount.

Cervicofacial Subcutaneous Emphysema

Subcutaneous emphysema is defined as the penetration of air or other gases beneath the skin and submucosa, resulting in soft-tissue distention. A specific type, termed *cervicofacial subcutaneous emphysema* (CFSE), is a relatively rare occurrence and may be limited to traumatic, iatrogenic, or spontaneous events.

The first recorded instance was reported by Turnbull in 1900.[342] It details the account of a musician inducing CFSE by playing his bugle immediately after the extraction of a mandibular tooth; the subsequent swelling resolved a few days after he ceased playing the instrument. Indeed, most early reports of CFSE usually follow tooth extractions and were the result of activities the patient engaged in that raised intraoral pressures.[300] With the introduction of air-driven handpieces, there was an increased risk of CFSE.[11,67,370] Although tooth extraction, especially of mandibular third molars, remains the most commonly reported reason for CFSE,[53,67,145,276,366,370] it can also result from restorative treatment,[52,104,158,320,344] periodontal surgery,[99,309] crown and bridge treatments,[116,374] and root canal therapy.[29,149,161,237,323,344,361]

Compressed air, delivered through an air syringe, a high-speed handpiece, or a combination of the two during the operative procedure is associated with 71% of the cases.[139] A more recent study[14] reviewing 47 computed tomography (CT)–documented cases of CFSE found the exclusive cause was the high-speed handpiece in 66% of these patients. Other means of introducing air into the facial spaces include the following:

- Use of the dental laser (Er:YAG) in the gingival pocket: compressed air is used to cool the tip but is of sufficient pressure to dissect the sulcular wound.[208]
- Hydrogen peroxide beyond the canal space: most often reported when the fluid is introduced through a perforation in the canal[29] or overinstrumented apex.[161]
- Lack of rubber dam isolation: a compressed air syringe was implemented to increase visibility in these molar retreatments, with air infiltration through the gingival sulcus.[76,169]
- Barotrauma secondary to Boerhaave syndrome: spontaneous rupture of the distal esophagus as a result of intractable vomiting, causing an increase in intraluminal esophageal pressure.[123]

The deposition of air beyond the canal space is a function of both the diameter of the canal at the apex[81] and the position of the syringe tip.[307] An air emphysema has the potential to spread along the facial planes in a similar manner (Figs. 20.46 and 20.47). Air entering the parapharyngeal and retropharyngeal spaces can lead to soft-tissue infections,[91] airway compromise,[263] optic nerve damage,[41] and even death.[68,264] The majority of the cases, however, present as a soft, skin-colored swelling without redness that occurs during or shortly after the dental appointment. The cervicofacial swelling is always associated with crepitus (a crunchy feeling to the tissues upon palpation), which may not be palpable before a latency period of several hours. The more serious presentation of pneumomediastinum is characterized by dyspnea with a brassy voice, chest or back pain, and Hamman sign (a bubbling or crunching sound caused by air or movement accompanied by cardiac pulsation).[130] Secondary complications such as pyogenic mediastinitis or necrotizing fasciitis must be considered, and appropriate means taken to evaluate and manage them (blood cultures, broad-spectrum intravenous antibiotics).[76] Differential diagnosis should include an allergic reaction (more pronounced dermal signs prior to cardiac/respiratory symptoms), hematoma (a rapid accumulation of fluid without initial discoloration and without crepitation), and angioneurotic edema (a defined, circumscribed area of edema, often preceded by a burning sensation in the mucous membranes or skin).[99]

Treatment is largely empirical and is dependent on the extent of the CFSE. Antibiotics are usually prescribed due to the nonsterility of the air supply and the possibility of necrotic debris or microorganisms forced into surrounding tissues.[263] Administration of 100% oxygen via a nonrebreather mask is reported to hasten the resolution of the emphysema because the oxygen, which replaces the trapped air, is more readily absorbed at the site.[169] More severe presentations warrant hospitalization with intravenous antibiotics, where the extent and regression of the CFSE can be closely monitored using CT scans and complications can be managed more effectively.[36] Resolution, barring any severe complications, is characteristically rapid and uneventful,



Fig. 20.46 A, Facial presentation after the rubber dam was removed. **B,** Close-up view reveals marked distention of the left suborbital region, with ptosis of the eye and loss of the nasolabial fold. There was marked crepitus upon palpation, but the patient was pain-free. **C,** She was placed on an antibiotic regime, as per the protocol, 2 days after the incident, the emphysema has resolved and the tissues are normal in color and texture.

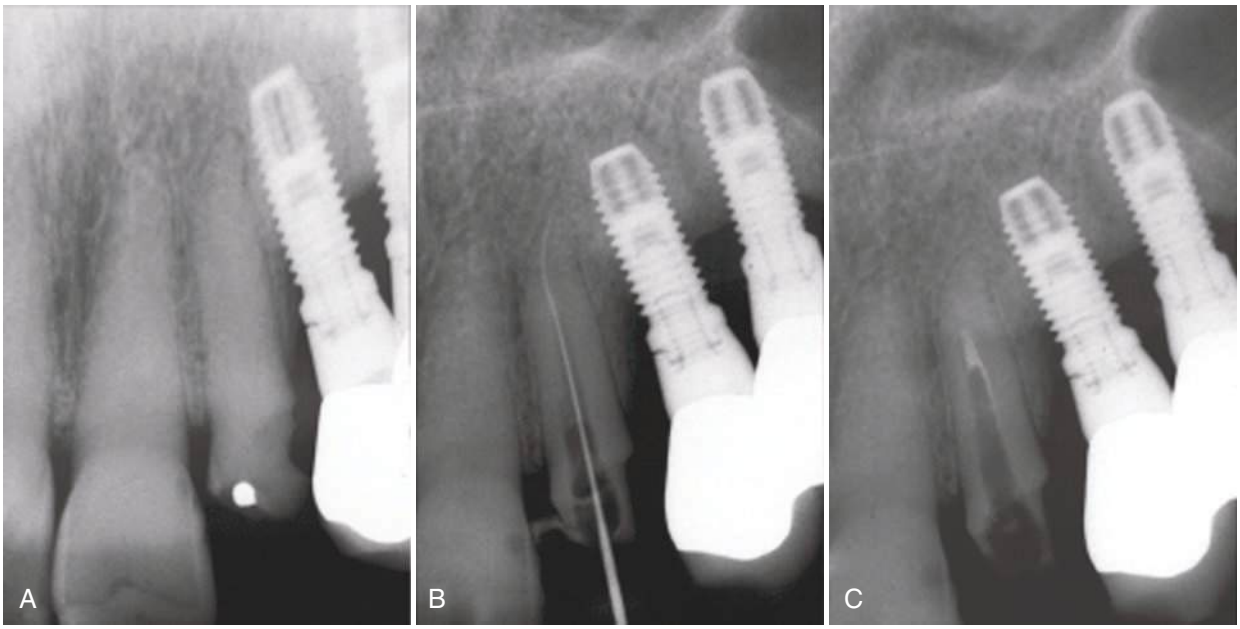


Fig. 20.47 A, Preoperative radiograph tooth #10[158]; dystrophic calcification has obliterated the pulp chamber and the coronal half of the canal space. **B,** A #10 file inserted to estimated working length; the file is actually through a facial perforation of the root at the radicular extent of the ultrasonic excavation. Repeated drying of the canal space using an air syringe forced air through the perforation site and precipitated the emphysema seen in Fig. 20.26, *A* and *B*. **C,** The radicular perforation site was sealed with mineral trioxide aggregate (MTA) before the canal space was obturated. The tooth has remained asymptomatic since the incident.

with the patients returning to their normal appearance within 4 to 7 days. A reevaluation of the cause of the emphysema should be examined and corrected before treatment resumes, especially in instances of a perforation.

From an endodontic perspective, minimizing compressed air at the site, from either a high-speed handpiece or an air syringe, is key to preventing this consequence. Isolation with a rubber dam, the use of paper points or high-volume aspiration for drying canal spaces, and avoiding the use of hydrogen peroxide during the procedure will also reduce the risk of introducing or entrapping air beneath the overlying tissues. For endodontic microsurgery, the use of a rear-exhausting high-speed handpiece for root resection as well as ultrasonics for root end preparation would decrease the likelihood of precipitating a CFSE during root end procedures.[19]

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